Small soft core uP Inventory

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Opencore and other soft core processors

_uP_all_soft folder	opencores or prmary link	status	author	style / clone	data	inst sz	FPGA	repor ter	com ents	LUTs ALUT	Dff	LUT?	mults	blk ram	F max	date	tool ver	MIPS /inst	clks/ inst	KIPS /LUT	ven dor	SOC	src code	#src files	top file	doc	tool chai	fltg pt	Hav'd	max dat	max inst	byt ead	# inst	adr mo	# reg	pip e	start year	last revis	secondary web link	note worthy	comments
1802-soc	https://github.com/scottll		Scott Baker	1802	8	8															L	Y	vhdl	14	soc	Y		N		64K	64K	Y	~60		16		2016	2026	https://en.wikiped	1802 CPU + UART + Timer + I/O Ports	
8051	https://opencor	alpha	Simon Teran, Jakas	8051	8	8	kintex-7-3	James	tunrec	1744	617	6	1		111	##	14.7	0.33	4.0	5.3	ALX		verilog	32	oc8051_tc	Y	yes	N		64K	64K	Y					2001	2016		8051 core includes several on-chip peripherals, like timers and counters	
8051	https://opencor	alpha	Simon Teran, Jakas	8051	8	8	**zu-3e**	James	area o	1424	645	6			242	##	v21.1	0.33	4.0	14.0	ALX		verilog	32	oc8051_tc	Y	yes	N		64K	64K	Y					2001	2016		8051 core includes several on-chip peripherals, like timers and counters	
16bit_processc	https://github.com/prant		Md Badiuzzaman Pran	MIPS	16	16												0.67	1.0				schematic														2018	2018	https://prantoamt	course project, schematics only	simple up with well done schematics
16bit_relay_up	https://relaiscor	WIP	Peter Prikasky	accum	16	16												0.67	4.0				schematic					N		64K	64K	N	16	3	4			2023	https://hackaday.i	min Accum uP: PC, Accum, SR & IR reg	Excel macro simulator; imm, abs & indirect adr'
16bit_up	https://github.com/anijjar		Aman Nijjar	risc	16	16	artix7_35	James	Unable to create vivad			6				##	v23.2	0.67	1.0		X	Y	vhdl	21	system	Y	asm	N		8K	8K	N	16		8		2021	2021		~std 16-bit, 8 reg RISC, course project	uses latches? Primitive RTL
16bit_verilog	https://github.com/vprab		Vinay Prabhu	accum	32	16	artix-7	James	verilog error			6				##	v23.2	1.00	2.0		A		verilog	17	cpu			N	Y		2K	N	16		3			2019		educational, distinct from previous 16	**combinatorial** multiply and **divide**
16bit_vhdl	https://github.com/vprab		Vinay Prabhu	accum	32	16	artix-7	James	incomplete port map			6				##	v23.2	1.00	2.0				vhdl	16	processor			N	Y	64K	64K	N	12		3			2018		educational	did both VHDL & verilog, different ISAs
1802-pico-basi	https://github.c	beta	Steve Teal	1802	8	8	**zu-3e**	James	area o	247	136	6		2	427	##	v21.1	0.33	12.0	47.6	LX		vhdl	6	pico_basic	Y	yes	N		64K	64K	Y	52		16		2016	2016	https://wiki.forth-	VHDL 1802 Core with TinyBASIC	tiny Basic in ROM, Interrupts & DMA not impler
24bit_up	https://github.c	alpha	Harshal Mittal	RISC	24	24	**zu-3e**	James	area o	3535	2166	6	1		187	##	v21.1	0.80	1.0	42.2	X		verilog	17	processor			N		16M	16M	N	17		32		2019	2019		basic 24-bit RISC, course work	big Dff count, multiple writes to register file
32-bit_MIPS	https://sourcefc	beta	Cairo University	MIPS	32	32	**zu-3e**	James	very slow synthesis			6	1		100	##	v21.1	1.00	1.0				vhdl	18	mips_moc	Y	yes	N		4G	4G	Y			32		2011	2018		Cairo University EE dept	stopped run in synthesis
486tang	https://github.com/nand2		nand2mario	x86	32	8															G		verilog		ao486_to	Y	yes	N	N	4G	4G	Y			8			2025	https://hackaday.i	486 on a credit-card-sized FPGA boarc	uses ao486-MiSTer
495_cpu	https://github.com/Totor		Brian Cheng	accum	8	16															A		vhdl	6	top_level	Y		N		256	256	Y	14				2019	2020		very basic	simple & complete doc
6809_6309	https://opencor	beta	Alejandro Paz Schmidt	6809	8	8	arria-2	James Brakef		1680		A			145	##	q18.0	0.33	3.0	9.5	ALX	B	verilog	5	MC6809_	Y	yes	N	N	64K	64K	Y	44	13	8		2012	2015		6309 op-codes not implemented	
6809_6309	https://opencor	beta	Alejandro Paz Schmidt	6809	8	8	kintex-7-3	James Brakef		1996	370	6			175	##	14.7	0.33	3.0	9.7	ALX	B	verilog	5	MC6809_	Y	yes	N	N	64K	64K	Y	44	13	8		2012	2015		6309 op-codes not implemented	
6809_6309	https://opencor	beta	Alejandro Paz Schmidt	6809	8	8	stratix-5	James Brakef		1711		A			133	##	q14.0	0.33	3.0	8.6	ALX	B	verilog	5	MC6809_	Y	yes	N	N	64K	64K	Y	44	13	8		2012	2015		6309 op-codes not implemented	
6809_6309	https://opencor	beta	Alejandro Paz Schmidt	6809	8	8	**spartan7-**	James	vivadc	1592	366	6			100	##	v23.2	0.33	3.0	6.9	ALX	B	verilog	5	MC6809_	Y	yes	N	N	64K	64K	Y					2012	2015		6309 op-codes not implemented	does not match timing results of zynq+
6809_6309	https://opencor	beta	Alejandro Paz Schmidt	6809	8	8	**kintex-u3**	James	vivadc	1656	367	6			185	##	v23.2	0.33	3.0	12.3	ALX	B	verilog	5	MC6809_	Y	yes	N	N	64K	64K	Y					2012	2015		6309 op-codes not implemented	does not match timing results of zynq+
6809_6309	https://opencor	beta	Alejandro Paz Schmidt	6809	8	8	**zu-3e**	James	vivadc	1716	367	6			370	##	v21.1	0.33	3.0	23.7	ALX	B	verilog	5	MC6809_	Y	yes	N	N	64K	64K	Y					2012	2015		6309 op-codes not implemented	does not match timing results of zynq+
6809_6309	https://opencor	beta	Alejandro Paz Schmidt	6809	8	8	**kintex-u3**	James	vivadc	1595	367	6			200	##	v23.2	0.33	3.0	13.8	ALX	B	verilog	5	MC6809_	Y	yes	N	N	64K	64K	Y					2012	2015		6309 op-codes not implemented	does not match timing results of zynq+
6809_6309	https://opencor	beta	Alejandro Paz Schmidt	6809	8	8	c-7	James Brakef		1074	460	A			190	##	q25.1	0.33	3.0	19.5	ALX	B	verilog	5	MC6809_	Y	yes	N	N	64K	64K	Y	44	13	8		2012	2015		6309 op-codes not implemented	
68hc05	https://opencor	stable	Ulrich Riedel	6805	8	8	kintex-7-3	James Brakef		1112		6			300	##	14.7	0.33	4.0	22.2	X		vhdl	1	6805		yes	N	N	64K	64K	Y					2007	2009			
68hc05	https://opencor	stable	Ulrich Riedel	6805	8	8	**zu-3e**	James	vivadc	1106	117	6			485	##	v21.1	0.33	4.0	36.2	X		vhdl	1	6805		yes	N	N	64K	64K	Y					2007	2009			68c05 & 68c08 very different Fmax
68hc08	https://opencor	stable	Ulrich Riedel	6808	8	8	kintex-7-3	James Brakef		2290		6			101	##	14.7	0.33	4.0	3.6	X		vhdl	1	x68ur08		yes	N	N	64K	64K	Y					2007	2009			
68hc08	https://opencor	stable	Ulrich Riedel	6808	8	8	**zu-3e**	James	vivadc	1875	128	6			164	##	v21.1	0.33	4.0	7.2	X		vhdl	1	x68ur08		yes	N	N	64K	64K	Y					2007	2009			68c05 & 68c08 very different Fmax
8bit_chapman	http://www.ece	beta	Rob Chapman, Steven	forth	8	8	kintex-7-3	James Brakef		176	64	6			131	##	14.7	0.33	1.0	245.5	ALX		vhdl	10	stack_pro	Y		N		256	256	Y	24				1998	1998		course work	
8bit_chapman	http://www.ece	beta	Rob Chapman, Steven	forth	8	8	**zu-3e**	James	vivadc	132	63	6			305	##	v21.1	0.33	1.0	762.2	ALX		vhdl	10	stack_pro	Y		N		256	256	Y	24				1998	1998		course work	
8bit_piped_prc	https://opencor	stable	Mahesh Sukhdeo Palve	RISC	8	16	kintex-7-3	James	swapp	1049		6		1	370	##	14.7	0.33	1.0	116.4	X		verilog	28	top	Y							20		16		2013	2017	https://github.com	uses Perl as assembler	use Perl to generate ROM file
8bit_piped_prc	https://opencor	stable	Mahesh Sukhdeo Palve	RISC	8	16	**zu-3e**	James	vivadc	1500	1822	6		1	500	##	v21.1	0.33	1.0	110.0	X		verilog	28	top	Y							20		16		2013	2017	https://github.com	uses Perl as assembler	use Perl to generate ROM file
8bit-verilog_mcu		stable	Josh Friend	accum	8	8	**zu-2e**	James	timing	392		6		1	500	##	v20.1	0.33	2.0	210.5	X		verilog	11	cpu					512	512	Y	16				2012	2012		for class project, small data stack	PB clock, students to add features
8-bit-cpu	https://github.com/Poojar		Poojan-84	accum	8	16																	verilog	4	top	Y		N		256	256	Y	16					2026		very simple 8-bit cpu, single stepping	control via RP2040 running python
a_tiny_up	https://www.quora.com/\		Simon Moore, Frankie	RISC	32	32	arria-5	James	tiny LU	35		A				##	q18.0	0.67	1.0				system	1	TinyComp	Y	asm	N	Y	1K	1K	N	13		128		2007	2011	https://www.cl.ca	from Thacker's version, Un Cambridge course	
a2z	https://hackada	stable		RISC	16	24	cyclone-4	James Brakef		1524		4	1	12	62	##	q17.0	0.67	1.0	27.4	A		verilog		top_a2z												2016	2018			
a2z	https://hackada	errors		RISC	16	24	kintex-7-3	James	replace Altera RAM wi			6					14.7	0.67	1.0		A		verilog														2016	2018		runs on Cyclone IV	
aap	https://github.c	stable	Simon Cook	RISC	16	16	arria-2	James Brakef		7193		A			393	##	q18.0	0.67	1.0	36.6	A		verilog	7	de0_nano	Y	yes		Y	64K	16M	Y			64		2015	2016	http://www.embe	includes Altera project	4 to 64 reg, 24-bit pc, no status reg
aap	https://github.c	stable	Simon Cook	RISC	16	16	cyclone-4	James Brakef		10630		4			306	##	q18.0	0.67	1.0	19.3	A		verilog	7	de0_nano	Y	yes		Y	64K	16M	Y			64		2015	2016	http://www.embe	includes Altera project	4 to 64 reg, 24-bit pc, no status reg
acc	https://github.c	stable	Juan Gonzalez-Gomez	accum	15	15	kintex-7-3	James	rom &	89	96	6		1	227	##	14.7	0.67	2.0	855.5	AX		verilog	1	acc2	Y	yes	N			4K						2016	2016	https://github.com	26 chptr course using Apollo Comman	??why LUT count different from agcnorm
ae18	https://opencor	beta	Shawn Tan	PIC18	8	16	arria-2	James Brakef		1084		A	1		207	##	q13.1	0.33	1.0	63.1	ALX		verilog	1	ae18_core		yes	N	Y	4K	1M						2003	2009	https://hackaday.i	not 100% compatable	negative edge reset "clock"
ae18	https://opencor	beta	Shawn Tan	PIC18	8	16	**zu-3e**	James	vivadc	954	501	6			208	##	v21.1	0.33	1.0	72.1	ALX		verilog	1	ae18_core		yes	N	Y	4K	1M						2003	2009	https://hackaday.i	not 100% compatable	negative edge reset "clock"
aeMB	https://opencor	beta	Shawn Tan	uBlaze	32	32	kintex-7-3	James Brakef		1018	354	6	3		131	##	14.7	1.00	1.0	128.5	ALX		verilog	7	aeMB_cor	Y	yes	N		4G	4G	Y					2004	2009		not 100% compatable	
aeMB	https://opencor	beta	Shawn Tan	uBlaze	32	32	**zu-3e**	James Brakef		997	434	6	3		250	##	v21.1	1.00	1.0	250.8	ALX		verilog	7	aeMB_cor	Y	yes	N		4G	4G	Y					2004	2009		not 100% compatable	
af65k	https://github.c	alpha	Andre Fachat	6502	32	8	kintex-7-3	James Brakef		4424	873	6			69	##	14.7	1.00	4.0	3.9	X		vhdl	13	gecko65k	Y		N	N								2011	2019	http://www.6502	extended 6502 AKA 65K with 16, 32 or 64 bit data	
ag_6502	https://opencor	beta	Oleg Odintsov	6502	8	8	kintex-7-3	James Brakef		824	105	6			176	##	14.7	0.33	4.0	17.7	ALX		verilog	2	ag_6502		yes	N	N	64K	64K	Y					2012	2012		verilog code generation, "phase level accurate"	
agcnorm	https://opencor	beta	Dave Roberts	accum	15	15	spartan-3	James Brakef		3732	1115	4		2	20	##	14.7	0.66	1.0	3.5	X		vhdl	5	AGC	Y		N	Y	4K	72K	N	11		1		1962	2012	http://klabs.org/h	Apollo Guidance Computer via 3-input NOR gate emulation	
agcnorm	https://opencor	beta	Dave Roberts	accum	15	15	kintex7-3	James Brakef		2252	1110	6		2	32	##	14.7	0.66	1.0	9.3	X		vhdl	5	AGC	Y		N	Y	4K	72K	N	11		1		1962	2012	http://klabs.org/h	Apollo Guidance Computer via 3-input NOR gate emulation	
ahmes	https://github.c	stable	Fabio Pereira	accum	8	8	kintex-7-3	James Brakef		186	100	6			476	##	14.7	0.33	3.0	281.6	X	B	vhdl	3	ahmes			N	N	256	256	Y	15	1			2016	2017	http://embeddedsystems.io/ahmes-a-simple-8-bit-cpu-ir		bare CPU with no RAM
aizup/aizup_m	instruct1.cit.cor	stable	Yamin Li, Wanming Ch	RISC	8	16	arria-2	James Brakef		129		A			298	##	q13.1	0.17	2.0	192.6	AX		vhdl	1	cpu			N	N	64K	64K		16		4		1996	1998		used in Cornell EE475 course	MIPS/inst reduced due to few inst
aizup/aizup_ov	instruct1.cit.cor	stable	Yamin Li, Wanming Ch	RISC	8	16	kintex-7-3	James Brakef		138	51	6			318	##	14.7	0.17	3.0	128.3	AX		vhdl	1	cpu		asm	N	N	64K	64K	Y	16		4		1996	1998		used in Cornell EE475 course	MIPS/inst reduced due to few inst
aizup/aizup_pi	instruct1.cit.cor	stable	Yamin Li, Wanming Ch	RISC	8	16	kintex-7-3	James Brakef		198	52	6			375	##	14.7	0.17	2.0	157.9	AX		vhdl	1	cpu		asm	N	N	64K	64K	Y	16		4		1996	1998		used in Cornell EE475 course	MIPS/inst reduced due to few inst
aizup/aizup_se	instruct1.cit.cor	stable	Yamin Li, Wanming Ch	RISC	8	16	kintex-7-3	James Brakef		136	90	6			313	##	14.7	0.17	8.0	48.1	AX		vhdl	1	cpu		asm	N	N	64K	64K	Y	16		4		1996	1998		used in Cornell EE475 course	MIPS/inst reduced due to few inst
aizup/simple_8	instruct1.cit.cor	stable	Yamin Li, Wanming Ch	RISC	8	8	kintex-7-3	James Brakef		136	55	6			313	##	14.7	0.17	8.0	48.1	AX		vhdl	1	cpu		asm	N	N	64K	64K	Y	16		4		1996	1998		used in Cornell EE475 course	similar to mica
altium/TSK165	http://techdocs.	proprietar	Altium	PIC16	8	12	spartan-3	Altium		416	211	4			50			0.33	2.0	19.8	ALX		proprietary			Y	yes	N	Y	256	4K	Y					2004	2017	CR0140.pdf, CR01	frozen, asm, C, C++, schem, VHDL & V	clock is 50MHz, #s for other fpgas
altium/TSK300	http://techdocs.	proprietar	Altium	RISC	32	32	spartan-3	Altium		3326	1776	4		4	50			1.00	1.0	15.0	ALX		proprietary			Y	yes	N	N	4G	4G	Y					2004	2017	CR0140.pdf, http:/	frozen, asm, C, C++, schem, VHDL & V	clock: 50MHz, opt mult/div, #s for other fpgas
altium/TSK51A	http://techdocs.	proprietar	Altium	8051	8	8	spartan-3	Altium		1890	482	4		1	50			0.33	6.0	1.5	ALX		proprietary			Y	yes	N	N	64K	64K	Y					2004	2017	CR0140.pdf, CR01	frozen, asm, C, C++, schem, VHDL & V	clock is 50MHz, #s for other fpgas
altium/TSK80x	http://techdocs.	proprietar	Altium	Z80	8	8	spartan-3	Altium		2558	300	4			50			0.33	3.0	2.2	ALX		proprietary			Y	yes	N	N	64K	64K	Y					2004	2017	CR0140.pdf, CR01	frozen, asm, C, C++, schem, VHDL & V	clock is 50MHz, #s for other fpgas
altor32	https://opencor	stable	Ultra Embedded	OpenRISC	32	32	kintex-7-3	James Brakef		2602	1578	6		5	192	##	14.7	1.00	1.0	73.9	ALX		verilog	16	altor32	Y	yes	N	Y	4G	4G	Y					2012	2015	https://openrisc.io	simplified OpenRISC 1000	xilinx S3 primitives
altor32_lite	https://opencor	stable	Ultra Embedded	OpenRISC	32	32	c5	James Brakef		1394	1304	A			230	##	q25.1	1.00	1.0	165.0	ALX		verilog	16	altor32	Y	yes	N	Y	4G	4G	Y					2012	2015	https://openrisc.io	simplified OpenRISC 1000	xilinx S3 primitives
altor32_lite	https://opencor	stable	Ultra Embedded	OpenRISC	32	32	kintex-7-3	James Brakef		1928	1250	6			236	##	14.7	1.00	2.0	61.3	ALX		verilog	7	altor32	Y	yes	N	Y	4G	4G	Y					2012	2014	https://openrisc.io	simplified OpenRISC 1000, no pipeline	xilinx S3 primitives
alwcpu	https://opencor	alpha	Andreas Hilvarsson	RISC	16	16	kintex-7-3	James Brakef		298	78	6			194	##	14.7	0.67	1.0	437.1	ALX		vhdl	7	top		some	N	N	64K	64K	Y			8		2009	2010		lightweight CPU	minimal features, **uses generics for configura**
alwcpu	https://opencor	alpha	Andreas Hilvarsson	RISC	16	16	kintex-7-3	James Brakef		377	77	6			194	##	14.7	0.67	1.0	345.5	ALX		vhdl	7	top		some	N	N	64K	64K	Y			16		2009	2010		lightweight CPU	maximal features (additional inst)
amber	https://opencor	stable	Conor Santifort	ARM7	32	32	kintex-7-3	James Brakef		6409	2351	6		2	82	##	14.7	0.75	1.0	9.6	ALX		verilog	25	a23_core	Y	yes	N		4G	4G	Y	80		16	3	2010	2017	https://en.wikiped	no MMU, shared cache	2048 LUTs used as single port RAM
amber	https://opencor	stable	Conor Santifort	ARM7	32	32	**zu-3e**	James	area o	3105	1857	6		10	168	##	v21.1	0.75	1.0	40.7	ALX		verilog	25	a23_core	Y	yes	N		4G	4G	Y	80		16	3	2010	2017	https://en.wikiped	no MMU, shared cache	
amber	https://opencor	stable	Conor Santifort	ARM7	32	32	kintex-7-3	James Brakef		12450	3502	6		2	98	##	14.7	1.05	1.0	8.2	ALX		verilog	25	a25_core	Y	yes	N		4G	4G	Y	80		16	3	2010	2017	https://en.wikiped	no MMU	4330 LUTs used as RAM
amber	https://opencor	stable	Conor Santifort	ARM7	32	32	kintex-7-3	James Brakef		6103		6		18	127	##	v18.2	1.05	1.0	21.8	ALX		verilog	25	a25_core	Y	yes	N		4G	4G	Y	80		16	3	2010	2017	https://en.wikiped	no MMU	
amber	https://opencor	stable	Conor Santifort	ARM7	32	32	**zu-3e**	James	area o	5066	2382	6		20	175	##	v21.1	1.05	1.0	36.4	ALX		verilog	25	a25_core	Y	yes	N		4G	4G	Y	80		16	5	2010	2017	https://en.wikiped	no MMU	
amic-0	https://github.c	stable	Alberto Moriconi	stack	32	8	**zu-3e**	James	vivadc	622	357	6			250	##	v21.1	1.00	6.0	67.0	X		vhdl	12	processor	Y		N		4G	4G	Y					2010	2021	https://en.wikiped	based on mic-1 by Andrew Tanenbaur	36-bit uCode, usually Java virtual machine
an-noc-mpsoc	https://opencor	mature	Alireza Monemi	uBlaze	32	32	kintex-7-3	James Brakef		1164		6	3	1	192	##	i14.7	1.00	1.0	165.2	X	Y	verilog	90	aeMB	Y	yes	N		4G	4G	Y					2014	2023		choice of lm32, aeMB, mor1kx or or12	full system has network of cores
an-noc-mpsoc	https://opencor	mature	Alireza Monemi	uBlaze	32	32	**zu-3e**	James	vivadc	1079		6	3	1	333	##	v21.1	1.00	1.0	308.9	X	Y	verilog	90	aeMB_top	Y	yes	N		4G	4G	Y					2014	2023		choice of lm32, aeMB, mor1kx or or12	full system has network of cores
any-1	https://github.c	defined	Robert Finch	RISC	64	36	**zu-3e**	James	errors							##	v21.1	2.00	1.0		X		system	83	any1base	Y		Y					128		64		2021	2021	http://anycpu.org	Cray-1 like with full set of vector instr	three versions with different ISAs, inst sz, reg sz
ao486	https://opencor	beta	Aleksander Osman	x86	32	8	cyclone-4	James Brakef		36094		4	4	47	46	##	q13.1	1.00	1.0	1.3	A	Y	system	85	ao486	Y	yes			4G	4G	Y					2014	2014	https://www.stuff	complete 486, SoC configuration	Henry Wong thesis at U.Toronto, also youtube
ao486	https://opencor	beta	Aleksander Osman	x86	32	8	**zu-2e**	James Brakef	altera avalon IC			6				##	v20.1	1.00	1.0		A	Y	system	85	ao486	Y	yes			4G	4G	Y					2014	2014	https://github.com	complete 486, SoC configuration	non-SoC, no MMU, not superscalar
ao486_mister	https://github.c	beta	Sorgelig	x86	32	8	**zu-3e**	James Brakefield				6						1.00	1.0		A	Y	system	85	ao486	Y	yes			4G	4G	Y					2020	2025	https://hackaday.i	complete 486, SoC configuration	mister version of ao486: reworked with many n
ao486_mister	https://github.c	beta	nand2mario	x86	32	8	gowin					4						1.00	1.0		G	Y	verilog		ao486	Y	yes			4G	4G	Y					2020	2021	https://hackaday.i	486 on a credit-card-sized FPGA boarc	reworked for for tang 136K LUTs
ao68000	https://opencor	beta	Aleksander Osman	68000	16	16	arria-2	James Brakef		3479		A		6	169	##	q13.1	0.67	4.0	8.1	A	Y	verilog	1	ao68000	om	yes	N		4G	4G	Y					2010	2012		uses microcode, instruction prefetch buffer	
aoocs	https://github.c	beta	Aleksander Osman	68000	16	16	arria-2	James Brakef		17852		A	2	43	57	##	q18.0	0.67	4.0	0.5	A	Y	verilog	22	aoOCS	om	yes	N		4G	4G	Y					2010	2011		uses ao68000 core, Amiga chip set em	Wishbone Amiga OCS SoC
aoocs	https://github.c	beta	Aleksander Osman	68000	16	16	cyclone-1	James Brakef		26009		4	2	67	45	##	q18.0	0.67	4.0	0.3	A	Y	verilog	22	aoOCS	om	yes	N		4G	4G	Y					2010	2011		uses ao68000 core, Amiga chip set em	Wishbone Amiga OCS SoC
aoocs	https://github.c	beta	Aleksander Osman	68000	16	16	cyclone-2	Aleksander O		26227		4	2	65		##	q10.1	0.67	4.0		A	Y	verilog	22	aoOCS	om	yes	N		4G	4G	Y					2010	2011		uses ao68000 core, Amiga chip set em	Wishbone Amiga OCS SoC
aoocs	https://github.c	beta	Aleksander Osman	68000	16	16	kintex-7-3	James	altera pimitives			6				##	14.7	1.00	1.0		A	Y	verilog	22	aoOCS	om	yes	N		4G	4G	Y					2010	2011		uses ao68000 core, Amiga chip set em	Wishbone Amiga OCS SoC
aor3000	https://opencor	beta	Aleksander Osman	MIPS	32	32	kintex-7-3	James Brakef		5307		6	4	9	129	##	14.7	1.00	1.0	24.2	AX		verilog	19	aoR3000	Y	yes	N		4G	4G	Y			32	5	2014	2015		MIPS R3000A compatible, has MMU	moved declarations forward
aor3000	https://opencor	beta	Aleksander Osman	MIPS	32	32	**zu-3e**	James	high F	4199	2520	6	4	8	175	##	v21.1	1.00	1.0	41.8	AX		verilog	19	aoR3000	Y	yes	N		4G	4G	Y			32	5	2014	2015		MIPS R3000A compatible, has MMU	moved declarations forward
apollo_68080	http://www.apc	proprieta	Gunnar von Boehn	68000	8	16	cyclone-V	Gunnar von Boehn															vhdl			Y	yes	N		4G	4G	Y			32		2012	2022	http://www.apollo	sells Amiga card, "68080" with 64-bit	claims very fast FPGA versions
apollo11_fpga	https://github.c	stable	Michael Kohn	apollo11	16	16															L	Y	verilog	7	agc	Y				64K	64K	Y						2024	https://www.mike	Apollo11 guidance uP, some differences	
apple2fpga	http://www.cs.c	stable	Stephen A Edwards	6502	8	8	kintex7-3	James Brakef		1416	654	6		8.5	159	##	14.7	0.33	4.0	9.2	AX	Y	vhdl	19	de2_top	Y	yes	N	Y	64K	64K	Y					2007	2022		emulation of Apple II computer	replaced Altera PLL with stub
apple2fpga	http://www.cs.c	stable	Stephen A Edwards	6502	8	8	**zu-3e**	James	vivadc	1238	706	6		7	195	##	v21.1	0.33	4.0	13.0	AX	Y	vhdl	19	de2_top	Y	yes	N	Y	64K	64K	Y					2007	2022		emulation of Apple II computer	replaced Altera PLL with stub
aquarius	https://opencor	stable	Thorn Aitch	SuperH-2	32	16	kintex-7-3	James Brakef		4071		6	2	10	97	##	14.7	1.00	1.0	23.7	ALX		verilog	21	top	Y	yes	N		4G	4G	Y					2003	2015	http://0pf.org/j-cc	clone of Hitachi SH-2	project seems to have stalled
aquarius	https://opencor	stable	Thorn Aitch	SuperH-2	32	16	**zu-3e**	James	vivadc	3563	1384	6	2	16	147	##	v21.1	1.00	1.0	41.2	ALX		verilog	21	top	Y	yes	N		4G	4G	Y					2003	2015	http://0pf.org/j-cc	clone of Hitachi SH-2	project seems to have stalled
ARC	https://www.sy	proprietar	Synopsys	ARC	32	16	proprietary												1.0				proprietary			Y	yes			4G	4G								https://www.sync	several families each with options	for ASIC use, FPGA versions avail
ARM_Cortex_A	https://develop	ASIC	ARM	ARM A53	64	32	asic	Xilinx		6000		A			####			2.00	0.5	1000			asic			Y	yes	Y				Y							https://en.wikiped	uses pro-rated LC area	dual issue, includes fltg-pt & MMU & caches
ARM_Cortex_A	https://develop	ASIC	ARM	ARM A9	32	16	arria V	altera		4500		A			####			2.50	1.0	583.3			asic			Y	yes	Y		4G	4G	Y	80		16	10		2012	https://en.wikiped	uses pro-rated LC area	dual issue, includes fltg-pt & MMU & caches
ARM_Cortex_M	http://www.arm	proprietar	ARM	ARM M1	32	16	virtex-5	ARM	65nm	1900		6			200			1.00	1.0	105.3	AX		proprietary			Y	yes	N		4G	4G	Y			16	3	2007		https://en.wikiped	ARM Cortex M0, M1 & M3 avail for FF	see xilinx Xcell64
ARM_Cortex_M	https://www.ar	proprietar	ARM	ARM M1	32	16						6						1.00	1.0		X		encrypted			Y	yes	N		4G	4G	Y			16	3		2019	https://www.arm.	free use on Xilinx Vivado, encrypted RTL, uses Digilent A7 or S7 board, AIX bus interfac	
ARM_Cortex_R	https://develop	ASIC	ARM	ARM R5	32	16	asic	Xilinx				A			600				1.0				asic			Y	yes	Y		4G	4G	Y	80		16				https://en.wikiped	uses pro-rated LC area	real-time interrupt handling
arm_cpu_ddca	https://github.com/nguye		Evan Nguyen	arm	32	32	**zu-3e**	James	LUT RAM for inst & dat			6				##	v21.1	1.00	1.0				system	23	top	Y	yes	Y		4G	4G	Y			16			2021		from "Digital design and computer arc	single cycle, empty synthesis
arm_harris	http://booksite.	simulatior	David Harris	ARM	32	32																	system	49	arm_singl	Y	yes	N	Y	4G	4G	Y					2014	2015	https://booksite.e	courseware to go with book	both VHDL & System Verilog
arm_harris	http://booksite.	simulatior	David Harris	ARM	32	32																	vhdl	46	arm_singl	Y	yes	N	Y	4G	4G	Y					2014	2015	https://booksite.e	only a few op-codes	also has book figures & course slides
arm_m0_zach	https://github.com/ZackB		Zack Boyle	arm	32	32																	verilog	10	core	Y	yes	N		4G	4G	Y			32			2025	https://scholarcor	Open Source ARM Cortex M0 Processor, senior thesis	

_uP_all_soft folder	opencores or prmary link	status	author	style / clone	data	inst sz	FPGA	repor ter	com ents	LUTs ALUT	Dff	LUT?	mults	blk ram	F max	date	tool ver	MIPS /inst	clks/ inst	KIPS /LUT	ven dor	SOC	src code	#src files	top file	doc	tool chai	fltg pt	Hav'd	max dat	max inst	byt ead	# inst	adr mo	# reg	pip e	start year	last revis	secondary web link	note worthy	comments
arm_rusian	https://github.com/0xD5C		ruslan	arm	32	32	**zu-3e**	James	LUT R/	2360	4815	6			200	##	v21.1	1.00	1.0	84.7			system	6	ARM_Mul	Y	yes	Y		4G	4G	Y			16			2019		from "Digital design and computer arc	single cycle,
arm_rusian	https://github.com/0xD5C		ruslan	arm	32	32	**zu-3e**	James	LUT R/	3563		6			147	##	v21.1	1.00	1.0	41.2			system veril		ARM_Sing	Y	yes	Y		4G	4G	Y			16			2019		from "Digital design and computer arc	multi-cycle
arm4u	https://opencores.org/prc		Jonathan Masur	arm	32	32	**zu-3e**	James	altera primitives			6				##	v21.1	1.00	1.0		A		vhdl	12	cpu	Y	yes	Y		4G	4G	Y	80		16		2014	2014		ARMv3 ISA, clones early ARM processors in functionality	
arm9-soft-cpu	https://github.com/risclite		Li Xinbing	ARM9	32	32	**zu-3e**	James Brakef		1807	736	6			357	##	v21.1	1.00	1.0	197.6			verilog	4	risclite_m	Y	yes	Y		4G	4G	Y						2020		ARMv4-compatible CPU core	no mult, interrupts or reg banks
arm9-soft-cpu	https://github.com/risclite		Li Xinbing	ARM9	32	32	**zu-3e**	James Brakef		2098	778	6	4		238	##	v21.1	1.00	1.0	113.5			verilog	4	risclite_m	Y	yes	Y		4G	4G	Y						2020		ARMv4-compatible CPU core	no interrupts or reg banks
arm9-soft-cpu	https://github.com/risclite		Li Xinbing	ARM9	32	32	**zu-3e**	James Brakef		3914	1257	6	4		167	##	v21.1	1.00	1.0	42.6			verilog	4	arm9_con	Y	yes	Y		4G	4G	Y						2020		ARMv4-compatible CPU core	Dhrystone value: 1.2 DMIPS/MHz
arm-cpu	https://github.com/navida		Navid Adelpour	ARM	64	32	**spartan7**	James	clocks issue														verilog	14	cpu	Y	yes	N		4G	4G	Y			32		2018	2018		both single cycle & pipelined versions,	64-bit registers & memory interface
arm-cpu	https://github.com/navida		Navid Adelpour	ARM	64	32	**spartan7**	James	clocks issue														verilog	14	cpu_pipel	Y	yes	N		4G	4G	Y			32		2018	2018		both single cycle & pipelined versions,	64-bit registers & memory interface
armv2a_vlsi	https://github.com/syatl/		Samy Attal	arm	32	32	**spartan7**	James	multic	1069	778	6			86	##	v23.2	1.00	1.5	53.8	X		vhdl	21	arm_core	Y	yes	N	Y	4G	4G	Y			27	3		2022		ASIC project, has vcc & vss connection	original ARM/Acorn, external caches
armv4_uarch	https://github.com/grantw		Grant Wilk	ARM9	32	32	**max10**	Grant Wilk		2860		4			50	##	q18.0	1.00	1.0	17.5	A		vhdl	18		Y	yes	N		4G	4G	Y			16			2020	https://grantwilk.	custom uarch for the ARMv4 ISA on In	course work, top level is schematic
armv4_uarch	https://github.com/grantw		Grant Wilk	ARM9	32	32	**zu-3e**	James Brakefield				6				##	v21.1	1.00	1.0		A		vhdl	18		Y	yes	N		4G	4G	Y			16			2020	https://grantwilk.	custom uarch for the ARMv4 ISA on In	course work, Quartus project
armv5-cpu	https://github.com/takah		hooray	arm	32	32	**spartan7**	James	does not synthesize, removed sync reset							##	v24.1	1.00	1.0		X		system	48	top	Y	yes	N		4G	4G	Y			16			2021		Armv5 single-cycle processor	reg file: 2 we's, 4 read ports, refs Harris & Harri
arm-w-branchp	https://github.com/marsl		Mehmet ARSLAN	arn	32	32															AX		verilog	37	topmodul	Y	yes	Y		4G	4G	Y			32	5		2026		5-stage pipe ARM uP w advanced haza	branch pred, target buf, neg edge reg write
artemis	https://github.com/solder		Sudharshan Sundaram	RISC	16	16	**zu-3e**	James	incomplete source code							##	v21.1	1.00	1.0				verilog	9	main_test	Y	asm	N				N	18		8		2018	2020	https://www.yout	simple, educational uP with decent vi	vivado project
artiq	https://m-labs.hk/experim		Sébastien Bourdeaudu	risc-v	32	32	**spartan7**	James	no RTL, python only												X	Y	rusthdl			Y	yes			4G	4G	Y					2016	2025	https://github.com	ctrl sys 4 quantum info experiments	rust 2 verilog; vex-riscv, mor1kx & lm32 suppor
asip38	https://aaltodoc.aalto.fi/b		Lauri Isola	accum	32	38	**zu-3e**	James	xilinx l	2962	1056	6	4	35	100	##	v22.2	1.00	1.0	33.8	X	Y	vhdl	14	top	Y	asm	N	Y	16K	16K	N	31	4	4		2018	2021	http://www.kolun	Application-Specific Instruction set Pr	missing prog & data mem, missing mult
asip38	https://aaltodoc.aalto.fi/b		Lauri Isola	accum	32	38	**zu-3e**	James	xilinx l	2962	1056	6	4	35	100	##	v22.2	1.00	1.0	33.8	X	Y	vhdl	14	asip38	Y	asm	N	Y	16K	16K	N	31	4	4		2018	2021	http://www.kolun	Application-Specific Instruction set Pr	missing prog & data mem, missing mult
aspida	https://opencor	stable	Sotiriou	DLX	32	32	kintex-7-3	James	dated	3586		6			257	##	14.7	1.00	1.0	71.7	X		verilog	10	DLX_top	Y	yes			4G	4G						2002	2009		DLX	compiled sync version
aspida	https://opencor	stable	Sotiriou	DLX	32	32	**zu-2e**	James	dated xilinx primitives			6				##	v20.1	1.00	1.0		X		verilog	10	DLX_top	Y	yes			4G	4G						2002	2009		DLX	compiled sync version
atlas_2K	https://opencor	beta	Stephan Nolting	RISC	16	16	kintex-7-3	James	48 LU	1595	1108	6	1	4.5	151	##	14.7	0.80	1.0	75.9	ALX		vhdl	19	ATLAS_2K	Y	asm	N	Y	64K	64K	M	80		8		2013	2015	https://www.allat	ARM thumb like inst set	has MMU & full SOC features
atlas_2K	https://opencor	beta	Stephan Nolting	RISC	16	16	**zu-3e**	James	40 LU	1222	1160	6	1	4.5	262	##	v21.1	0.80	1.0	171.4	ALX		vhdl	19	ATLAS_2K	Y	asm	N	Y	64K	64K	M	80		8		2013	2015		ARM thumb like inst set	has MMU & full SOC features
atlas_core	https://opencor	beta	Stephan Nolting	RISC	16	16	spartan3	Stephan Nolt		2406	1091	4	1	11	81	##	14.7	0.80	1.0	27.0	ALX		vhdl	19	ATLAS_2K	Y	asm	N	Y	64K	64K	M	80		8		2013	2015	https://www.allat	ARM thumb like inst set	has MMU & full SOC features
atlas_core	https://opencor	beta	Stephan Nolting	RISC	16	16	cyclone4	Stephan Nolt		2967	1364	4	1	32	99	##	q18.0	0.80	1.0	26.7	ALX		vhdl	19	ATLAS_2K	Y	asm	N	Y	64K	64K	M	80		8		2013	2015		ARM thumb like inst set	has MMU & full SOC features
atlas_core	https://opencor	beta	Stephan Nolting	RISC	16	16	kintex-7-3	James Brakef		559	269	6	1		200	##	v14.1	0.80	1.0	286.2	AX		vhdl	8	ATLAS_CP	Y	asm	N	Y	64K	64K	Y	80		8		2013	2015	https://www.allat	ARM thumb like inst set	non-MMU version
atlas_core	https://opencor	beta	Stephan Nolting	RISC	16	16	**zu-3e**	James Brakef		611	285	6	1		333	##	v21.1	0.80	1.0	436.4	AX		vhdl	8	ATLAS_CP	Y	asm	N	Y	64K	64K	Y	80		8		2013	2015		ARM thumb like inst set	non-MMU version
atmega8_pong	https://fr.wikive	stable	Juergen Sauermann	AVR	8	16	spartan-3	James	clock (	2767		4	1	10	53	##	14.7	0.33	1.0	6.3	X	Y	vhdl	37	avr_fpga_	Y	yes	N		64K	64K	Y	17		4		2017	2017		several projects using avr core	uses Sauermann core
atmega8_pong	https://fr.wikive	stable	Juergen Sauermann	AVR	8	16	spartan-3	James	clock (	2898		4	1	11	53	##	14.7	0.33	1.0	6.0	X	Y	vhdl	37	pacman_N	Y	yes	N		64K	64K	Y	17		4		2017	2017		several projects using avr core	uses Sauermann atmega16 core
attiny_atmega	https://opencor	beta	Gheorghiu Iulian	AVR	8	16	**zu-3e**	James Brakef		1366	116	6			179	##	v21.1	0.33	1.0	43.1	X	Y	verilog	9	mega_cor	Y	yes	N		64K	128K	Y	72		32		2018	2019	https://git.morgot	configurable AVR processor w/8 configurations	
avion	https://github.com/Yusuf		Yusuf Sait	accum	10	10																	verilog	3	avion_tb	Y		N		64	64		9					2025	https://www.yout	educational	also a video
avr_core	https://opencor	stable	Rusian Lepetenok	AVR	8	16	kintex-7-3	James Brakef		2135		6			127	##	14.7	0.33	1.0	19.7	X		verilog	15	avr_core	Y	yes	N		64K	128K	Y	72		32		2002	2017		VHDL core also	
avr_core	https://opencor	stable	Rusian Lepetenok	AVR	8	16	**zu-3e**	James Brakef		1624	519	6			250	##	v21.1	0.33	1.0	50.8	X		verilog	70	avr_core	Y	yes	N		64K	128K	Y	72		32		2002	2017		VHDL core also	
avr_cpu	https://github.com/nanan		nanamake Nanamaru	avr	8	16	**cyclone4**	nanamake Na		1845	520	4	1		64	##	q14.0	0.33	1.0	11.4	A		verilog			Y	yes	N		64K	128K	Y	72		32			2018		quartus project & report files	2nd version with data & prog mems
avr_fpga	https://opencor	stable	Juergen Sauermann	AVR	8	16	kintex-7-3	James Brakef		1606		6	1	6	120	##	14.7	0.33	1.0	24.7	X		vhdl	20	cpu_core	Y	yes	N		64K	128K	Y	72		32		2009	2010		extended lecture on FPGA uP design	
avr_fpga	https://opencor	stable	Juergen Sauermann	AVR	8	16	kintex-7-3	James Brakef		1877		6	1	6	115	##	14.7	0.33	1.0	20.2	X	Y	vhdl	20	avr_fpga	Y	yes	N		64K	128K	Y	72		32		2009	2010	https://fr.wikivers	extended lecture on FPGA uP design	missing module in atmega8_pong_vga
avr_fpga	https://opencor	stable	Juergen Sauermann	AVR	8	16	**zu-3e**	James Brakef		1606		6	1	6		##	v21.1	0.33	1.0		X		vhdl	20	cpu_core	Y	yes	N		64K	128K	Y	72		32		2009	2010		extended lecture on FPGA uP design	
avr_fpga	https://opencor	stable	Juergen Sauermann	AVR	8	16	**zu-3e**	James Brakef		1877		6	1	6		##	v21.1	0.33	1.0		X	Y	vhdl	20	avr_fpga	Y	yes	N		64K	128K	Y	72		32		2009	2010	https://fr.wikivers	extended lecture on FPGA uP design	missing module in atmega8_pong_vga
avr_hp	https://opencor	stable	Strauch Tobias	AVR	8	16	kintex-7-3	James	2 slot	1554		6			223	##	14.7	0.33	1.0	47.4	X		vhdl	10	avr_core_om		yes	N		64K	128K	Y	72		32		2010	2012		hyper pipelined (eg barrel) AVR	
avr_singlecycle	https://github.com/ykhr7		ykhr7121rkhy	AVR	8	16	max10	ykhr7	not en	3619	975	4	1	128		##	q17.0	0.33	1.0		A		vhdl	6	avr_core	Y	asm	N		64K	128K	Y	72		32		2018	2025	https://qiita.com/	full set of Quartus files	needs larger part or less RAM
avr8	https://opencor	beta	Nick Kovach	AVR	8	16	kintex-7-3	James Brakef		174		6			418	##	14.7	0.33	1.0	792.2	X		verilog	1	rAVR	Y	yes	N		64K	64K	Y	17		4		2010	2010		Reduced AVR Core for CPLD	not a full clone, doc is opencores page
avr-cpu	https://github.c	stable	Sung Hoon Choi	AVR	8	16	**zu-3e**	James Brakefield				6				##	v21.1	0.33	1.0				vhdl	15	avr_cpu	Y	yes	N		64K	128K	Y	72		32			2019			
avrtinyx61core	https://opencor	beta	Andreas Hilvarsson	AVR	8	16	kintex-7-3	James Brakef		1243		6			194	##	14.7	0.33	1.0	51.5	X		vhdl	1	mcu_core		yes	N		64K	128K	Y	72		32		2008	2009			
ax8	https://opencor	stable	Daniel Wallner	AVR	8	16	spartan-6	James	missin	1549		6		1	213	##	14.7	0.33	1.0	45.3	X		vhdl	14	A90S1200		yes	N		64K	128K	Y	72		32		2002	2010		both A90S1200 & A90S2313	inserted fake inst ROM
a-z80	https://opencor	stable	Goran Devic	Z80	8	8	cyclone-2	Goran Devic		2084		4		29	19	##	q11.1	0.33	1.0	3.0	AX		verilog	24	z80_top_c	Y	yes	N	N	64K	64K	Y					2014	2020	https://github.com	gate level reverse eng'd Z80	Complete implementation of a Sinclair ZX Spect
a-z80	https://opencor	stable	Goran Devic	Z80	8	8	kintex-7-3	James Brakef		1186		6			24	##	14.7	0.33	1.0	6.8	AX		verilog	24	z80_top_c	Y	yes	N	N	64K	64K	Y					2014	2020	https://github.com	gate level reverse eng'd Z80	Complete implementation of a Sinclair ZX Spect
a-z80	https://opencor	stable	Goran Devic	Z80	8	8	spartan-6	Goran Devic		1819		6		8		##	14.7	0.33	1.0		AX		verilog	24	z80_top_c	Y	yes	N	N	64K	64K	Y					2014	2020	https://github.com	gate level reverse eng'd Z80	Complete implementation of a Sinclair ZX Spect
a-z80	https://opencor	stable	Goran Devic	Z80	8	8	**zu-3e**	James	timing	1761	365	6			41	##	v21.1	0.33	1.0	7.7	AX		verilog	24	z80_top_c	Y	yes	N	N	64K	64K	Y					2014	2020	https://github.com	gate level reverse eng'd Z80	Complete implementation of a Sinclair ZX Spect
b16	www.bernd-pay	stable	Bernd Paysan	forth	16	5	spartan-6	James Brakef		554		6			134	##	14.7	0.67	1.0	161.7	AX		verilog	15	b16	Y	yes	N		64K	64K	N					2002	2017	https://github.com	two versions: one/15 source files, derived from c18	
b16	www.bernd-pay	stable	Bernd Paysan	forth	16	5		James Brakefield				6						0.67	1.0		AX		verilog	1	b16-small	Y	yes	N		64K	64K	N					2002	2019	https://github.com	two versions: one/15 source files, derived from c18	
b8008	https://github.com/robert		Robert Rico	8008	8	8															L		vhdl	28	b8008_to	Y	yes	N		64K	64K	Y						2026		Cycle-accurate Intel 8008 up with two-phase clocking & muxed bus	
baby8	https://github.com/jeceljr		Jecel de Assumpcao Jr	RISC	8	8	cyclone5	Jecel de Assu		29		A		16	58	##		0.17	4.0	84.5	AGLX		schem	17	baby8cpu	Y	asm	N		64K	64K	Y			16			2024	https://mdpi-res.c	minimal 8-bit uP with 16-bit adrs	schematic, verilog & system verilog
baby8	https://github.com/jeceljr		Jecel de Assumpcao Jr	RISC	8	8	ecp5	Jecel de Assu		77		4		4	58	##		0.17	4.0	31.8	AGLX		schem	17	baby8cpu	Y	asm	N		64K	64K	Y			16			2024	https://mdpi-res.c	minimal 8-bit uP with 16-bit adrs	relatively low uniform Fmax
baby8	https://github.com/jeceljr		Jecel de Assumpcao Jr	RISC	8	8	gowin	Jecel de Assu		48		4		4	58	##		0.17	4.0	51.1	AGLX		schem	17	baby8cpu	Y	asm	N		64K	64K	Y			16			2024	https://mdpi-res.c	minimal 8-bit uP with 16-bit adrs	micro-coded; mcpu has best figure of merit
baby8	https://github.com/jeceljr		Jecel de Assumpcao Jr	RISC	8	8	ice40	Jecel de Assu		285		4			58	##		0.17	4.0	8.6	AGLX		schem	17	baby8cpu	Y	asm	N		64K	64K	Y			16			2024	https://mdpi-res.c	minimal 8-bit uP with 16-bit adrs	ASIC & FPGA stats for risc-v, baby8 & soft uP
baby8	https://github.com/jeceljr		Jecel de Assumpcao Jr	RISC	8	8	spartan7	Jecel de Assu		31		6		4	58	##		0.17	4.0	79.1	AGLX		schem	17	baby8cpu	Y	asm	N		64K	64K	Y			16			2024	https://mdpi-res.c	minimal 8-bit uP with 16-bit adrs	stats for several soft uP 4 FPGA/ASIC versions
babyrisc	http://www.san	stable	John Rible	RISC	8	16	kintex-7-3	James Brakef		468		6			141	##	14.7	0.33	2.0	49.7	X		verilog	1	qs5_mix	Y		N		64K	64K	Y	15		8		1997	1999	http://www.sandp	part of a three class course	memory rd/wt & ALU per clock
babyrisc	http://www.san	stable	John Rible	RISC	8	16	**zu-3e**	James Brakef		249		6			286	##	v21.1	0.33	2.0	189.3	X		verilog	1	qs5_mix	Y		N		64K	64K	Y	15		8		1997	1999	http://www.sandp	part of a three class course	memory rd/wt & ALU per clock
basic-cpu	https://embedd	stable	Justin Rajewski	RISC	8	16	**zu-3e**	James	syntax errors			6				##	v21.1	0.33	2.0				verilog	1									16				2018	2018		16 inst, scrapped web page, 98 lines of verilog, no call/rtn, bare core, excellent exampl	
basic-simd-up	https://github.c	stable	Tingyuan Liang	RISC	16	18	**spartan7_**	James Brakef		1369	259	6			71	##	v23.2	0.75	1.0	39.1			verilog	5	cputop	Y		N	Y	1K	1K		47		8		2018	2022		simple SIMD processor in Verilog	compiled via Cadence to ASIC layout
bc6502	http://finitron.c	beta	Robert Finch	6502	8	8	kintex-7-3	James Brakef		619		6			197	##	14.7	0.33	4.0	26.2	X		verilog	18	bc6502		yes	N	N	64K	64K	Y					2012	2012			bare source
bc6502	http://finitron.c	beta	Robert Finch	6502	8	8	**zu-3e**	James	vivadc	583		6			286	##	v21.1	0.33	4.0	40.4	X		verilog	18	bc6502		yes	N	N	64K	64K	Y					2012	2012			bare source
ben_eater_up	https://github.com/ajithc		Ajith Thomas	accum	8	8																	vhdl	27	test_cpu	Y	asm	N		256	16	Y	10					2020	https://eater.net/	based on Ben Eater's tutorial on building an 8-Bit breadboard computer	
ben_eater_up	https://github.com/hsnav		Humberto Silva Naves	accum	8	8															A		verilog	14	computer		asm	N		256	16	Y					2015	2019	https://eater.net/	Ben Eater's 8-bit breadboard compute	microcode?
ben_eater_up	https://github.com/XarkL		Ken Jordan	accum	8	8	spartan7-	James Brakef		164	137	6			100	##	v23.2	0.33	2.0	100.6	X		vhdl	6	system	Y	asm	N		256	16	Y					2015	2019	https://eater.net/	Ben Eater's 8-bit breadboard computer	
ben_eater_up	https://github.com/hneen		Paul Kappmeyer	accum	8	8	spartan-7	James	more than one clock														schem	5														2021		Digital schematic, Ben Eater uP	TTL components
ben_eater_up	https://github.com/JetSta		XarkLabs	accum	8	8																	vhdl	38	computer	Y	asm	N		256	16	Y					2015	2019	https://eater.net/	Ben Eater's 8-bit breadboard computer	
beri	https://www.cl.	mature	Gregory Chadwick	MIPS	64	32																	bluesp	34	mipstop	Y	yes								32		2012	2017	https://github.com	Bluespec Extensible RISC Implementat	CHERI (Capability Hardware Enhanced RISC Inst
bfcpu	http://www.clif	stable	Clifford Wolf	Turing	8	3	kintex-7-3	James Brakef		422		6			345	##	14.7	0.01	4.0	2.0	X	B	vhdl	4	cw6671	Y	yes	N	N	64K	64K	Y	8				2003	2003	https://en.wikiped	no accum, data pointer and bracketed	current version & earlier version
bfcpu	http://www.clif	stable	Clifford Wolf	Turing	8	3	**zu-3e**	James Brakef		387		6			500	##	v21.1	0.02	4.0	6.5	X	B	vhdl	4	cw6671	Y	yes	N	N	64K	64K	Y	8				2003	2003	https://en.wikiped	no accum, data pointer and bracketed	internal 1-byte data cache doubles performanc
bfcpu	http://www.clif	stable	Clifford Wolf	Turing	8	3	**zu-3e**	James Brakef		303		6			500	##	v21.1	0.01	4.0	4.1	X	B	vhdl	4	cw6670	Y	yes	N	N	64K	64K	Y	8				2003	2003	https://en.wikiped	no accum, data pointer and bracketed	first implementation, no data cache
bit-serial	https://github.com/howe		Richard Howe	accum	16	16	spartan6	James Brakef		324	210	6		8	100	##	14.7	0.67	51.0	4.1	X		vhdl	6	top	Y		N		2K	2K	N	15				2020	2024	https://hackaday.	bit serial, 16-bit uP, very simple	supports Forth
bit-serial	https://github.com/howe		Richard Howe	accum	16	16	spartan6	James	area o	226	181	6		8	100	##	14.7	0.67	51.0	5.8	X		vhdl	6	top	Y		N		2K	2K	N	15				2020	2024	https://hackaday.	bit serial, 16-bit uP, very simple	supports Forth
bit-serial	https://github.com/howe		Richard Howe	accum	16	16	spartan6	James	speed	137	86	6			100	##	14.7	0.67	51.0	9.6	X		vhdl	6	cpu	Y		N		2K	2K	N	15				2020	2024	https://hackaday.	bit serial, 16-bit uP, very simple	supports Forth
bit-serial	https://github.com/howe		Richard Howe	accum	16	16	spartan6	James	area o	89	66	6			100	##	14.7	0.67	51.0	14.8	X		vhdl	6	cpu	Y		N		2K	2K	N	15				2020	2024	https://hackaday.	bit serial, 16-bit uP, very simple	supports Forth
bit-serial	https://github.com/howe		Richard Howe	accum	16	16	spartan7	James	errors init bkRAM			6				##	v23.2	0.67	51.0		X		vhdl	6	top	Y		N		2K	2K	N	15				2020	2024	https://hackaday.	bit serial, 16-bit uP, very simple	supports Forth
bit-serial	https://github.com/howe		Richard Howe	accum	16	16	**zu-3e**	James	errors init bkRAM			6				##	v21.1	0.67	51.0		X		vhdl	6	top	Y		N		2K	2K	N	15				2020	2024	https://hackaday.	bit serial, 16-bit uP, very simple	supports Forth
bjx1	https://github.c	alpha	Brendan Bohannon	RISC	32	16	kintex-7-3	James	syntax errors			6				##	14.7	1.00	2.0				verilog	34	exunit	Y		Y	N	4G	4G	Y		9	16		2017	2018		128-bit memory path	based on SH-4, work suspended
blue	https://opencor	stable	Al Williams	accum	16	16	spartan-3	James	remov	1025		4			63	##	14.7	0.67	1.0	41.1	X	Y	verilog	16	topbox	Y		N		4K	4K	N	18		2		2009	2010		derived from Caxton Foster's Blue	http://www.youtube.com/watch?v=dt4zezZP8
blue_fpga	https://github.com/Gecko		Jaime Centeno	accum	16	16	spartan7	James	need to run V2022			6				##	v24.1	0.67	1.0		X		vhdl	49	system	Y		N		4K	4K	N	18		2		2020	2023		gate level png's, simulator exe	
bobcat		beta	Stan Drey	DSP	16	24	kintex-7-3	James Brakef		1622		6	1		107	##	14.7	0.67	1.0	44.0	X		vhdl	30	bobcat_cc	Y		N		64K	64K						1998	2000			dead web links
brainfuckcpu	https://opencor	beta	Aleksander Kaminski	mem	8	3	kintex-7-3	James Brakef		110		6			432	##	14.7	0.08	2.0	157.2	X		verilog	1	brainfuck_cpu			N	Y				8		0		2014	2015	http://www.cliffo	Touring machine like, 2ndary link is ar	adj prog & data mem size, terrible name
bst-cpu	https://github.c	stable	Yichun Ma	RISC	32	32	arria-2	James Brakef		1439		A		2	58	##	q18.0	1.00	1.0	40.2	A		verilog	26	sc_computer			N		4G	4G				32		2016	2016		learning, single cycle uP	
bst-cpu	https://github.c	stable	Yichun Ma	RISC	32	32	kintex-7-3	James	altera primitives			6				##	14.7	1.00	1.0		A		verilog		sc_computer			N		4G	4G				32		2016	2016		learning, pipeline uP	
btsr1arch	https://github.c	alpha	Brendan Bohannon	CISC	64	16	artix-7	James Brakef		55967	#####	6	52	112	75	##	v23.2	1.00	2.0	0.7	X		verilog	149	topunit	Y	yes	Y	N	256T	256T	Y	64		32		2018	2024	https://www.yout	64-bit regs, 16x inst, 48-bit VM	BJX2 is superset of BtSR1, 4 data sizes
btsr1arch	https://github.c	alpha	Brendan Bohannon	CISC	64	16	artix-7	James Brakef		29964	#####	6	26	108	58	##	v24.2	1.00	2.0	1.0	X		verilog	149	topunit	Y	yes	Y	N	256T	256T	Y	64		32		2018	2024	https://www.yout	64-bit regs, 16x inst, 48-bit VM	BJX2 is superset of BtSR1, 4 data sizes
btsr1arch	https://github.c	beta	Brendan Bohannon	CISC	32	16	kintex-7-3	James Brakef		4762		6		10	167	##	14.7	1.00	1.5	23.3	X		verilog	11	bsrexunit	Y	yes	Y	N	64K	64K	Y	64		32		2018	2024		is BtSR1, msp430 like, fltg-pt defined	3 data sizes, no (R++) or (--R) modes
bugs18	https://drive.google.com/		Myron Plichota	forth	16	18	spartan-7	Myron Plichota				6			48						X	Y	verilog	18	Bugs18_S	Y	asm	N		64K	64K	N	19					2022		Four bit op-codes, Python assembler &	full set of RTL SOC devices
bytemachine	https://github.c	mature	c0pperdragon	forth	8	8	kintex-7-3	James Brakef		319		6		1	250	##	14.7	0.33	2.0	129.3	AX		vhdl	7	bytemach	ome		N	N		4K	Y	30				2016	2017		top is Altera schematic	results are for 2016 bare core
c16	https://opencor	stable	Jsauermann	accum	16	8	spartan-3	James Brakef		1751		4		16	57	##	14.7	0.33	1.0	10.7	X		vhdl	22	Board_cp	mi	yes	N		64K	64K	Y			5		2003	2012		8080 derivative, optional UART, 8-bit	xilinx 4K RAM primitives
c16too	https://www.sc	stable	Cole Design and Devel	RISC	16	16	kintex-7-3	James Brakef		510		6			271	##	14.7	0.67	4.0	88.9	X		vhdl	1	core	Y	asm	N		64K	64K	N	20		8		2003		coledd.com/electr	graphics capability	clock/2 and six phases
c2650_mister	https://github.com/Grabu		Grabulosaure	c2650	8	8															A	Y	vhdl, V	39	sys_top	Y		N		32K	32K	Y					2018	2020	https://en.wikiped	clone of Signetics 2650 uP	based on the IBM 1130, Altera project & PLL
c3pu	https://github.com/isovic		Ivan Sovic	accum	16	16	spartan3	Ivan Sovic		580	268	4					14.6	0.67	3.0		X		vhdl	17	mc3pu	Y	asm	N		64K	64K		22		8		2013	2015		Spartan3: 268FF, 580 4LUT; 22 inst, 8 reg, 3clks/inst, 65K wds, asm	
c3pu	https://github.com/isovic		Ivan Sovic	accum	16	16	kintex7	James	no los	303	234	6			250	##	14.7	0.67	3.0	184.3	X		vhdl	17	Cpu_black	Y	asm	N		64K	64K		22		8		2013	2015		large state enumeration	uses internal tri-states
c88	https://github.c	alpha	Daniiel Bailey	accum	8	8	kintex-7-3	James Brakef		3088		6	2		167	##	14.7	0.33	2.0	8.9	X		vhdl	25	C88	Y	asm	N		8	256	Y	10		8		2015	2015	https://www.yout	only 8 memory locations	used 3658 Dff, doesn't infer block or LUT RAM
c88	https://github.c	alpha	Daniiel Bailey	accum	8	8	spartan-3	James	Dff ge	2664		4	2		54	##	14.7	0.33	1.0	6.7	X		vhdl	25	C88	Y	asm	N		8	256	Y	10		8		2015	2015	https://www.yout	only 8 memory locations	used 3785 Dff, doesn't infer block or LUT RAM
cardiac	https://opencor	mature	Al Williams	accum	13	12	spartan-3	James Brakef		557		4			71	##	14.7	0.30	1.0	38.5	X		verilog	16	vtach	Y	asm	N		100	100	N	10				2013	2019	https://www.cs.d	CARDboard Illustrative Aid to Comput	3 digit BCD arithmetic
cast_8051	http://www.cas	proprietar	CAST Inc	8051	8	8	virtex-6	CAST I	820 sl	1800		6		2	81	##	12.1	0.33	3.0	5.0	X		proprietary			Y	yes	N		64K	64K	Y			32				http://www.cast-i	Cast has uP related IP	several versions, FPGA kits
cast_ba22	http://www.cas	proprietar	CAST Inc	RISC	32	16	spartan-6	CAST Inc		1800		6		32	72			1.00	1.0	40.0	X		proprietary			Y	yes			4G	4G				32				http://www.cast-i	Cast has uP related IP	several versions, FPGA kits

_uP_all_soft folder	opencores or prmary link	status	author	style / clone	data	inst sz	FPGA	repor ter	com ents	LUTs ALUT	Dff	LUT?	mults	blk ram	F max	date	tool ver	MIPS /inst	clks/ inst	KIPS /LUT	ven dor	SOC	src code	#src files	top file	doc	tool chai	fltg pt	Hav'd	max dat	max inst	byt ead	# inst	adr mo	# reg	pip e	start year	last revis	secondary web link	note worthy	comments
cbox16	https://github.com/Engine		engineersbox	arm	16	16	spartan-7	James Brakefield				6				##	v22.1						schem	10	manual_cpu			N		64K	64K				8			2022		very little	Digital schematic, VHDL & verilog
cd16	http://anycpu.o	stable	Brad Eckert	forth	16	16	spartan-3	James Brakef		681		4			83	##	14.7	0.67	2.0	41.0	AX	B	vhdl	16	cd16			N		128K	8M						2003	2003	http://web.archiv	Spartan-3 block RAM	bare core
cd16	http://anycpu.o	stable	Brad Eckert	forth	16	16	spartan-3	James Brakef		618		4		7	31	##	14.7	0.67	2.0	16.9	AX	Y	vhdl	16	demosocext			N		128K	8M						2003	2003	http://web.archiv	Spartan-3 block RAM	includes stack RAMs & some inst RAM
cdc160	https://github.com/jadels		Jan Adelsbach	cdc160	12	12																	scala	2	cdc160	Y		N		4K	4K		64					2015			
cfm	https://github.com/cbiffle		Cliff L. Biffle	forth	16	16																	haskell	23				N		64K	64K						2018	2018	https://clash-lang	Forth-inspired processor targeting the	alu inst is ucoded, some missing ops
chad	https://github.com/bradle		Brad Eckert	forth	18	16	atrix-7-1	James	DFF ex	1982		6		5	127	##	v21.1	0.80	1.0	51.4	AXML		verilog	33	mcu_arty	Y	yes	N		64K	64K	N	23		16			2021		verilog, .f &.c code; fpga project files	max SOC, -1 speed grad
chad	https://github.com/bradle		Brad Eckert	forth	18	16	atrix-7-3	James	optior	1972		6		3	196	##	v21.1	0.80	1.0	79.5	AXML		verilog	33	mcu	Y	yes	N		64K	64K	N	23		16			2021		verilog, .f &.c code; fpga project files	min SOC, -3 speed grade
chad	https://github.com/bradle		Brad Eckert	forth	18	16	atrix-7-3	James	DFF ex	1995		6		5	175	##	v21.1	0.80	1.0	70.4	AXML		verilog	33	mcu_arty	Y	yes	N		64K	64K	N	23		16			2021		verilog, .f &.c code; fpga project files	max SOC, -3 speed grade
chad	https://github.com/bradle		Brad Eckert	forth	18	16	**zu-3e**	James Brakef		2196	2211	6		5	250	##	v21.1	0.80	1.0	91.1	AXML		verilog	33	mcu_arty	Y	yes	N		64K	64K	N	23		16			2021		verilog, .f &.c code; fpga project files	
chip_6502	http://www.aholme.co.uk		Andrew Holme	6502	8	8	spartan7-	James Brakef		514	767	6			200	##	v23.2	0.33	4.0	32.1	X	Y	verilog	5	chip_6502	Y	yes	N		64K	64K	Y						2016	http://www.aholn	cycle accurate generated from transis	also author of two Forth TTL machines
chip8	https://bitbucke	errors	Carsten Elton Sørenser	RISC	8	16	kintex-7-3	James	missing modules							##	14.7						verilog	28	chip8	Y		N									2013	2018	https://en.wikipec	Verilog implementation of the SuperC	https://www.zophar.net/pdroms/chip8/chip-8-
chiselwatt	https://github.c	stable	anton blanchard	ppc	32	32															ALX	Y	scala	28	core	Y	yes			4G	4G	Y			32			2023	https://en.wikipec	tiny POWER Open ISA uP written in Scala	
classic_HP_cal	https://github.c	stable	Brian Nemetz	accum	56	10	kintex-7-3	James Brakef		1750		6		3	233	##	14.7	0.17	10.0	2.2	X		vhdl	15	classichp_	Y		N		30	4K	N	40		7		2012			processor & ROMs for HP-55, 45 & 35	includes LED display driver & UART, for Papilio
classic_mips	https://github.com/Kazaw		Xavier Yuhan Liu	mips	32	32	artix-7	Xavier	some	2463	1289	6				##	v2017	1.00	1.0		X		verilog	30	top	Y	yes	N		4G	4G	Y			32	5	2017	2024		Minisys-1 ISA, pipelined and non-pipel	English RTL, comments in Chinese
classy_core_17	https://github.com/classy		Andreas Schweizer	AVR	8	16	spartan-3	Andreas Schv		358		4			164	##	14.7	0.33	1.0	151.2			vhdl	8	top	Y	yes	N		64K	128K	Y	72		32			2019	https://blog.classy	adjuct to some custom logic	Implementing a CPU in VHDL parts 1..3
cmips	https://github.c	mature	Roberto Hexsel	MIPS	32	32	cyclone4	Roberto Hexs		6347	2596	4		22	50	##		1.00	1.0	7.9	A	Y	vhdl	22	core	Y	yes	N	N	4G	4G	Y			32	5	2017	2019	http://www.inf.uf	5-stage pipeline, MIPS32r2 core	
c-nit	http://www.c-n	stable	Sumit	RISC	16	16	spartan-3	James	xilinx I	752		4		3	100	##	14.7	0.67	2.0	44.5	X		verilog	6	soc	or	asm	N	N	64K	64K	Y	22		15		2003	2004		RISC with several load/store modes	
coco3fpga	https://github.c	mature	Gary Becker	6809	8	8	spartan7	James	Kent's	1536	195	6			78	##	v23.2	0.33	3.0	5.6	X		verilog	39	cpu09l_12	Y	yes	N		64K	64K		44	13	8		2007	2015	http://www.davel	uses John Kent's 6809 & adds color co	altera project with 6809 & 6502 uPs
coco3fpga	https://github.c	mature	Gary Becker	6809	8	8												0.33	3.0		A	Y	verilog	39	coco3fpga	Y	yes	N		64K	64K		44	13	8		2007	2015	http://www.davel	uses John Kent's 6809 & adds color co	altera project with 6809 & 6502 uPs
coen_316_cpu	https://github.c	alpha	G.K Yvann Monny	RISC	32	32	kintex-7-3	James	does r	897		6			127	##	14.7	1.00	3.0	47.0	X		vhdl	8	cpu_dp			N		32	32	N	20		32		2018	2018		MIPS based, simulation DO files, I&D c	very small caches do not infer any RAM
cole_c16	https://www.sc	beta	Cole Design & Develop	RISC	16	16	spartan-6	James Brakef		554		6			298	##	14.7	0.67	7.0	51.4	X		vhdl	1	core	Y	asm	N		64K	64K	N	20		8		2002	2012	https://blog.classy	(7) clks per inst, complete SOC	
complete_8bit	https://www.qu	stable	Van-Lei Le		8	8	kintex-7-3	James	modifi	208		6		1	260	##	14.7	0.33	3.0	137.5	X		vhdl	6	computer	N		N		96	128	Y					2016				memory_unit uses block RAM, IO ports pruned
complete-arm-	https://github.com/Vedar		Vedant Raval	arm	32	32	spartan7	James	single cycle			6				##	v23.2	1.00	1.0		X		vhdl	33	main	Y	yes	N		4G	4G	Y	80		16			2021		Single-cycle & multi-cycle ARM uP	wants distributed ROM & RAM?
complete-arm-	https://github.com/Vedar		Vedant Raval	arm	32	32	spartan7	James	multi-cycle #1			6				##	v23.2	1.00	1.0		X		vhdl	33	multi_cycl	Y	yes	N		4G	4G	Y	80		16			2021		Single-cycle & multi-cycle ARM uP	missing memory IP
complete-arm-	https://github.com/Vedar		Vedant Raval	arm	32	32	spartan7	James	multi-cycle #2			6				##	v23.2	1.00	1.0		X		vhdl	33	multi_cycl	Y	yes	N		4G	4G	Y	80		16			2021		Single-cycle & multi-cycle ARM uP	missing memory IP
cookie	https://github.com/pento		pentolope	RISC	16	16															A	Y	system	46	top_cooki	N	yes	N									2020	2022	https://github.con	OoO and parallel processing	also C compiler
copro6502	https://github.c	stable	David Banks	6502	8	8	kintex7-3	James	bare c	636	144	6			258	##	14.7	0.33	3.0	44.7	X	Y	vhdl, Verilog		T65	Y	yes			64K	64K	Y					2014	2019	https://stardot.or	65C102	
copro6502	https://github.c	stable	David Banks	6809	8	8	kintex7-3	James	bare c	1958	206	6			107	##	14.7	0.33	3.0	6.0	X	Y	vhdl, Verilog		cpu09	Y	yes			64K	64K	Y					2014	2019	https://stardot.or	6809	
copro6502	https://github.c	stable	David Banks	32032	32	16	kintex7-3	James	bare core			6				##	14.7	1.00	3.0		X	Y	vhdl, Verilog		32016.xis	Y	yes			64K	64K	Y					2014	2019	https://stardot.or	32016	floating-point is separate co-processor
copro6502	https://github.c	stable	David Banks	68000	32	16	kintex7-3	James	bare core			6				##	14.7	1.00	3.0		X	Y	vhdl, Verilog		68000.xis	Y	yes			64K	64K	Y					2014	2019	https://stardot.or	68000	
copro6502	https://github.c	stable	David Banks	arm	32	32	kintex7-3	James	bare core			6				##	14.7	1.00	2.0		X	Y	vhdl, Verilog		ARM2.xis	Y	yes			64K	64K	Y					2014	2019	https://stardot.or	ARM2	
copro6502	https://github.c	stable	David Banks	CISC	8	8	spatan6-9		ISE projects for each c			6				##	14.7				X	Y	vhdl, Verilog			Y											2014	2019	https://stardot.or	65C102, Z80, 80286, 6809, PDP11, AR	19 ISE projects on 8 uP, various clock speeds
copro6502	https://github.c	stable	David Banks	CISC			spatan6-9					6					14.7				X	Y	vhdl, Verilog			Y	yes			64K	64K	Y					2014	2019	https://stardot.or	65C102, Z80, 80286, 6809, PDP11, AR	various uP RTL authors
copro6502	https://github.c	stable	David Banks	pdp11	16	16	spatan6-9	James Brakef		3689	947	6	1	16	33	##	14.7	0.67	3.0	2.0	X	Y	vhdl, Verilog		PDP11.xis	Y	yes			64K	64K	Y	70	13	8		2014	2019	https://stardot.or	PDP11	max'd out block RAM
copro6502	https://github.c	stable	David Banks	pdp11	16	16	kintex7-3	James	bare c	4978	1587	6	1		204	##	14.7	0.67	3.0	9.1	X	Y	vhdl, Verilog		pdp2011_	Y	yes			64K	64K	Y	70	13	8		2014	2019	https://stardot.or	PDP11	
copro6502	https://github.c	stable	David Banks	x86	16	8	kintex7-3	James	bare core			6				##	14.7	0.67	3.0		X	Y	vhdl, Verilog		x86.xise	Y	yes			64K	64K	Y					2014	2019	https://stardot.or	80286	
copro6502	https://github.c	stable	David Banks	z80	8	8	kintex7-3	James	bare core			6				##	14.7	0.33	3.0		X	Y	vhdl, Verilog		z80.xise	Y	yes			64K	64K	Y					2014	2019	https://stardot.or	Z80	
copyblaze	https://opencor	stable	Abdallah Ellbrahimi	picoBlaze	8	18	kintex-7-3	James	missin	622		6			217	##	14.7	0.33	2.0	57.5	AX		vhdl	16	cp_copybl	Y	asm	N		256	2K	Y					2011	2016		wishbone extras	
core_arm	https://opencor	beta	Konrad Eisele	ARM	32	16	kintex-7-3	James Brakef		1239		6		3	250	##	14.7	1.00	1.0	201.8	X	Y	vhdl	151	arm_proc	Y	yes	N		256M	256M				16		2004	2009	http://cfw.sourcef	very large project with many unused s	missing files found in sourceforge dir, very little
core9900	https://github.com/dnotq		Matthew Hagerty	TMS9900	8	8	spartan7-	James	incom	41	30	6			122	##	v23.2	0.33	5.0		X	Y	vhdl	7	top	Y	yes	N		64K	64K				16			2017		MSP 9900	LUT counts don't match those of a 8bit uP
cortex_m3	http://www.clo	proprietar	Tobias Strauch	ARM	32	16																	proprietary												16		2013		cortex M3 data sh	claims to be mature	various academic papers, several projects
cosmac	https://github.c	beta	Eric Smith	1802	8	8	kintex-7-3	James Brakef		244		6			270	##	14.7	0.33	1.0	365.5	X		vhdl	1	cosmac	Y	asm	N	N	64K	64K	Y	100		16		2009	2020		AKA COSMAC ELF of 1976	Fmax is for bare core, runs CamelForth
cosmac	https://github.c	beta	Eric Smith	1802	8	8	kintex-7-3	James	inferre	598		6		17	87	##	14.7	0.33	1.0	48.0	X	X	vhdl	14	elf	Y	asm	N	N	64K	64K	Y	100		16		2009	2020		uses PIXIE graphics core	modified to use block RAM
cosmacELF	https://hackada	stable	Winston Lowe	1802	8	8												0.33	1.0		X		scala	8	toplevel	Y	asm	N	N	64K	64K	Y	100		16			2020	https://hackaday.i	AKA COSMAC ELF of 1976	instructions on using Scala
cpu_32	https://github.com/aslak3		Lawrence Manning	RISC	16	16																	vhdl	10	cpu	Y	asm	N		64K	64K	Y	32		8			2020	https://www.yout	educational, DIY, VHDL, youtube video, uses **customasm**, doc in readme.md	
cpu_32	https://github.com/aslak3		Lawrence Manning	RISC	32	32	spartan7	James Brakef		2281	785	6	2		104	##	v23.2	1.00	1.0	45.7	X		vhdl	16	cpu32	Y	asm	N		64K	64K	Y	32		16			2022	https://www.yout	**uses customasm,** doc in readme.m	VGA pattern generator youtube video
cpu_basic	https://github.com/vhdlf/		vhdlf	x86	32	8	cyclone-4	vhdlf		3558	715	4	4			##	q19.1				A		vhdl	7	top	Y		N		64K	64K	Y	26		16		2020	2025	https://github.con	32-bit CPU with x86 inst. format	readme has screen shots, **very readable RTL**
cpu_takagi	https://github.com/takagi		Masayuki Takagi	RISC	8	15	spartan7	James Brakef		643	1017	6			111	##	v23.2	0.67	1.0	115.8	X		verilog	3	cpu					64K	64K	Y	16		16		2016	2016		no use of LUT RAM or block ROM	not much documentation
cpu0	https://jonathar	errors	Chen Zhong-Cheng	RISC	32	32	spartan7	James	errors			6				##	v23.2	1.00	1.0		X		verilog	4	cpu0	Y	yes	N		4G	4G	Y	60		16		2012	2025	https://github.con	**700 page tutorial on LLVM**	**LLVM Backend for the Cpu0 Architecture**
cpu11	https://github.com/1801E		1801BM1	PDP11	16	16		James	large directory tree									0.67	3.0		AX		verilog	11		Y	yes		N	64K	64K	Y	70	13	8		2014	2025		PDP-11 uP reverse engineering	USSR uP, no DEC prototype, proprietary die des
cpu11	https://github.com/1801E		1801BM1	PDP11	16	16	cyclone3	1801B	sampl	2843	752	4		7	115			0.67	3.0	9.0	AX		verilog	11		Y	yes		N	64K	64K	Y	70	13	8		2014	2025		11/34 arch in 3+ chips, Russian clone	all branches have FPGA resourse & fmax
cpu11	https://github.com/1801E		1801BM1	PDP11	16	16	cyclone2	1801B	sampl	2753	728	4		11	64			0.67	3.0	5.2	AX		verilog	11		Y	yes		N	64K	64K	Y	70	13	8		2014	2025		11/34 arch in 3+ chips, Russian clone	pdp11/23 clone, die photos
cpu11	https://github.com/1801E		1801BM1	PDP11	16	16	cyclone4	1801B	sampl	2794	720	4		7	97			0.67	3.0	7.8	AX		verilog	11		Y	yes		N	64K	64K	Y	70	13	8		2014	2025		11/34 arch in 3+ chips, Russian clone	pdp11/23 clone, die photos
cpu11	https://github.com/1801E		1801BM1	PDP11	16	16	max10	1801B	sampl	2691	736	4		7	86			0.67	3.0	7.1	AX		verilog	11		Y	yes		N	64K	64K	Y	70	13	8		2014	2025		11/34 arch in 3+ chips, Russian clone	pdp11/23 clone, die photos
cpu11	https://github.com/1801E		1801BM1	PDP11	16	16	cyclone5	1801B	sampl	2114	1099	A		6	106			0.67	3.0	11.2	AX		verilog	11		Y	yes		N	64K	64K	Y	70	13	8		2014	2025		11/34 arch in 3+ chips, Russian clone	pdp11/23 clone, die photos
cpu11	https://github.com/1801E		1801BM1	PDP11	16	16	cyclone1(	1801B	sampl	2677	724	4		7	79			0.67	3.0	6.6	AX		verilog	11		Y	yes		N	64K	64K	Y	70	13	8		2014	2025		11/34 arch in 3+ chips, Russian clone	pdp11/23 clone, die photos
cpu11	https://github.com/1801E		1801BM1	PDP11	16	16	eagle	1801B	sampl	3821	830	4		9	86			0.67	3.0	5.0	AX		verilog	11		Y	yes		N	64K	64K	Y	70	13	8		2014	2025		11/34 arch in 3+ chips, Russian clone	pdp11/23 clone, die photos
cpu11	https://github.com/1801E		1801BM1	PDP11	16	16	artix7	1801B	sampl	3798	1536	6		9	100			0.67	3.0	5.9	AX		verilog	11		Y	yes		N	64K	64K	Y	70	13	8		2014	2025		11/34 arch in 3+ chips, Russian clone	pdp11/23 clone, die photos
cpu11	https://github.com/1801E		1801BM1	PDP11	16	16	kintex7	1801B	sampl	3073	1380	6		9	133			0.67	3.0	9.7	AX		verilog	11		Y	yes		N	64K	64K	Y	70	13	8		2014	2025		11/34 arch in 3+ chips, Russian clone	pdp11/23 clone, die photos
cpu16	http://www.ultr	stable	C.H. Ting	forth	16	5	kintex-7-3	James Brakef		347		6			364	##	14.7	0.67	1.0	702.1	X		vhdl	1	cpu16			N	N	64K	64K	N	28				2000	2000		P16 in VHDL	CPU24.vhd with width=16
cpu-16	https://opencores.org/prc		Yvo Zoer	RISC	16	16												0.67	3.0		A		verilog	5	cpu16			N	N	64K	64K	N	32		8		2019	2021		no LUT RAM, uses block RAM	Altera register file
cpu-16-bit	https://github.com/Vedar		Vedang Asgaonkar	RISC	16	16	spartan7	James	8 latch	468	195	6			147	##	v23.2	0.67	3.0	70.2	X		vhdl	5	cpu	Y		N		64K	64K	N	17		8			2022		ld/st multiple & predication insts	trimming of inst reg
cpu2	https://github.com/cas-m		Craig Shannon	RISC	32	32	artix7														X		vhdl	12	cpu	Y	asm	N		4K	4K							2024		used to verify know how to use FPGAs	**example of SLL & SRL, very slow**
cpu6502_true_	https://opencor	stable	Jens Gutschmidt	6502	8	8	kintex-7-3	James Brakef		1678		6			159	##	14.7	0.33	4.0	7.8	X		vhdl	7	r6502_tc		yes	N	N	64K	64K	Y					2008	2024		cycle accurate	web page update only
cpu65c02_true	https://opencor	stable	Jens Gutschmidt	6502	8	8	spartan-6	James	latch v	4794		6			47	##	14.7	0.33	4.0	0.8	X		vhdl	8	core		yes	N	N	64K	64K	Y					2008	2018		cycle accurate	
cpu7	https://github.com/robert		Robert Östling	forth	16	16															L	Y	verilog	3	cpu7	Y	asm	N		64K	64K		16	3	8			2019		combination stack/RISC hybrid	RISC ISA w/stack address modes, return bit
cpu8080	https://opencor	stable	Scott Moore	8080	8	8	kintex-7-3	James Brakef		1179		6			299	##	14.7	0.33	9.0	9.3	X		verilog	1	m8080	Y	yes	N	N	64K	64K	Y					2006	2016		includes VGA display generator, three variants	
cpu86	http://www.ht-l	beta	Hans Tiggeler	x86	8	8	kintex-7-3	James Brakef		3421		6	1		127	##	14.7	0.17	2.0	3.1	X		vhdl	23	cpu86_top	Y	yes	N	N	1M	1M	Y					2002	2018	http://www.ht-lat	8088 clone	ht-labs offers several uP cores
cpugen	https://opencor	stable	Giovanni Ferrante	RISC	32	16	kintex-7-3	James Brakef		474		6			192	##	14.7	0.67	1.0	271.8	AX		vhdl	14	cpu	Y	asm	N	N								2003	2009		x86 .exe generates VHDL RISC uP	using 16 bit example
cpugen	https://opencor	stable	Giovanni Ferrante	RISC	32	16	kintex-7-3	James Brakef		1597		6	8		154	##	14.7	1.00	1.0	96.3	AX		vhdl	14	cpuc	Y	asm	N	N								2003	2009		x86 .exe generates VHDL RISC uP	using 32 bit example
cpus-caddr	https://github.com/lisper/		Brad Parker	lisp	32	48	spartan6	Brad F	ran his	4223	2794	6		102	50	##	14.7	1.00	1.0	11.8	X		verilog	70	top	Y	lisp		Y	16M	16K						2011	2016	https://dspace.mi	Verilog FPGA re-implementation of M	uses 48-bit u-code, multiple clocks
cpus-pdp11	https://github.com/lisper/		Brad Parker	PDP11	16	16	spartan3	Brad Parker				4				##					X	Y	verilog	35	top2	Y	yes		N	64K	64K	Y			8		2006	2016		A working PDP-11 cpu with an RK11 disk emulator which uses a IDE disk as a backing s	
cpus-pdp8	https://github.com/lisper/		Brad Parker	PDP8	12	12	spartan3	Brad Parker		1605	481	4		1	50	##	14.7	0.40	2.0	6.2	X	Y	verilog	15	top	Y	yes	N	N	4K	4K						2004	2016		A working PDP-8/i cpu with an RF08 disk emulator which uses a IDE disk as a backing s	
cqpic	http://www002	stable	Sumio Morioka	PIC16	8	14	arria-2	James	ROM parameter errors			A				##	q13.1	0.67	1.0		A		vhdl, v	5	CQPIC	Y	yes	N	Y	256	4K	Y					1999	2022		LPM macros	
cray1	www.chrisfento	alpha	Christopher Fenton	CRAY1	64	16	kintex-7-3	James Brakef		13463	7358	6	19	10	127	##	14.7	6.00	1.0	56.6	X		verilog	46	cray_sys_	Y	yes	Y	N	4M	4M	N	128		536		2010	2015	https://www.chris	homebrew Cray1	24-bit address registers
cray1	www.chrisfento	alpha	Christopher Fenton	CRAY1	64	16	**zu-3e**	James	undefi	11510		6	15	1		##	v21.1	6.00	1.0		X		verilog	46	cray_sys_	Y	yes	Y	N	4M	4M	N	128		536		2010	2015	CRAY data sheets	homebrew Cray1	24-bit address registers
cray1	www.chrisfento	alpha	Christopher Fenton	CRAY1	64	16	spartan7-	James	undefi	11554	8305	6	15	1		##	v24.1	6.00	1.0		X		verilog	46	cray_sys_	Y	yes	Y	N	4M	4M	N	128		536		2010	2015	CRAY data sheets	homebrew Cray1	24-bit address registers
cray1	https://www.ch	alpha	Christopher Fenton	CRAY1	64	16	artix-35t	Christopher Fenton				6			105		v24.1	3.00	1.0		X		system veril		cray_sys_	Y	yes	Y	N	4G	4G	N	128		536		2010	2015	https://hackaday.	homebrew Cray-YMP-c90-j90	32-bit adr regs, **wrist watch, no RTL**
cray2_reboot	https://opencor	beta	John Kula	CRAY2	64	16																	non-EDIF gate & modul			Y	yes	Y	N	256M	256M	N	128		528		2016	2017	Cray 1, 2 & 3 docs	gate level code	32-bit address registers
crisv32_axis_e	http://develope	asic	Axis Communications	RISC	32	16																Y	proprietary			Y	yes			4G	4G	Y			16			2007	http://developer.a	embedded comm	very dated product
dalton_8051	www.cs.ucr.edu	stable	Tony Givargis	8051	8	8	kintex-7-3	James	see i8	2725	1350	6	1	0.5	105	##	14.7	0.33	1.0	12.7	X		vhdl	7	i8051_all	Y	yes	N	N	64K	64K	Y					1999	2003	https://ics.uci.edu	ASIC	
dapzipi8	https://github.com/ehsan		Ehsan Ali	picoBlaze	8	18	zu-5e	Ehsan	conve	305	49	6		2	224	##	v22.1	0.33	1.0	242.4	X		vhdl	20	top	Y	asm	N		256	2K	Y						2022		Deterministic Branch Prediction for Re	also zipi8 starting point, PhD thessis
darfpga	https://github.com/darfpg		darfpga	z80	8	8															A	Y	vhdl, verilog			Y	yes	N	N	64K	64K	Y						2022	https://github.con	games ported to MiSTer and DE10-lite	
dataflow_chap	https://opencor	alpha	Rob Chapman, Steven	forth	16	16	kintex-7-3	James	file WebCase report			6					14.7	0.33	1.0				vhdl	27	DataFlowI	Y		N		256	256						2003			course work	
dcpu16	https://github.c	beta	Shawn Tan, Marcus Pe	RISC	16	16	kintex-7-3	James Brakef		662		6	1		318	##	14.7	0.67	4.0	80.4	X		vhdl, v	5	dcpu16_c	Y	asm	N	N	64K	64K	N	37		8		2009	2012	https://en.wikipec	for the 0X10c game	4+ addressing modes, 4 & 5-bit reg /modefields
DE9000	https://www.sir	stable	Michael Freeman	accum	##		spartan-3														X		vhdl	19	de9000	Y												2025		31 digits, polynomial evaluation	patterned after Babbage's Diference Engine
df6805	www.hitechglob	proprietar	Hitech Global	6805	8	8	stratix-1	Hitech Global		1690		4			83			0.33	4.0	4.1	A		proprietary			Y	yes	N	N	64K	64K	Y							6805 data sheets		
dfp	https://opencor	stable	Ron Chapman	forth	8	8	kintex-7-3	James Brakef		297		6			192	##	14.7	0.33	1.0	213.2	X		vhdl	25	DataFlowI	Y											2003	2009		8-bitter, generates a custom VHDL stack machine, compiler is in Forth	
dgb16	see FISA64	stable	Robert Finch	RISC	16	16	kintex-7-3	James Brakef		780		6			313	##	14.7	0.67	1.0	269.0	X		verilog	1	dbg16	Y		N	Y						8				https://github.con	inside FISA64 project	debug uP for fisa64
digital_up	https://github.com/hneen		Helmut Neemann	mips	16	16	spartan-7	James	clockii	716	309	6	1		182	##	v22.1	0.67	1.0	170.1	X		schem	46	processorHD		asm	N	Y	64K	64K		60		16		2016	2022	https://github.con	uP implemented as schematic	has assembler and ISA pdf, 2Kx16 RAM?
digital_up	https://github.com/hneen		Helmut Neemann	mips	16	16	zu-5e	James	clockii	709	310	6	1		250	##	v22.1	0.67	1.0	236.2	X		schem	46	processorHD		asm	N	Y	64K	64K		60		16		2016	2022	https://github.con	uP implemented as schematic	has assembler and ISA pdf, 2Kx16 RAM?
diogenes	https://opencor	beta	Fekknhifer	RISC	16	16	kintex-7-3	James Brakef		807		6		1	297	##	14.7	0.67	1.0	246.3	X		vhdl	11	cpu	Y	asm	N			1K						2008	2009		"student RISC system"	
dlx_calvino	https://github.com/aleten		Alessandro Calvino	DLX	32	32																	vhdl	66	a-dlx	Y	yes	N		4G	4G				32			2019		masters thesis, gate level design	also supports Synopsys Design Compiler
dlx_chiara	https://github.c	stable	Alessandro Di Chiara	DLX	32	32	kintex-7-3	James Brakef		2915		6			90	##	14.7	1.00	1.0	30.9	X		vhdl	32	a-dlx	Y	yes	N		4G	4G				32	5	2017	2017		Course project, no RTL comments, VHDL via instructor?	
dlx_nicola	https://github.c	stable	Nicola Vianello	DLX	32	32																	vhdl	37	a-dlx	Y	asm	N		4G	4G				32			2019		masters thesis	five stage pipeline, forwarding, automatic haza
dlx_palmiero	https://github.c	ASIC	Christian Palmiero	DLX	32	32	kintex-7-3	James	design heiarchy proble			6				##	14.7	1.00	1.0				vhdl	41	a-dlx	Y	yes	N		4G	4G				32	5	2015	2017		Course project, VHDL to netlist (STM 4	ASIC design
dlx_superscala	https://www.rs.	errors	Joachim Horch	DLX	32	32	kintex-7-3	James	degnerate			6				##	14.7	1.00	1.0				vhdl	4	dlx	Y	yes	N		4G	4G				32		1997	1998		Course project, Two inst/clock, doc in	collapses for no apparent reason
dme	https://github.c	stable	ErwinM	RISC	16	16	kintex-7-3	James Brakef		1755		6			53	##	14.7	0.67	1.0	20.4	X		verilog	49	cpu	Y	yes	N		64K	64K	Y	40		8		2016	2017		based on magic-16	computer & computer2 null dsgns: no outputs

_uP_all_soft folder	opencores or prmary link	status	author	style / clone	data	inst sz	FPGA	repor ter	com ents	LUTs ALUT	Dff	LUT?	mults	blk ram	F max	date	tool ver	MIPS /inst	clks/ inst	KIPS /LUT	ven dor	SOC	src code	#src files	top file	doc	tool chai	fltg pt	Hav'd	max dat	max inst	byt ead	# inst	adr mo	# reg	pip e	start year	last revis	secondary web link	note worthy	comments
dp32		errors	Peter Ashenden	RISC	32	32	kintex-7-3	James	errors			6				##	14.7	1.00	1.0				vhdl												32		2001	2001	book, CDROM	from The Designers Guide to VHDL	timing delays in source code
dp8051	https://www.dc	proprietar	Digital Core Design	8051	8	8	virtex-5	Digital Core D		1699		6			200	##	14.7	0.30	1.0	35.3	ALX		proprietary			Y	yes	N		64K	64K						1999	1999		also PIC, HC11, 68000, 680x, d32pro	full system with RAM
dragonfly	http://www.leo	beta	LEOX team	MISC	8	16	kintex-7-3	James Brakef		788		6			164	##	14.7	0.33	1.0	68.6	X		vhdl	6	dgf_core	Y	asm	N	Y	256	2K	Y	42	6	7		2001	2003		unusual, uses FIFOs	
drv16	https://github.com/jecelj		Jecel de Assumpção	risc	16	16	gowin	Jecel de Assu		282	33	4			95	##		0.67	2.0	112.9	AGLX		schem	8	drv16.v	Y		N		64K	64K				16			2024	https://www.mdp	educational, LUT count comparisons t	Digital schematic, RISC-V 16-bit ISA
drv16/mcpu16	https://github.com/jecelj		Jecel de Assumpção	risc	8	8	gowin	Jecel de Assu		69	48	4			313	##		0.17	2.0	378.8	AGLX		schematic			Y		N		256	256				16			2024	https://www.mdp	very simple accumulator based 8 bit uP with four instructions	
drv16/ncpu	https://github.com/jecelj		Jecel de Assumpção	risc	8	8	gowin	Jecel de Assu		264	182	4			127	##		0.33	2.0	79.4	AGLX		schematic			Y		N		256	256				16			2024	https://www.mdpi.com/2674-0729/3/4/20		
dsp16	https://github.com/jotegc		Jose Tejada	dsp	16	16	cyclone5	Jose Tejada		2471	612	A		12							A		verilog	12	jtdsp16	Y	asm	N	Y	64K	64K	N	29		16		2020	2021		compatible with ATT WE DSP16	
dspuva16	http://www.DTI	stable	Santiago de Pablo	DSP	16	16	kintex-7-3	James Brakef		332		6			317	##	14.7	0.67	1.0	640.7	X		verilog	1	dspuva16		asm	N	Y	256	4K		40		16		2001	2004	www.1-core.com/	16 bit data memory, 24 bit regs	broken web link
e0c6200	https://github.com/agg23		Adam Gastineau	accum	4	12															A	Y	system	54	cpu	Y		N			4K						2023	2023		Tamagotchi P1 for Analogue-Pocket/MiSTer, based on Epson E0C6200 uP	
e80	https://github.com/Stokp		Panos Stokas	risc	8	8	spartan7	James	did no	2585		6	1		50	##	v25.2	0.67	1.0	13.0	AGX	Y	vhdl	12	computer	Y	asm	N		256	256		39		8			2026	https://apothesis.	thesis: dsgn impl n eval of E80 8-bit uP	stack ld-st, Constructionism & Microworlds gan
EC16_on_ICE	https://github.com/Edgar		Edgar Conzen	accum	16	16	ice40	Edgar Conzer		940	257	4		1	20		r23.1	0.67	2.0	7.1	L	Y	vhdl	54	ec16_top	Y		N		64K	64K		50				2023	2024		designed FPGA board, Lattice Radiant	
eco32	https://opencor	stable	Hellwing Geisse	RISC	32	32	kintex-7-3	James Brakef		2339		6		1	160	##	14.7	1.00	1.5	45.5	ALX	Y	verilog	14	cpu	Y	yes	N		512M	256M	Y	61		32		2003	2022	https://github.com	MIPS like, slow mul & div	
eco32	https://opencor	stable	Hellwing Geisse	RISC	32	32	kintex-7-3	James Brakef		3367		6		5	147	##	14.7	1.00	1.5	29.1	ALX	Y	verilog	24	eco32	Y	yes	N		512M	256M	Y	61		32		2003	2022	homepages.thm.d	MIPS like, slow mul & div	
eco32f	https://github.c	stable	Stefan Kristiansson	RISC	32	32	kintex-7-3	James Brakef		3845		6	3	4	123	##	14.7	1.00	1.0	32.1	X		verilog	12	eco32f	Y	yes	N		512M	256M	Y	61		32	6	2014	2014		pipelined version of the eco32 CPU	cache & mmu
edge	https://opencor	alpha	Hesham ALMatary	MIPS	32	32	spartan-6	James Brakef		5345		6	7	1	8	##	14.7	1.00	1.0	1.5	X		verilog	30	edge_core	Y	yes	N	N	4G	4G	Y			32	5	2014	2014		Edge Processor (MIPS)	MIPS1 clone
edu-8	https://github.com/Sweiry		SweiryDev	risc	8	16															G		vhdl	5	top	Y	asm	N	Y	256	256	Y	12	2	4			2025	https://www.linke	SoC 8-Bit Educational RISC CPU	complete Gowin project
egpu	https://arxiv.org/pdf/240		Martin Langhammer	RISC	32	40	agilex	Langh	no RTI	10697	#####	A	32	259	771	##	q22.4	8.00	1.0	576.6	A							Y		4G	4G		63		32			2024	https://ar5iv.labs.	800MHz in Agilex FPGA, word size and	ISA configured for each task
eight_bit_uc		stable	Synplicity	RISC	8	12	kintex-7-3	James	signal/variable mixup			6					14.7	0.67	1.0				vhdl	10	eight_bit_uc						2K	Y			32		2000	2000		part of Amplify documentation	
eight32	https://github.com/robins		Alastair M. Robinson	accum	32	8	cyclone-4	Alasta	appro	1300		4			133			1.00	1.0	102.3			vhdl	17	eightthirty	Y	yes	N		500M	500M	Y	28		8		2019	2023	https://retroramb	5-bit op-code & 3-bit reg #	full tool set, see github page for ISA description
ejrh_cpu	https://github.c	stable	Edmund Horner	RISC	16	16	kintex-7-3	James Brakef		928		6	1	2	196	##	14.7	0.67	1.0	141.6	X		verilog	17	machine	Y									16		2015	2015		see web archive for doc	
electronfpga	https://github.c	mature	David Banks	6502	8	8															AX	Y	vhdl			Y	yes	N	N	64K	64K	Y					2014	2020	https://en.wikiped	Acorn Electron ULA in various FPGAs	uses T65 core
ensilica	http://www.ens	proprietar	ensilica.com	eSi-1600	16	16	virtex-5	ensilica		1100		6			160			1.00	1.0	145.5	AX		verilog		eSi-1600	Y	yes			64K	64K	Y	92	10	16	5	2001	2016		verilog source included with license	room for 90 user inst, also as ASIC
ensilica	http://www.ens	proprietar	ensilica.com	eSi-1600	16	16	virtex-5	ensilica		1100		6			160			1.00	1.0	145.5	AX		verilog		eSi-1650	Y	yes			64K	64K	Y	92	10	16	5	2001	2016		verilog source included with license	room for 90 user inst, also as ASIC
ensilica	http://www.ens	proprietar	ensilica.com	eSi-3200	32	16	stratix-4	ensilica		2200		A			200			2.00	1.0	181.8	AX		verilog		eSi-3250	Y	yes			4G	4G	Y	104	10	16	5	2001	2016		verilog source included with license	room for 90 user inst, also as ASIC
ensilica	http://www.ens	proprietar	ensilica.com	eSi-3200	32	16	stratix-4	ensilica		1800		A			200			1.50	1.0	166.7	AX		verilog		eSi-3200	Y	yes			4G	4G	Y	104	10	16	5	2001	2016		verilog source included with license	room for 90 user inst, also as ASIC
ep16	https://github.c	beta	C.H. Ting	forth	16	5	kintex-7-3	James Brakef		837		6			254	##	14.7	0.67	1.0	203.6	X		vhdl	5	ep16	Y	yes	N	N	32K	32K	N	32				2005	2012	PDF files	initialized Lattice memory blocks	5-bit instructions
ep24		stable	C.H. Ting	forth	24	6	kintex-7-3	James	substi	1020		6		3	167	##	14.7	0.83	1.0	135.6	X		vhdl	1	ep24	Y	asm	N	N		4K		27				2002	2002		room for 37 additional op-codes	removing stack clear: 503 LUT6 & 143MHz
ep32	https://www.an	proprietar	C.H. Ting	forth	32	6	XP2	C.H. Ting		3368		4					ispL	1.00	1.0		L		proprietary										29				2007	2018	https://wiki.forth-	kindle book & RTL available: EP32 RISC	RTL: $25 from C.H. Ting
ep32	http://forth.org	mature	C.H. Ting	forth	32	5	spartan7	James Brakef		1147	2309	6			100	##	v23.2	1.00	1.0	87.2	X		vhdl	7	ep32	Y	forth	N		4G	4G		29					2012		has eForth binary & source	now free
ep32	http://forth.org	mature	C.H. Ting	forth	32	5	spartan7	James	XP2 block RAM primiti			6				##	v23.2	1.00	1.0		X	Y	vhdl	7	ep32_chip	Y	forth	N		4G	4G		29					2012		has eForth binary & source	now free
ep8080	https://github.c	beta	C.H. Ting	8080	8	8	kintex-7-3	James Brakef		1276		6			184	##	14.7	0.33	9.0	5.3	X		vhdl	4	ep80	Y	yes	N	N	64K	64K	Y					2002	2016	8080 data sheets	initialized Lattice memory blocks	work related to eP16
ep994a	https://github.c	stable	Erik Piehl	9900	16	16	kintex-7-3	James Brakef		1340		6		5	286	##	14.7	0.83	3.0	59.0	X		vhdl	10	ep994a	Y	yes	N	N	64K	64K	Y			16		2016	2019	https://hackaday.i	TI 990 emulation	also tms9902 (uart) core by Paul Urbanus?
ep994a/icy99	https://github.c	stable	Erik Piehl	9900	16	16												0.83	3.0		L	Y	verilog	29	tms9900	Y	yes	N	N	64K	64K	Y			16		2016	2023	https://hackaday.i	TMS9900 compatible CPU core in VHD	ECP5 supported, TI-99/4A
eric5	http://www.ent	proprietar	Thomas Entner	forth	9	8	cyclone-4	entner-electr		110		4	opt		60			0.42	1.0	229.1	A		proprietary							512	1K				3-4		2005	2011		25 MIPS: ERIC5xs, ERIC5Q	
erp	https://opencor	stable	Shahzadjk	RISC	8	16	spartan-3	James Brakef		366		4	1	1	70	##	14.7	0.33	1.0	63.5	X		verilog	1	ERPverilog	Y							15		6		2004	2014		two report PDFs & one Verilog file	
ez8	https://github.c	stable	Howard Mao	accum	8	16	kintex-7-3	James	replac	644		6		2	233	##	14.7	0.33	2.0	59.6	X		verilog	13	ez8_cpu					256	4K						2014	2014	http://zhehaomao.com/		not sure inferred RAM correct?
f100l_fpga	https://github.c	stable	Michael Kohn	fl100	8	8	spartan-7	James	not a xilinx design												L	Y	verilog	10	f100l	Y						Y						2024	https://www.mike	Ferranti F100-L CPU, derived from int	ISA likes and dislikes
f14_cadc	https://github.com/ATayl		Adam Taylor	risc	20	20															X		vhdl	13	cadc-top	Y		N		128	####				64			2026	https://www.tom	F-14A Central Air Data Computer (CAD	desgnd before i4004 (1968), secret until 1998
f18a	http://www.gre	asic	Chuck Moore	forth																			proprietary			Y	yes													AKA G144A12: 12x12 array	family of parallel processors
f21	http://www.ultr	asic	Jeff Fox	forth	21	5																	proprietary														1997	2011	http://www.ultrat	"machine forth", crazy address space	chip & simulator, AKA MuP21 or F21
f32c	https://github.c	beta	marko zec, vordah, Da	risc-v	32	32	atrix-7-3	zec & vordah		1048		6	4	33	185	##	14.7	1.00	1.0	176.5	X		vhdl	50		Y	yes	N	Y	4G	4G	Y	30		32	5	2014	2025	http://www.nxlab	MIPS or RISC-V ISA, Arduino support	https://www.youtube.com/watch?v=55MzMH
f8	https://fosdem.org/2025/		Phillip Krause	cisc	8	8															GL		verilog	37	cpu	Y	yes	N		64K	64K	Y	200	5			2023	2025	https://hackaday.	8 bit arch designed for C and memory	single-cycle & multi-cycle versions
falcon	https://github.com/FalconCpu/falcon			RISC	32	32															A		verilog	38	cpu	Y	yes			4G	4G				32		2023	2025	https://sdcc.sourc	Design my own CPU ISA (Falcon 32), th	Krause a major contributor to SDCC
fc16		paper	Richard Haskell	forth	16																		proprietary																	PDF papers	chpt 11: VHDL By Example: Fundamentals of Di
fgpu	https://github.c	stable	Muhammed al Kadi	SIMT	32	32	zynq7045	Muhammed a		128K		6	##	167		##	v17.2				X		vhdl	34	fgpu	Y	yes	Y		4G	4G	Y			32		2016	2017	https://dl.acm.org	eigth cores, reviews comparable proje	vivado fltg-pt IP, benchmarks, wikipedia: GPGPU
fisa32	https://github.c	beta	Robert Finch	RISC	32	32	kintex-7-3	James Brakef		3479		6	3	2	152	##	14.7	1.00	1.0	43.7	X		verilog	1	FISA32	Y		N	Y						32		2014	2014	https://github.com/robfinch/Cores		
fisa64	https://github.c	beta	Robert Finch	RISC	64	32	kintex-7-3	James Brakef		10404		6	12	7	65	##	14.7	1.50	1.0	9.4	X		verilog	1	FISA64	Y		N	Y								2015	2015	https://github.com/robfinch/Cores		need to use multi-cycle on mult
fisc	https://github.c	stable	Miguel Santos	RISC	64	32	arria-2	James	errors			A				##	q18.0	2.00	1.0				vhdl	21		Y	yes	Y	N			Y	85	6	32	5	2018	2018	http://www.archf	Flexible Instruction Set Computer	caches, VHDL & System Verilog versions, altera
fisc	https://github.c	stable	Miguel Santos	RISC	64	32	cyclone-4	James Brakef		5036		4		21	66	##	q18.0	2.00	1.0	26.1	A		system	13	fisc_core	Y	yes	Y	N			Y	85	6	32	5	2018	2018	http://www.archf	Flexible Instruction Set Computer	caches, VHDL & System Verilog versions, altera
flare_cpu	https://github.com/fl4shk		Andrew Clark	RISC	32	32																	scala	14		Y		N		4G	4G	Y			16			2024		RISC in spinalHDL	
flexgrip	http://www.ecs	paper	Kevin Andryc	GPGPU	32	32	atrix-7	James Brakef		72649		6	##	119	100	##	14.7	1.00	0.1	11.0	X		vhdl	46	gpgpu_ml505_top_level												2013	2016	http://www.ecs.u	eight GPU processors	requested & received source files
flexgripplus	https://github.c	mature	Josie Condia	GPGPU	32	32																	vhdl															2020	https://opencores	GPGPU based on G80 architecture of NVIDIA, heavly based on flexgrip	
fluid_core	https://opencor	alpha	Stephen Nolting	RISC	8	12	kintex-7-3	James Brakef		956		4			381	##	14.7	0.33	1.0	131.7	X		verilog	17	FluidCore			N	Y						8		2015	2020	https://www.allat	data width adj., mem sizes adj.	2020 version requires registration
forth_kf532	https://github.c	stable	Tarasov Ilia	forth	32	6	kintex-7-3	James	no *.c	1719		6	4	4	172	##	14.7	1.00	1.0	100.3	X		vhdl	1	kf532	N		N	Y	1K	16K						2013	2013		no trace of source code on web	
forth-cpu	https://anycpu.org/forum		Richard Howe	forth	16	16	spartan3	James Brakef		782	91	4	1	2	90	##	14.7	0.67	1.0	77.5	X		vhdl	10	cpu			N									2013	2024	http://www.aholn	forth uP for GPS reciever, 32-bit stack	Spartan3 block RAM & multiplier
forth-cpu/h2	https://opencor	stable	Richard Howe	forth	16	16	kintex-7-3	James Brakef		1858		6		9	149	##	14.7	0.67	1.0	53.8	X	Y	vhdl	11	top	Y		N		64K	64K		25				2017	2020	https://github.com	H2 Forth SoC, VHDL reads *.hex & *.bi	derived from J1, hex & bin files in 2/16/2018 ta
forwardcom	https://github.c	stable	Agner Fog	cisc	32	32	atrix-7	Agner Fog		13248	4990	6			64	##	v20.1	1.00	1.0	4.8	X		system	18	top	Y	asm	Y		64K	32K	Y			64		2016	2023	https://www.forw	x86 like, complete ISA, MMX & vector	x86 adr modes, vector inst use width of vect reg
forwardcom	https://github.c	stable	Agner Fog	cisc	64	32	atrix-7	Agner Fog		21121	7392	6			56	##	v20.1	2.00	1.0	5.3	X		system	18	top	Y	asm	Y		64K	32K	Y			64		2016	2023	https://www.forw	x86 like, complete ISA, MMX & vector	x86 adr modes, vector inst use width of vect reg
fpag4_risc16_1	http://www.fpg	errors	Van Loi Le	RISC	16	16	kintex-7-3	James	degenerate design			6				##	14.7	0.66	1.0				verilog	15	Risc_16_b	Y		N	Y	64K	64K		13	4	16		2017	2017		similar to mips16_16_1cycl	incomplete Risc_16_bit module
fpg1	https://github.com/hrvacl		Hrvoje Čavrak	PDP1	18	18	cycloneV		altera primitives												A	Y	verilog	31	pdp1	Y	yes	N		4K	4K							2019		video display of PDP-1 console, a mister core, retro gaming	
fpga_arm	https://github.c	ncomplet	Robert Dunne	arm	32	32															A		verilog			Y	yes	N		4G	4G	Y						2021	https://www.ama	from book:Computer Arch Tut Using an FPGA: ARM & Verilog Intros	
fpga_x86	https://github.c	ncomplet	Robert Dunne	x86	32	8															A		verilog			Y	yes	N		4G	4G	Y						2023	https://www.ama	from book: Intro to X86 Machine Code Asm Lang: Using an FPGA with Verilog	
fpga4_8bit_up	http://www.fpg	stable	Van Loi Le	accum	8	8	kintex-7-3	James Brakef		258		6		1	200	##	14.7	0.33	3.0	85.3	X		vhdl	9	computer	ome		N		96	128	Y	10		2		2016	2016	book: LaMeres Int	educational	16 input & 16 output ports fill out 256 byte adr
fpga4_mips_5p	http://www.fpg	errors	Van Loi Le	MIPS	32	32	kintex-7-3	James	degenerate design			6				##	14.7	1.00	1.0				verilog			Y	yes	N	N	4G	4G	Y			32	5	2017	2017		educational, full pipelined MIPS	incomplete
fpga4_mips16_	http://www.fpg	stable	Van Loi Le	RISC	16	16	kintex-7-3	James Brakef		369		6			200	##	14.7	0.67	1.0	363.1	X		verilog	8	mips_16			N		65K	65K		13		8		2017	2017		educational, no block RAM inferred	same prog & data mem and alu as mips16_16_
fpga4_mips16_	http://www.fpg	stable	Van Loi Le	RISC	16	16	kintex-7-3	James Brakef		352		6			213	##	14.7	0.67	1.0	405.0	X		vhdl	8	mips_vhdl			N		65K	65K		8		8		2017	2017		educational, no block RAM inferred	actual prog sz=16, actual data mem sz=256
fpga4_up8_12	http://www.fpg	errors	Van Loi Le	accum	8	12	kintex-7-3	James	degenerate design			6				##	14.7	0.33	1.0				verilog	7	microcontroller			N									2016	2016		educational, simplified PIC12	incomplete
fpga_firendly_(	https://past.dat	paper onl	Mathieu Rosière	OpenRISC	32	32	virtex-5					6														Y	yes	N	N	4G	4G	Y			32			2012		Out-of-Order Superscalar Processor on FPGA: The ReOrder Buffer Design	
fpga-64	http://www.syn	stable	Peter Wendrich	6502	8	8	kintex-7-3	James Brakef		2210		6		2	156	##	14.7	0.33	4.0	5.8	X	Y	vhdl	26	fpga64_cc	Y	yes	N	N	64K	64K	Y			26		2005	2008		Rendition of Commodore 64	altera top level schematic
fpga-bbc	https://github.com/mikes		Mike Stirling	6502	8	8															A		vhdl					N		65K	65K						2011	2016	https://www.mike	BBC micro, uses t65 uP	also ZX-spectrum retro project
fpga-calculator	https://github.com/gdevic		Goran Devic	risc	4	12	cyclone2					4									A	Y	system	15	top	Y		Y	Y	4K	4K				8			2026	https://baltazarst	fully functional scientific calculator in	custom soft CPU, microcode firmware, and sup
fpga-chip8	https://bitbucket.org/csor		Carsten Sørensen	chip8	8	16	spartan6		xise project			6					14.7				X	Y	verilog	10	chip8	Y		N		4K	4K	Y	35		16			2016	https://en.wikiped	Implementation of the SuperChip virtu	part of MiST
fpgacomputer	https://github.c	errors	Milan Vidakovic	RISC	16	8	arria-2	James	errors			A				##	q18.0	0.67	4.0			Y	verilog	10	computer	Y	asm	N	N	64K	64K	Y	25		8		2018	2018	https://mvidakovi	16-bit CPU, 64KB, UART (115200 bps), and VGA	
fpgacomputer	https://github.c	errors	Milan Vidakovic	RISC	16	8	kintex-7-3	James	erros			6				##	14.7	0.67	4.0			Y	verilog	10	computer	Y	asm	N	N	64K	64K	Y	25		8		2018	2018	https://mvidakovi	16-bit CPU, 64KB, UART (115200 bps), and VGA	
fpgammix	https://github.c	stable	Tommy Thorn	MMIX	64	32	arria-2	James Brakef		11605		A	8	10	94	##	q13.1	1.50	4.0	3.0	A		system	3	core	Y	yes	Y	Y	16Q	16Q	Y	256		288		2006	2014	https://en.wikiped	clone of Knuth's MMIX	micro-coded
fpz8	https://opencor	stable	Fabio Pereira	Z8	8	8	cyclone-4	James Brakef		5184		4	1	16		##	14.7	0.33	4.0		A		vhdl	4	fpz8_cpu_	Y		N	Y	2K	16K	Y					2016	2016		Zilog Z8 encore (eZ8) 8-bit core	Altera megafunctions (mem)
free_risc8	https://web.arcl	stable	Thomas Coonan	PIC16	8	14	kintex-7-3	James Brakef		355	118	6			142	##	14.7	0.33	1.0	132.2	X		verilog	8	cpu	Y	yes	N		256	4K	Y					2002	2011	https://web.archive.org/web/20120309123835/http://www.mindspring.com/~tcoonan/index.html		
free6502	http://web.arch	stable	David Kessner	6502	8	8	kintex-7-3	James Brakef		646	144	6			193	##	14.7	0.33	4.0	24.6	X		vhdl	5	free6502	Y	yes	N	N	64K	64K	Y					1999	2000	http://www.sprov	microcoded	
ft64	https://github.c	alpha	Robert Finch	RISC	64	32																	verilog		FT64v3b	Y	yes	Y		16E	16E	Y					2017	2018	https://www.ama	4th attempt at 64-bit core (raptor64, t	amazon kindle book, L1 & L2 icaches & L1 dcach
fusa_cpu	https://github.com/Jeries		Jeries9947	risc	32	32																	system	13	locksetro_top			N		256	256	N	11		32			2026		dual-core lockstep RISC uP	compact hardware fault-detection arch
fx68k	http://fx68k.fxatari.com/f		Jorge Cwik	68000	16	16	spartan7	James Brakef		3162	1504	6		1	100	##	v23.2	1.00	3.0	10.5	X		system	3	fx68k	Y	yes	N		4G	4G	Y			16		2018	2021	https://github.com	Cycle accurate, see http://atari-forum.com/viewtopic.php?f=28&t=34730#p358139	
gameboy	https://github.com/blazer		Raphael Stäbler	accum	8	8															X	Y	verilog	45	top	Y		N		64K	64K	Y					2021	2021	https://gekkio.fi/f	independent of Mister	z80-8080 hybrid, see pdf file
gameboy	https://github.com/zephr		Wenting Zhang	accum	8	8															X	Y	verilog	23	boy	Y		N		64K	64K	Y					2021	2022	https://gekkio.fi/f	directory structure diagram	lists sources, https://github.com/nightslide7/G
ghostchip	https://github.com/jacobl		Jacob Lister	chip8	8	16	spartan7	James	verilog problems													Y	verilog	13	ghostchip	Y		N		4K	4K	Y	35		16			2025	https://en.wikiped	virtual ISA for 1802 uP with many system (gaming) level functions	
gigatron	https://github.com/micha		michalin	accum	8	16	spartan7	James	deleted Gowin RTL, em			6				##	v23.2	0.33	3.0		G	Y	verilog	19	gigatron	Y	yes	N		32K	64K	Y	17					2024	https://hackaday.i	based on TTL version: gigtaron.io, use	uses sweet16 style interpreter for aps
gl85	http://simlab.ec	stable	Alex Miczo	8085	8	8	kintex-7-3	James	gate level design			6					14.7	0.33	4.0		X		vhdl	1	i8085	Y	yes	N	N	64K	64K	Y					1993		http://www.fpga.	also a TTL implementation in VHDL	complete 8-bit uP software & games
gpu	https://opencor	stable	Diego A. Idarraga				kintex-7-3	James	errors in source			6				##	14.7	1.00	1.0				vhdl	21	gpu			Y									2015	2015		graphic processing unit	coding errors
gumnut	https://github.c	stable	Oswaldo Hernandez	RISC	8	18															AX		system	16	gumnut-c	Y	asm	N	Y	256	4K	Y	22		8			2021		coursework using Ashenden's design	
gumnut	http://digitaldes	stable	Peter Ashenden	RISC	8	18	kintex-7-3	James Brakef		388		6			259	##	14.7	0.33	1.0	220.7	AX		verilog	6	gumnut-rt	Y	asm	N	Y	256	4K	Y			8		2007			see Digital Design: An Embedded Systems Approach Using VHDL	
gup	https://opencor	stable	Kevin Phillipson	68HC11	8	8	arria-2	James Brakef		925		A	1	1	127	##	q13.1	0.33	4.0	11.3	A		vhdl	25	gator_upr	Y	yes	N	N	64K	64K	Y					2008	2011	https://www.mil.u	top level is schematic	
hack	https://github.com/jopdo		Jegor van Opdorp	accum	16	16																	system verilog			Y	asm	N	Y	32K	32K	N	18		2			2021	https://www.nand	SystemVerilog version of the course materials on hardware design	
hack	https://github.com/menny		Menny Thankachan	accum	16	16												0.67	2.0				verilog	14	cpu	Y	asm	N	Y	32K	32K	N	18		2			2020		CPU used to run Tetris	book: Elements of Computing Systems
hack	https://gitlab.com/x653/r		Michael Schroder	accum	16	16															A		verilog	24	cpu	Y	asm	N	Y	32K	32K	N	18		2			2023	https://www.nand	CPU used to run Tetris	book: Elements of Computing Systems
hack	https://github.com/theap		Peter Clarke	accum	16	16	spartan7	James	block	4803	287	6			83	##	v23.2	0.67	2.0	5.8	X		verilog	22	cpu	Y	asm	N	Y	32K	32K	N	18		2			2016	https://www.nand	CPU used to run Tetris	book: Elements of Computing Systems
hack	https://github.com/philzo		Philip Zucker	accum	16	16	spartan7	James	block	4705	287	6			82	##	v23.2	0.67	2.0	5.8	X		verilog	22	computer	Y	asm	N	Y	32K	32K	N	18		2			2021	https://www.nand	educational formally verified version of the Nand 2 Tetris course using Coq	
	https://github.com/almaz		Rafael Almazar	accum	16	16																																			
hack	https://github.com/wuha		Wu Han	accum	16	16		Wu Han		267	152	4		4		##		0.67	2.0		L		verilog	22	hack	Y	asm	N	Y	32K	32K	N	18		2			2020	https://www.nand	CPU used to run Tetris	book: Elements of Computing Systems
hamblen_scom	http://hamblen.	stable	James O. Hamblen	accum	16	16	cyclone-1	James	altera	80		4		1	204	##	q18.0	0.67	2.0	852.7	A		verilog	1	scomp			N	N	256	256	N	4					2008	http://hamblen.ed	from Hamblen 2008 "Rapid prototypir	tiny edu, high IO count

_uP_all_soft folder	opencores or prmary link	status	author	style / clone	data	inst sz	FPGA	repor ter	com ents	LUTs ALUT	Dff	LUT?	mults	blk ram	F max	date	tool ver	MIPS /inst	clks/ inst	KIPS /LUT	ven dor	SOC	src code	#src files	top file	doc	tool chai	fltg pt	Hav'd	max dat	max inst	byt ead	# inst	adr mo	# reg	pip e	start year	last revis	secondary web link	note worthy	comments
hamblen_scom	http://hamblen.	stable	James O. Hamblen	accum	16	16	cyclone-1	James	altera	196		4		1	166	##	q18.0	0.67	2.0	283.5	A		verilog	2	DE2_TOP			N	N	256	256	N	4					2008	http://hamblen.ec	from Hamblen 2008 "Rapid prototypir	tiny edu, high IO count
harvard_arch_	https://github.com/omare		omarelhedaby	RISC	32	32																	vhdl	135	harvard_pro		asm	N	Y									2021		ASIC design	many source files
hc11core	http://www.gm	stable	Green Mountain Comp	68HC11	8	8	kintex-7-3	James Brakef		2190		6			127	##	14.7	0.33	4.0	4.8	X		vhdl	1	hc11rtl	Y	yes	?	N	64K	64K	N	53		8	2	2000		6811 data sheets	restricted use license, with corrections	
hc12	https://www.ac	ncomplet	Ibrahim Hazmi	hc12	8	8																	vhdl			Y	yes	N		64K	64K	Y						2009		book/masters-thesis on his implemen	CD attached to book has full RTL
hd63701	https://opencor	planning	Tsuyoshi Hasegawa	6801	8	8	spartan-6	James Brakef		1412		6	1	3	31	##	14.7	0.33	4.0	1.8	X		verilog	6	HD63701_CORE			N	N	64K	64K	Y					2014			Used in Atari game console, 6801 clone?	
hf-risc	https://opencor	stable	Sergio Johann Filho	MIPS	32	32	kintex-7-3	James Brakef		1446		6		4	115	##	14.7	1.00	1.0	79.2	X		vhdl	9	spartan3e_n		yes	N	N	4G	4G	Y	41		32		2016		https://github.co	MIPS I subset, no multiplier	
hicovec	https://opencor	beta	Harald Manske, Gundc	RISC	32	32	kintex-7-3	James	compiler errors			6					14.7	1.00	1.0				vhdl	28	cpu	Y	asm	N				Y					2008	2010		hybrid scalar & vector processor	
hive	https://opencor	stable	Eric Wallin	stack	32	16	arria-2	James Brakef		1420		A	8	24	283	##	q13.1	1.00	1.0	199.4	ALX		verilog		hive_core	Y		N				N	40		10	8	2013	2015		4 symetrical stacks, eight threads via pipeline barrel	
hp86b	https://sites.goo	errors	Olivier De Smet	Capricorn	8	8	spartan-3	James	unresolved xilinx inter			4				##	14.7	0.33	2.0				verilog	85	cpu										64		2010		https://en.wikipe	uses PicoBlaze, emualtes HP86B	picoblaze uart uses LUT4s
hpc-16	https://opencor	beta	Umair Siddiqui	RISC	16	16	kintex-7-3	James Brakef		871		6			152	##	14.7	0.67	1.0	116.6	X		vhdl	20	cpu	Y	asm	N		64K	64K				16		2005	2015			
hvhdl	https://github.c	WIP	johonkanen																		AGLX		vhdl														2022	2025	https://hardware	kit for putting together a uP using his floating-point VHDL, ambitious project	
i8051	https://github.c	stable	Tony Givargis	8051	8	8	kintex-7-3	James	see da	2690	1325	6	1	0.5	105	##	14.7	0.33	4.0	3.2	X		vhdl	9	i8051_all	Y	yes	N		64K	64K	Y					1999	2016	https://github.co	author has book & course	Embedded System Design: A Unified Hardware/
i8051	https://github.c	stable	Tony Givargis	8051	8	8	spartan7	James	see da	1960	1339	6		0.5	48	##	v24.1	0.33	4.0	2.0	X		vhdl	9	i8051_all	Y	yes	N		64K	64K	Y					1999	2016	https://ics.uci.edu	author has book & course	Embedded System Design: A Unified Hardware/
i8080-vhdl	https://github.com/bfenn		Brendan Fennell	8080	8	8	kintex7	James Brakef		1508	308	6			124	##	14.7	0.33	8.0	3.4			vhdl	14	cpu8080_	Y	yes	N		64K	64K	Y						2018		implemented invaders game	
i8086up	https://github.com/fallah		Ali Fallah	x86	8	8	spartan3	Ali Fallah		3132	529	4	1	12	98	##	14.7	0.67	4.0	5.2	X	Y	vhdl	8	processor	Y	yes	N		512	64K	Y			7			2019		simple x86 with VGA, SD, uart	case stmt, one branch per inst, xilinx IP
ibm1410	https://github.com/cube1		Jay Jaeger	1401	6	66															X		vhdl	###		Y	yes										2021	2025		gate level version of ibm1410	also console simulator
ice_mk2	https://gitlab.cc	alpha	Mario Hoffmann	RISC	16	16		James	ECP5 primitives												L		verilog	8	top	Y		N		4K	4K	N	16		16		2020	2020	https://hackaday.io/project/174049-ice-cpu-mk-ii		variant of fpga4student
iDEA	https://github.c	alpha	Hui Yan Cheah etal	RISC	16	32	virtex-6	Liu Ch	unable	321		6	1	2	405		13.2	0.67	1.0	845.3	X		verilog	22	cpu_top	Y	yes	N	Y	64K	64K	N	24		32	9	2011	2016	The iDEA DSP Bloc	uses DSP slice in barrel mode for ALU	from GitHub, rq'd NOPs lower actual results
ignite_ptsc		asic	George Shaw	forth	32	8													1.0				proprietary					N		4G	4G						1995	2002		ShBoom clone, fast ASIC with high coc	PTSC web site had full documentation
igor	https://github.c	errors	Lykkebø	lisp			kintex-7-3	James	missing files			6				##	14.7	0.33	1.0				vhdl	25	leval												2010	2010		IGOR - A microprogrammed LISP mach	two versions, spartan3 LUT4
iitb-proc	https://github.com/preeta		Preetam Pinnada	RISC	16	16	spartan7	James Brakef		285	87	6			100	##	v25.1	0.67	3.0	78.4	X		vhdl	17	iitb_proc			N		64K	64K	N	15	3	8			2020	https://www.scrib	course project for EE224 @EE.IITB, fou	very little doc, sizeable state machine
inst_list_proce	https://opencor	planning	Mahesh Palve	accum	8	15	kintex-7-3	James	using :	786		6		1	340	##	14.7	0.33	1.0	142.6	X		verilog	34	top	Y		N		128	1K		32				2014			pipelined, state machine	UART, SPI & timer included
instant-soc	https://www.fp	stable		risc-v	32	32															AGLX		vhdl					N		4G	4G	Y			32		2020	2023	https://github.co	converts C++ into VHDL, risc-v CPU &	very large download
intel_8008	https://github.c	stable	Michael Kohn	8008	8	8															L	Y	verilog	8	i8008	Y		N		64K	64K	Y					2022	2025	https://www.mike	minor differences from 8008	ISA likes and dislikes
intel-8008-vhd	https://github.com/robert		Robert Rico	8008	8	8															L		vhdl	96	b8008_to	Y		N		64K	64K	Y						2026	https://forum.vcfe	Cycle-accurate VHDL impl of I8008 uP	three versions, b8008 latest
ion	https://opencor	mature	Jose Ruiz	MIPS	32	32	kintex-7-3	James Brakef		1533		6			163	##	14.7	1.00	1.0	106.0	AX		vhdl	12	mips_soc	Y	yes	N		4G	4G	Y			32		2011	2018	https://github.co	new version: moving to MIPS32r1	new version not ready, keeping old numbers
iop16b	https://github.c	alpha	Doug Gilliland	RISC	8	16	cyclone4	Doug Gillilan		271	76	4		2	50	##		0.33	4.0	15.2	A	Y	vhdl	51	cpu_top	Y	asm	N		4K	4K	Y	18		8		2021	2022	https://hackaday.i	I/O Processor with minimal instructio	full set of perpherals, 2022 version is huge
ippro	https://github.com/fsiddi		Fahad Siddiqui	RISC	16	32	virtex-7	Fahad Siddiqu		484	447	6	1	1	372	##		0.80	1.0	614.9	X		verilog	31			asm	N		64K	64K		30		32	5	2013	2023		16-bit RISC using DSP48	image processing, several publications
isetta	https://hackaday.io/proje		Roelh	accum	8	8																Y	schematic			Y	yes			64K	64K	Y			10		2023	2024		In TTL with 6502 & Z80 ISA via ucode	includes audio & video out
J1	www.excamera.	stable	James Bowman	forth	16	16	kintex-7-3	James Brakef		335		6		1	180	##	14.7	0.80	1.0	431.0	X		vhdl	1	j1	Y	forth	N		64K	64K		20			2	2006	2023	https://github.co	uCode inst, dual port block RAM	16 deep data & return stacks
J1	www.excamera.	stable	James Bowman	forth	16	16	**zu-2e**	James	area o	253		6		1	336	##	v20.1	0.80	1.0	1061	X		vhdl	1	j1	Y	forth	N		64K	64K		20			2	2006	2023	https://github.co	uCode inst, dual port block RAM	16 deep data & return stacks
J1a	www.excamera.	stable	James Bowman	forth	16	16	kintex-7-3	James	DFF ex	518		6			412	##	14.7	0.80	1.0	636.1	X		verilog	3	j1	Y	forth	N		64K	64K		20			2	2006	2023	https://github.co	uCode inst, dual port block RAM	DFF used for 18 deep data & return stacks
J1a32	www.excamera.	stable	James Bowman	forth	32	16	kintex-7-3	James	DFF ex	930		6			358	##	14.7	1.00	1.0	384.4	X		verilog	3	j1	Y	forth	N		64K	64K		20			2	2006	2023	**https://pythonli**	uCode inst, dual port block RAM	DFF used for 18 deep data & return stacks
J1b	www.excamera.	stable	James Bowman	forth	32	16	kintex-7-3	James	DFF ex	2612		6			302	##	14.7	1.00	1.0	115.5	X		verilog	3	j1	Y	forth	N		64K	64K		20			2	2006	2023		uCode inst, dual port block RAM	DFF used for 32 deep data & return stacks
J1b_16	www.excamera.	stable	James Bowman	forth	32	16	kintex-7-3	James	DFF ex	1588		6			355	##	14.7	1.00	1.0	223.4	X		verilog	3	j1	Y	forth	N		64K	64K		20			2	2006	2023		uCode inst, dual port block RAM	DFF used for 16 deep data & return stacks
j1sc	https://github.c	scala	Steffen Reith	forth	32	16																	scala	11	j1	Y	forth	N		64K	64K		20				2017	2020	https://steffenreit	J1 reimplemented using Scala/Spinal to generate VHDL or Verilog	
j1vh	https://github.com/flamir		Theo Hussey	forth	32	16															A		vhdl	5	j1vh	Y	forth	N		64K	64K		20					2019		VHDL clone of J1 forth CPU	altera block RAM
j68	https://code.goo	stable	Frederic Requin	68000	32	16	stratix-2	Freder	speed	1900		4		4	180			1.00	6.0	15.8	A		verilog	1	j68	Y	yes	N		4G	4G	Y			16		2009	2014		for use with Minimig	micro-coded on stack machine
j68	https://github.com/fredre		Frédéric Requin	68000	16	16	cyclone3	Frédéric Requ		1900		4		9	90			1.00	6.0	7.9			verilog	38	soc_j68		yes	N	N	64K	64K	Y			16			2018		A Size-Optimized Microcoded 68000 C	Stack based CPU with Forth-like microcode
jam	https://github.c	stable	Johan Thelin etal	RISC	32	32	kintex-7-3	James Brakef		1396		6			159	##	14.7	1.00	1.0	113.7	X		vhdl	17	cpu_sys	Y		N	Y	128K	128K				32	5	2002	2014		serial multiply & divide	took out clock divider
jam	https://github.c	stable	Johan Thelin etal	RISC	32	32	kintex-7-3	James Brakef		1369		6			143	##	14.7	1.00	1.0	104.2	X		vhdl	17	cpu	Y		N	Y	128K	128K				32	5	2002	2014		serial multiply & divide	
jane_nn		stable	Suresh Devanathan	RISC	4	8	kintex-7-3	James Brakef		723		6			178	##	14.7	0.33	1.0	81.4	X		vhdl	3	Processor	Y							27		16		2002			neural network microprocessor, specialized registers	
jca		stable	John Cronin	RISC	8	32	kintex-7-3	James	replac	3287		6	3	3	157	##	14.7	0.33	1.0	15.8	AX	Y	verilog	17	soc										16					has VGA controller, plays Pong	altera memories
jcore_aka_sh2	http://www.j-cc	difficult	Jeff Dionne. Rob Landle	SH2	32	16			need to run make per README file														vhdl	136													2014	2020	https://www.yout	https://www.youtube.com/watch?v=	Americans in Japan
j-core_pi	https://github.c	stable		SH2	32	16			missing file													Y	vhdl	45	cpu	Y	yes			4G	4G	Y			16		2014	2020	https://www.cnx-	different from jcore_aka_sh2, schematic for Spartan-6 board	
jimmy	https://github.com/kuash		Eduardo Corpeño	RISC	8	8	artix7	James Brakef		382	120				125	##	v23.2	0.33	1.0	108.0	AX		verilog	2	jimmy	Y		N	Y	256	256	Y	16		4		2020	2022		educational, 4 regs, 8-bit adr spaces	vendor neutral source code
jop	https://opencor	stable	Martin Schoeberl etal	forth	16	16	cyclone-1	Martin Schoe		2000		4			100		q10.0	0.67	1.0	33.5	A		vhdl	11	core	Y	yes	N		256K	256K						2004	2014		https://github.com/jop-devel/jop	java app builds some source code files
jpu16	https://github.c	stable	Joksan Alvarado	RISC	16	26	kintex-7-3	James	missing RAM files			6					14.7	0.67	1.0				vhdl	9	JPU16	Y	asm	N		64K	64K				16		2012			32 deep call stack, 8 addressing modes	
jtkcpu	https://github.com/JTFPG		Jose Tejada Gomez	6809	8	8	cyclone3	Jose Tejada G		2290		4	3		70	##	q18.0	0.33	3.0	3.4	A		verilog	12	jtkcpu	Y	yes	N	N	64K	64K	Y	44	13	8		2023	2024	https://forums.arc	gaming uP, compatible with Konami's	docs have ISA comparison 6809/6309/this
k1	http://mcforth.net/		Klaus Kohl-Schoepe	forth	16	16										##	v23.2	0.33	1.0		A		verilog	11	K1	Y	forth	N		64K	64K		24					2020		based on J1, Quartus project file	
k68	https://opencor	alpha	Shawn Tan	68000	16	16	kintex-7-3	James Brakef		2392		6			24	##	14.7	0.67	4.0	1.7	X		verilog	15	k68_cpu	Y	yes	N	N	4K	4G	Y			16		2003	2009		68K binary compatible	
kcp53000	https://github.com/sam-fa		Samuel Falvo II	risc-v	64	32	kintex-7-3	James	trimm	2455		6			175	##	14.7	2.00	1.0	142.9	X	B	verilog	4	polaris	Y	yes	N	Y	16E	16E	Y			32		2016	2017	https://github.co	kestrel #3, basic 64-bit RISC-V	uses state machine RTL generator
kestrel-2	kestrelcomputer	stable	Samuel Falvo II	forth	16	16	kintex-7-3	James Brakef		735		6		8	172	##	14.7	0.67	1.0	157.2	X	Y	verilog	27	M_kestrel	Y	yes	N		64K	64K		20			2	2012	2015	https://hackaday.	J1 with wishbone bus	M_j1a runs at 244MHz & 368 LUTs
kgp-risc	https://github.com/kranti		Kiran & Aluru	MIPS	32	32	spartan7	James	sparta	1428	1572	6				##	v23.2	0.33	1.0		X		verilog	25	topmodul	Y		N		4G	4G						2018	2020		only two register fields + shift amount	need to use inferred block RAM
kiwih	https://github.com/kiwih/		Hammond Pearce	accum	8	8	artix-7	James	has AS	265	173	6			100	##	v23.2	0.20	1.0	75.5	X		verilog	8	kiwih_tt_t	Y	asm	N		32	256	Y	20					2023	https://github.co	uP design via chatGPT4, ASIC gate list	study using chatGPT4 for hdware synthesis
kiwih	https://github.com/kiwih/		Hammond Pearce	accum	8	8	artix-7	James	no LU	2030	2167	6			90	##	v23.2	0.20	1.0	8.9	X		verilog	14	accumulat	Y	asm	N		256	256	Y	24					2023	https://efabless.co	uP design via chatGPT4, ASIC gate list	Scan (JTAG) chain of all memory & FF
klc32	https://opencor	planning	Robert Finch	RISC	32	32	kintex-7-3	James Brakef		3790		6	4	1	200	##	14.7	1.00	4.0	13.2	X		verilog	25	KLC32	Y	asm	N		4G	4G	Y			32		2011	2012	https://github.co	single ported block RAM register file :(	heavy use of includes
kpu	https://github.c	alpha	Andrea Corallo	RISC	32	32	kintex-7-3	James	missin	6178		6	3		19	##	14.7	1.00	1.0	3.0	X	Y	verilog	19	kpu	Y	yes	N	Y	4G	4G				32		2016	2018	http://andreacora	KPU is a minimal system on chip written used as testbench for the KPU core	
kraken16	https://people.e	stable	Bruce R. Land	RISC	18	18	kintex-7-3	James Brakef		281		6		1	278	##	14.7	0.67	1.0	662.3	X		verilog	1	DE2_TOPk	Y	asm	N	N	256	256	N	22		16			2008	https://people.ece	Cornell course material	
ks10	http://www.tec	alpha	Rob Doyle	PDP10	36	36	spartan-6	Rob Doyle		4427		6		15	50	##	14.7	1.00	2.0	5.6	X		verilog	39	esm_ks10	Y	yes	Y	N			N					2011	2014		36-bit accum & 18-bit adrs	ucf file, most tests pass
ktc32	https://github.c	stable	kinpoko	RISC	32	16	spartan7	James	sparta	27408	6554	6		44	125	##	v23.2	1.00	1.0	4.6	X	Y	system	15	top	Y	yes	N		4G	4G	Y	37		32		2022	2023		full basic ISA, hobby 32-bit CPU	see also zktc, xdc file, 16 & 32-bit insts
ladybug	https://github.com/Arlet/		Arlet Ottens	6502	8	8	spartan7	James	sparta	476	111	6		4	106	##	v23.2	0.33	4.0	18.4			verilog	2	cpu		yes	N	N	64K	64K	Y						2016	http://ladybug.xs4all.nl/arlet/fpga/6502/		
lattice6502	https://opencor	beta	Ian Chapman	6502	8	8	kintex-7-3	James Brakef		4942		6			214	##	14.7	0.33	4.0	3.6	X		vhdl	3	ghdl_proc	Y	yes	N	N	64K	64K	Y					2010	2010		targeted to LCMXO2280	
latticemico32	http://www.latt	stable	Yann Siommeau, Mich	LM32	32	32	arria_2	James Brakef		2166		A	4	30	149	##	q13.1	0.80	1.0	55.0	LX		verilog	24	lm32_cpu	Y	yes	N	Y	4G	4G	Y			32	6	2006	2017	https://en.wikipe	optional data & inst caches	Diamond3.10; see lm32 & misoc folders
latticemico32	http://www.latt	stable	Yann Siommeau, Mich	LM32	32	32	ECP3	Lattice Semic		2370		4	4	30	115			0.80	1.0	38.8	LX		verilog	24	lm32_cpu	Y	yes	N	Y	4G	4G	Y			32	6	2006	2017	https://en.wikipe	optional data & inst caches	Diamond3.10; see lm32 & misoc folders
latticemico8	http://www.latt	stable	Lattice Semiconductor	RISC	8	18	LFE2	Lattice Semic		265		4		1	104			0.33	2.0	64.4	ALX		vhdl	10	isp8_core	Y	yes	N		256	4K	Y			32		2005	2010	https://en.wikipe	16 deep call stack, four configurations	tool kit: LMS for Diamond3.10
lc-2	http://www.cs.u	mature	Eric Frohnhoefer	CISC	16	16	kintex-7-3	James	gate level primitives			6				##	14.7	0.67	2.0				vhdl	13	lc2_all	Y	yes	N		64K	64K	N	16		8		2002	2002	https://en.wikipe	from book: 978-0072467505 by Patt &	educational, compiled via Synopsys
lc-3	https://github.com/Sacusa		Sudhanshu Gupta	RISC	16	16																	vhdl			Y	asm	N		64K	64K	Y	16		8			2017	https://en.wikipe	from book: 978-0072467505 by Patt &	apndx has schematic, uses latches
legv8	https://github.com/Guilhe		Guilherme Dias	AA64	64	32																	vhdl	11	polileg	Y	yes	N		4G	4G	Y	9		32			2021		Monocycle uP in VHDL, subset of the A	course project
legv8	https://github.com/mattc		Matthew Olsson	AA64	64	32	kintex-7-3	James Brakef		884		6		2	137	##	14.7	1.00	1.0	155.0			verilog			Y	yes	N		4G	4G	Y	9		32		2018	2019		another implementation	legv8 from Patterson & Hennessy 2017
legv8	https://github.com/nxbyte		nxbyte	AA64	64	32																	verilog	6	arm_cpu	Y	yes	N				Y	9		32			2018		ARM LEGv8 CPU written in Verilog, single-cycle, pipelined & with hazard detection and	
legv8	https://github.c	stable	Seninha phillbush	AA64	64	32																	verilog	28		Y	asm	N		4G	4G	Y	10		32		2018	2019		single cycle & pipeline versions	course project
legv8	https://github.com/nextse		Warren Seto	AA64	64	32	kintex-7-3	James Brakefield				6				##	14.7	1.00	1.0			B	verilog	2	arm_cpu	Y	yes	N		4G	4G	Y	9		32		2018	2019		coursework, limited ISA, 3 versions	single cycle, inst: LDUR, STUR, ADD, SUB, ORR,
legv8	https://github.c	stable	Warren Seto	AA64	64	32	kintex-7-3	James Brakef		723	280	6		2	156	##	14.7	1.00	1.0	215.1	X	B	verilog	2	arm_cpu	Y	yes	N		4G	4G	Y	9		32		2018	2019		coursework, limited ISA, 3 versions	pipelined, inst: LDUR, STUR, ADD, SUB, ORR, AN
legv8	https://github.c	stable	Warren Seto	AA64	64	32	kintex-7-3	James Brakef		884		6		2	137	##	14.7	1.00	1.0	155.0	X	B	verilog	2	arm_cpu	Y	yes	N		4G	4G	Y	9		32		2018	2019		coursework, limited ISA, 3 versions	inst: LDUR, STUR, ADD, SUB, ORR, AND, CBZ, B,
lem1_9	https://opencor	alpha	James Brakefield	accum	1	9	kintex-7-3	James	1 stag	75		6		1	171	##	14.5	0.04	1.0	91.2	AX		vhdl	2	lem1_9	Y		N	Y	32	2K	N	24			1	2016	2017		single bit at a time, absolute adrs	
lem1_9	https://opencor	alpha	James Brakefield	accum	1	9	spartan7	James	area e	79	17	6		1	105	##	v25.1	0.04	1.0	53.2	AX		vhdl	2	lem1_9	Y		N	Y	32	2K	N	24			1	2016	2017		single bit at a time, absolute adrs	
lem1_9min	https://opencor	stable	James Brakefield	accum	1	9	kintex-7	James	1 stag	63		6		1	358	##	14.5	0.04	1.0	227.2	ALX		vhdl	3	lem1_9mi	Y	asm	N	Y	64	2K	N	8		64	1	2003	2009		logic emulation machine	
lem1_9min	https://opencor	stable	James Brakefield	accum	1	9	spartan7	James	1 stag	21	36	6		1	213	##	v25.1	0.04	1.0	405.7	ALX		vhdl	3	lem1_9mi	Y	asm	N	Y	64	2K	N	8		64	1	2003	2009		logic emulation machine	
lem1_9ptr	https://opencor	beta	James Brakefield	accum	1	9	kintex-7-3	James	1 stag	147		6		1	176	##	14.5	0.06	1.0	72.0	AX		vhdl	2	lem1_9pt	Y		N	Y	512	2K	N	24			1	2016			use speed opt, logic emulation machir	4 index registers: (ix),(--ix),(ix++),(ix+off)
lem1_9ptr	https://opencor	beta	James Brakefield	accum	1	9	spartan7	James	area e	114	17	6		1	103	##	v25.1	0.06	1.0	54.2	AX		vhdl	2	lem1_9pt	Y		N	Y	512	2K	N	24			1	2016			use speed opt, logic emulation machir	4 index registers: (ix),(--ix),(ix++),(ix+off)
lem16_18		alpha	James Brakefield	accum	16	18	kintex-7-3	James Brakef		483	38	6		1	302	##	14.5	0.16	1.0	100.0	X		vhdl	2	lem16_18m			N		256	1K		77			1	2010	2018		variable bit-length memory read/writ	op-codes coded, untested
lem16_18		alpha	James Brakefield	accum	16	18	spartan3	James Brakef		1214	297	4		1	85	##	14.5	0.16	1.0	11.2	X		vhdl	2	lem16_18m			N		256	1K		77			1	2010	2018		variable bit-length memory read/writ	op-codes coded, untested
lem4_9	https://opencor	beta	James Brakefield	accum	4	9	kintex-7-3	James	1 stag	144		6		1	195	##	14.5	0.16	1.0	216.7	AX		vhdl	2	lem1_9	Y		N	Y	32	2K	N	24			1	2016			binary & BCD digit addition, speed mode	
lem4_9	https://opencor	beta	James Brakefield	accum	4	9	spartan7	James	area e	124	12	6		1	106	##	v25.1	0.16	1.0	136.8	AX		vhdl	2	lem1_9	Y		N	Y	32	2K	N	24			1	2016			binary & BCD digit addition, speed mode	
lem4_9ptr	https://opencor	beta	James Brakefield	accum	4	9	kintex-7-3	James	1 stag	151		6		1	151	##	14.5	0.24	1.0	240.0	AX		vhdl	2	lem1_9pt	Y		N	Y	512	2K	N	24			1	2016			binary & BCD digit addition, speed mo	4 index registers: (ix),(--ix),(ix++),(ix+off)
lem4_9ptr	https://opencor	beta	James Brakefield	accum	4	9	spartan7	James	area e	172	18	6		1	87	##	v25.1	0.24	1.0	121.4	AX		vhdl	2	lem1_9pt	Y		N	Y	512	2K	N	24			1	2016			binary & BCD digit addition, speed mo	4 index registers: (ix),(--ix),(ix++),(ix+off)
lem4_9ptr	https://opencor	beta	James Brakefield	accum	4	9	xczu-1	James	area e	136	18	6		1	179	##	v25.1	0.24	1.0	315.9	AX		vhdl	2	lem1_9pt	Y		N	Y	512	2K	N	24			1	2016			binary & BCD digit addition, speed mo	4 index registers: (ix),(--ix),(ix++),(ix+off)
leon2	https://github.c	stable	Jiri Gaisler	SPARC	32	32	kintex-7-3	James Brakef		5992		6	1	12	133	##	14.7	1.00	1.0	22.3	X		vhdl	82	leon	Y	yes	Y		4G	4G	Y			64	5	1999	2003	https://en.wikipe	large config file, rad-hard asic version	https://www.gaisler.com/index.php/products/
leon3	http://www.gai	stable	Jiri Gaisler, Jan Anders	risc-v	32	32						6						1.00	1.0		AILX	Y	vhdl	100s		Y	yes	Y		4G	4G	Y			64	7	2003	2021	https://en.wikipe	RTL for LEON3, LEON5 and NOEL-V	for microchip & xilinx RAD hard parts
leon3	http://www.gai	stable	Jiri Gaisler, Jan Anders	SPARC	32	32	kintex-7-3	Jiri Gaisler		2920		6			183			1.00	1.0	62.7	AILX	Y	vhdl	100s	leon3x	Y	yes	Y		4G	4G	Y			64	7	2003	2021	https://en.wikipe	**customized for ~50 FPGA boards,**	**xls with utilization for all targets**
leros	https://opencor	stable	Martin Schoeberl	accum	16	16	spartan-6	Martin Schoe		112		6		1	182			0.67	1.0	1089	AX		vhdl	5	leros	Y	yes	N	Y	256	64K				2	2	2008	2020	https://github.co	256 word data RAM, PIC like	short LUT inst ROM
lfsr-vhdl	https://github.c	stable	Richard Howe	accum	8	16	spartan-6	Richard Howe		144	101	6		4	100	##	14.7	0.33	2.4	95.5	X	Y	vhdl	6	top	Y	yes	N		256	4K	Y	8	2				2025		Uses Linear Feedback Shift Register as	educational complexity, runs forth
lgp30	http://www.e-b	stable	Stanley Frankel	accum	32	32	spartan6	Stanle	severa	3646	1411	6	1	15	100	##	14.7	1.00	3.0	9.1		Y	vhdl	42	LGP-30	Y	yes	N		4K	4K	N			3			2017		FPGA version of LGP30 drum computer, also LGP21, RPC4000, 65F02	
light52	https://opencor	beta	Jose Ruiz	8051	8	8	kintex-7-3	James Brakef		1022		6	1	1	154	##	14.7	0.33	6.0	8.3	AX	Y	vhdl	8	light52_m	Y	yes	N	N	64K	64K	Y					2012	2018		targeted to balanced	~ 6 clocks/inst
light8080	https://opencor	stable	Jose Ruiz, Moti Litoche	8080	8	8	kintex-7-3	James Brakef		154		6		1	247		14.7	0.33	9.0	58.9	AX		verilog	5	i80soc	Y	yes	N	N	64K	64K	Y					2007	2019	https://github.co	targeted to area, includes UART, inter	older versions have both VHDL & Verilog
limen	https://github.com/domir		Dominik Salvet	RISC	16	16	spartan7	James Brakef		333	63	6		1	250	##	v23.2	0.67	1.0	503.0	X		vhdl	12	limen_sys	Y		N	Y	256	256	N	24		8		2018	2023	https://github.co	highschool thesis in Czech	limen_alpha is dual core version

_uP_all_soft folder	opencores or prmary link	status	author	style / clone	data	inst sz	FPGA	repor ter	com ents	LUTs ALUT	Dff	LUT?	mults	blk ram	F max	date	tool ver	MIPS /inst	clks/ inst	KIPS /LUT	ven dor	SOC	src code	#src files	top file	doc	tool chai	fltg pt	Hav'd	max dat	max inst	byt ead	# inst	adr mo	# reg	pip e	start year	last revis	secondary web link	note worthy	comments
lion	https://github.com/lliont/		Theodoulos Liontakis	RISC	16																A	Y	vhdl	7	lionsysten	Y	yes	N		64K	64K	Y			8		2015	2023	https://hackaday.i	custom gaming CPU, mem segments	software in C#, has BASIC
lion	https://github.com/lliont/		Theodoulos Liontakis	RISC	16																A	Y	vhdl	7	lionsysten	Y	yes	N		64K	64K	Y			8		2015	2023	http://users.sch.g	custom gaming CPU, mem segments	new directory, same RTL, Mister project
lion	https://github.com/lliont/		Theodoulos Liontakis	RISC	32																A	Y	vhdl	7	lionsysten	Y	yes	N		1M	1M	Y			8		2015	2023	http://users.sch.g	custom gaming CPU, Altera BDF files	new 32-bit version, Mister project
lipsi	https://github.c	stable	Martin Schoeberl	accum	8	8	cyclone4	Martin Schoe		162		4		1	162			0.17	1.0	167.0	A		scala	2		Y		N	N	64K	64K	Y	9	3	16		2017	2024	https://github.con	goal is 100 LUTs, program mapped to	"Lipsi, a very tiny processor"
lispmicrocontr	http://nyuzi.org	errors	Jeff Bush	lisp	32	32	kintex-7-3	James	missing init file			6				##	14.7	1.00	1.0				verilog	10	ulisp	Y		N													program.hex missing
lm32	https://github.c	mature	Sebastien Bourdeaudu	LM32	32	32	spartan7	James	include file problem			6				##	v23.2	1.00	1.0				verilog	24	lm32-top	Y	yes	N	Y	4G	4G	Y			32	6		2014		cleaned up lattice micro32, see milkymist	
Lutiac		custom	David Galloway, David	reg	16	NA	stratix-4	David Gallow		140		A	4		198			0.67	1.0	947.6	A		vhdl, verilog								64	N	64		32	3		2010	Talks at Un. Toron	synthesis maps PC into ucode	no inst mem: small state machine, ~200 inst op
lwrisc	https://opencor	stable	Li Wu	accum	8	12	arria-2	James Brakef		88		A		1	230	##	q13.1	0.17	1.0	443.6	A		verilog	9	risc_core		asm	N	Y	256	2K	Y	16				2008	2009		ClaiRISC simplified PIC, 4 reg rtn stack	absolute addressing only, lowered MIPS/clk
lxp32	https://opencor	beta	Alex Kuznetsov	RISC	32	32	kintex-7-3	James Brakef		850	526	6	3	1	196	##	14.7	1.00	2.0	115.4	AIX		vhdl	20	lxp32u_to	Y	asm	N	N	4G	4G	Y	30		256	3	2016	2024	https://lxp32.githu	register file in block RAM	vendor neutral source code
lxp32	https://opencor	beta	Alex Kuznetsov	RISC	32	32	**zu-3e**	James Brakef		948	844	6	4	1.5	250	##	v21.1	1.00	2.0	131.9	AIX		vhdl	20	lxp32u_to	Y	asm	N	N	4G	4G	Y	30		256	3	2016	2024	https://lxp32.githu	register file in block RAM	vendor neutral source code
lxp32	https://opencor	beta	Alex Kuznetsov	RISC	32	32	**zu-3e**	James Brakef		743	844	6	3	1.5	278	##	v24.1	1.00	2.0	186.9	AIX		vhdl	20	lxp32u_to	Y	asm	N	N	4G	4G	Y	30		256	3	2016	2024	https://lxp32.githu	register file in block RAM	vendor neutral source code
m1_core	https://opencor	beta	Fabrizo Fazzino, Albert	MIPS?	32	32	arria-2	James Brakef		2101		A			190	##	q13.1	1.00	1.0	90.6	AX		verilog	9	m1_core		yes	N		4G	4G	Y			32		2007	2012		GCC target?	
m16c5x	https://github.com/Morri		Michael Morris	PIC16	8	12	kintex-7-3	James	std library problems			6				##	14.7	0.33	2.0				verilog	32	m16c5x	Y	yes	N	Y	256	4K	Y					1998	2018		pipelined and non-pipelined versions	
m16c5x	https://opencor	mature	Michael Morris	PIC16	8	14	spartan-3	Michael Mor		1217		4		3	60	##		0.33	1.0	16.3	X	Y	verilog	3	m16C5x	Y	yes	N	Y	256	4K	Y					2013	2014		SOC LUT count	
m17	http://users.ece	asic	Philip Koopman	stack																			proprietary																https://users.ece.	chapter 4.3 in Koopman	6600 gate ASIC
m2cpu	https://github.com/ZakSN		Zakary Nafziger	cisc	8	8	max10	Zakary Nafzig		3504	1058	4		56	106	##	q22.1	0.33	6.0	1.7	A		vhdl	27	m2cpu_to	Y	asm	N		64K	64K	Y	75	4	7		2016	2018		micro-coded 8-bitter with 75 instructi	Quartus project files, vga output
m32632	https://opencor	stable	Udo Moeller	32032	32	8	kintex-7-3	James Brakef		10167		6	19	16	83	##	14.7	1.00	1.0	8.2	AX		verilog	18	example	Y	yes	Y	Y	4G	4G	Y	200		24	3	2009	2019	http://cpu-ns32k.net/		21.97 VAX Mips at 50MHz (Cyclone IV)
m65	www.ip-arch.jp/	stable	Naohiko Shimizu	6502	8	8	arria-2	James Brakef		483		A			110	##	q13.1	0.33	4.0	18.8	X		sfl & TI	8	m65cpu	Y	yes	N	N	4K	4K	Y					2001	2002			
m6502	https://github.c	ncomplet	Michael Kohn	6502	8	8															L		verilog	6	m6502	Y						Y					2023	2024		incomplete & obsolete, see w65c832	
m65c02	https://opencor	mature	Michael Morris	6502	8	8	spartan-6	James Brakef		466		6		3	118	##	14.7	0.33	4.0	20.8	X	Y	verilog	13	M65C02	Y	yes	N	N	64K	64K	Y					2013	2020	https://github.con	also a m65c02a version	micro-coded via F9408 soft sequencer
m65c02a	https://github.com/Morri		Michael Morris	6502	8	8	**zu-3e**	James	portmap mismatch			6				##	v21.1	0.33	4.0				verilog	61	M65C02A	Y	yes	N	N	64K	64K	Y						2021		enhanced 8/16-bit version of 65c02	PDFs on his figForth for M65C02A
magic-1	http://www.homebrewcp		Bill Buzbee	accum	8	8																	schematic			Y	yes	N		2M	2M	Y	256	5	7		2004	2014	https://hackaday.i	TTL computer, 6809ish, schematics or	magic-16 planning, 200 TTL chips
mais	https://www.do	stable	Rene Doss	MIPS	32	32	kintex-7-3	James Brakef		2760		6	4	5	245	##	14.7	1.00	1.0	88.7	X		vhdl	22	MAIS_soc	Y	yes	N	N	4G	4G				32	5	2013		use MIPS tools	register forwarding around ALU	license req'd for commercial use
mangomips32	https://github.c	stable	Ricky Tino	MIPS	32	32	spartan7	James Brakef		7923	4802	6	4		100	##	v23.2	1.00	1.0	12.6			verilog	25	mangomi	Y	yes	N		4G	4G	Y	100		32	5	2019	2023		cache support, runs linux	very percise specs, 100MHz on Artix7-2
manik	https://www.ds	stable	Sandeeo Dytta	RISC	32	32	kintex-7-3	James	needs editing to suppo			6					14.7	0.33	1.0				vhdl	45	manik2to	Y	yes	N		4K	4K	Y			16		2002	2006	www.niktech.com	optional data & inst caches	supports Xilinx, Altera, Actel, Lattice; broken we
mano_machine	https://github.c	stable	Susam Pal	accum	16	16	kintex-7-3	James	needs	364		6				##	14.7	0.67	2.0				vhdl	5	microproc	Y		N		4K	4K	N	25				2005	2016	https://en.wikiped	course project, bidir mem data	for XC9572 CPLD, large # of latches
mano-compute	https://github.com/Amin/		Amin Aliari	accum	16	16	spartan7	James	multi-	332	60	6			71	##	v23.2	0.67	1.0	144.1			vhdl	19	sayeh	Y		N		4K	4K	N	25					2020	https://en.wikiped	Mano uP implementation, course proj	different use of sayeh: *simple & yet enough*
marca	https://opencor	stable	Wolfgang Puffitsch	RISC	16	16	arria-2	James Brakef		1763		A		22	157	##	q13.1	0.67	6.0	10.0	A		vhdl	40	marca	Y		N		8K	16K		75		16	4	2007	2009		serial multiply & divide	clks/inst is approx
mark_ii	https://github.com/Vladis		Vladislav Mlejnecký	RISC	32	32			tri-sate net												A	Y	vhdl	24	mark_ii	Y	yes	Y		16M	16M	N			16		2017	2018		system on chip written in VHDL	custom PCB with MAX10
mark-1	http://www.aholme.co.uk		Andrew Holme	forth	8	8																	schematic			Y	yes	N		64K	64K	Y	22					2003		TTL forth uP	cloned by Vladislav Mlejnecký see mark_ii
mark-2	http://www.aholme.co.uk		Andrew Holme	forth	16	16																	schematic			Y	yes	N		64K	64K	Y	31					2003		TTL forth uP, PLD files	
maxicore32	https://github.c	WIP	Lawrence Manning	RISC	32	32	spartan7	James Brakef		1165	209	6	2		83	##	v23.2	0.50	1.0	35.8	LX		verilog	42	maxicore32		asm	N		4G	4G		12		16	2	2024	2025		standard risc	minimal ISA
mblite	https://opencor	beta	Tamar Kranenburg	uBlaze	32	32	kintex-7-3	James Brakef		941		6		2	227	##	14.7	1.00	1.0	240.9	AX		vhdl	18	core_wb	Y	yes	N		4G	4G	Y	86		32		2009	2017		not all instructions implemented	moved everything to work library
mb-lite_plus	http://www.late	stable	Huib Arriens	uBlaze	32	32	kintex-7-3	James Brakef		244		6		2	319	##	14.7	1.00	1.0	1308	X	B	vhdl	34	tumbl	Y	yes	N		4G	4G	Y			32		2010	2012		Delft Un. Of Tech. course work	use inferred RAM
mc1	https://github.com/mrisc3		Marcus Geelnard	RISC	32	32															A	Y	vhdl	33	top_level	Y		Y		4G	4G	Y					2020	2023	https://github.con	uP intended for FPGAs, based on MRIS	**video processor**, uses video RAM
mc6803	https://opencor	stable	Dukov	6803	8	8	spartan7	James Brakef		1618	1223	6			83	##	v23.2	0.33	3.0	5.7	X		system	2	mc6803	Y	yes	N	N	64K	64K	Y					1999			based on System68 and System01 by John E. Kent, translated CPU core from VHDL to S	
mc6809	https://github.c	stable	Greg Miller	6809	8	8	spartan6	James	errors	4510	454	6				##	v23.2	0.33	3.0		X		verilog	6	demo_roc	Y	yes	N	N	64K	64K	Y	44	13	8		2016	2020	https://shop.trenz	Cycle Accurate MC6809 Core	emphasis on cycle accuracy, DIP replacement
mc6809e		beta	Flint Weller	6809	8	8	kintex-7-3	James	gate level primitives er			6					14.7	0.33	3.0				vhdl	26	core_6809	Y	yes	N	N	64K	64K	Y	44	13	8		1999		https://www.linke	course work, ASIC orientation	
mc68kods	https://sites.goo	beta	Olivier De Smet	68000	32	16	kintex-7-3	James	errors	4617		6				##	14.7	1.00	8.0			Y	vhdl	10	mc68kods												2011			SOC for HP9816 computer emulation	
mc8051	http://www.ore	stable	Helmut Mayrhofer	8051	8	8	kintex-7-3	James Brakef		3022		6	1		83	##	14.7	0.33	4.0	2.3	X		vhdl	49	mc8051cc	Y	yes	N	N	256	64K	Y					1999	2013	www.oreganosyst	fast 8051, version available with floating-point by David Lundgren	
mcip_open	https://opencor	beta	Mezzah Jbrahim	PIC18	16	24	kintex-7-3	James Brakef		881		6	1		200	##	14.7	0.67	1.0	152.1	X		vhdl	23	MCIOopen_n		yes	N	Y	4K	1M	Y					2014	2015		light version of PIC18	
mcl51	http://www.mic	stable	Ted Fried	8051	8	8	artix-7-3	Ted Fried		312		6		2	180			0.33	8.0	23.8	X		verilog	3	mcl51_TO	Y	yes	N	N	64K	64K	Y					2016	2021	https://github.con	**micro-coded**	
mcl65	http://www.mic	stable	Ted Fried	6502	8	8	atrix-7-3	Ted Fried		252		6		2	196	##	14.7	0.33	4.0	64.2	X		verilog	1	mcl65	Y	yes	N	N	64K	64K	Y					2017	2021	https://github.con	**microcoded, cycle exact**	**excellent micro-coding LUT counts**
mcl65	http://www.mic	stable	Ted Fried	6502	8	8	kintex-7-3	James	insert	326		6		2	196	##	14.7	0.33	4.0	49.6	X		verilog	1	mcl65	Y	yes	N	N	64K	64K	Y					2017	2021	http://www.micro	**microcoded, cycle exact**	**excellent micro-coding LUT counts**
mcl86	https://github.c	stable	Ted Fried	x86	16	8	kintex-7-3	Ted Fried		308		6		4	180			0.67	20.0	19.6	X		verilog	3	EU	Y	yes	N	N	1M	1M	Y					2016	2021	http://www.embe	**microcoded, meets original 8088 timing@100MHz**	
mcpu	https://opencor	stable	Tim Boscke	accum	8	8	spartan-6	James Brakef		41	30	6			384	##	14.7	0.08	1.0	749.0	X		vhdl	1	tb02cpu2	Y	asm	N		64	64	Y	4				2007	2018	https://github.con	MCPU A minimal CPU for a CPLD	reduced MIPS/clk due to only 4 inst
mcs-4	https://opencor	alpha	Reece Pollack	4004	4	4	kintex-7-3	James Brakef		228		6			376	##	14.7	0.16	4.0	66.0	X		verilog	7	i4004			N		4K	4K	N					2012	2012		4004 was multi-chip	4004 CPU & MCS-4
mcu8	https://opencor	alpha	Dimo Pepelyashev	accum	8	8	kintex-7-3	James Brakef		274		6			299	##	14.7	0.33	1.0	360.1	X		vhdl	16	processor_E		asm			256	256	Y	17				2008	2009		asm, simulated, builds?	
mecrisp-ice	https://sourceforge.net/p		Matthias Koch	forth	16	16	spartan7	James Brakef		990	1180	6	1	4	100	##	v23.2	0.67	2.0	33.8	LX	Y	verilog	48	j1a	Y	forth	N		64K	64K	Y					2011	2023		16-bit data size, some comments in G	distinct j1a.v for each data size
mecrisp-ice	https://sourceforge.net/p		Matthias Koch	forth	32	16	spartan7	James Brakef		1976	2384	6	4	8	83	##	v23.2	1.00	2.0	21.1	LX	Y	verilog	48	j1a	Y	forth	N		4G	4G	Y					2011	2023		32-bit data size, some comments in G	distinct j1a.v for each data size
mecrisp-ice	https://sourceforge.net/p		Matthias Koch	forth	64	16	spartan7	James Brakef		6372	8860	6	16	16	63	##	v23.2	2.00	2.0	9.8	LX	Y	verilog	48	j1a	Y	forth	N		16E	16E	Y					2011	2023		64-bit data size, some comments in G	distinct j1a.v for each data size
mecrisp-quintu	https://sourceforge.net/p		Matthias Koch	risc-v	32	32	spartan7	James	added	572	89	6		1	100	##	v23.2	1.00	1.0	174.8	LX		verilog	24	FemtoRV32			N		4G	4G	Y			32		2011	2023		based on femtorv32, some comments	uls3s.v adds memory to femtorv32
mega65	https://github.com/mega6		Paul Gardner-Stephen	6502	8	8	kintex-7-3	James	bash script			6				##	14.7	0.33	2.0		X	Y	vhdl	114	machine	Y	yes	N	N	64K	64K	Y					2017	2024		Enhanced c65 running in FPGA	seeks high performance
mega65	https://github.com/mega6		Paul Gardner-Stephen	6502	8	8	spartan7	James	too m	431	296	6				##	v23.2	0.67	2.0		X	Y	vhdl	114		Y	yes	N	N	64K	64K	Y					2017	2024		Enhanced c65 running in FPGA	very large SOC with many builds & tests
mera400f	https://github.com/jakubf		jakubfi	RISC	16	16	spartan7	James	syntax errors			6				##	v23.2	0.67	2.0				verilog	77	mera400f	Y	yes	N		64K	64K	Y						2020		reimplementation of MERA-400 CPU, Polish, Mera400 was TTL uP	
mica	https://www.oocities.org/leon_heller/cpu.html			RISC	8	16																	vhdl	2				N		64K	64K	Y	10					2009		educational: ee475 Cornell	also 16-bit version
micro_nating	https://github.c	mature	Geoff Natin	RISC	16	16	spartan7	James	patched FS,SA,SB signa			6				##	v23.2	0.67	1.0				vhdl	56	processor_final			N	N	64K	64K	N	10		9		2016	2016		microcoded instruction set processor, educational	
micro16b	http://members	beta	John Kent	accum	16	16	kintex-7	James Brakef		205		6			434	##	14.7	0.33	2.0	349.0	X		vhdl	1	u16bcpu	Y	asm	N	N	64K	4K	Y	8				2002	2008	http://members.o	very limited inst set	MIPS/clk adj'd, 2 clks/inst
micro68k	https://github.c	stable	Michael Kohn	68000	32	16															L	Y	verilog	9	micro68k	Y						Y						2024	https://www.mike	"micro" 68000, unneeded instructions	ISA likes and dislikes
micro86	https://github.c	stable	Michael Kohn	x86	32	8															L	Y	verilog	7	micro86	Y				64K	64K	Y					2024	2025	https://www.mike	only implements the important instru	ISA likes and dislikes
micro8a	http://members	beta	John Kent	accum	8	16	kintex-7	James Brakef		531		6			204	##	14.7	0.33	3.0	42.3	X		vhdl	11	Micro8	Y		N	N	2K	2K	Y					2002	2002	http://members.o	derived from Tim Boscke's mcpu	also micro8 and micro8b variants
microblaze	https://www.xil	proprietar	Xilinx	uBlaze	32	32	kintex-7	Xilinx		546		6		1	320			1.03	1.0	603.7	X		proprietary			Y	yes	opt	Y	4G	4G	Y	86		32	3	2002			MicroBlaze MCS, smallest configurati	70 configuration options, MMU optional
microblaze	https://www.xil	proprietar	Xilinx	uBlaze	32	32	virtex ultr	Xilinx		563		6		1	682	##		1.03	1.0	1248	X		proprietary			Y	yes	opt	Y	4G	4G	Y	86		32	3	2002		https://en.wikiped	MicroBlaze MCS, smallest configurati	70 configuration options, MMU optional
microblaze-v	https://www.adiuvoengin		adiuvo	risc-v	32	32	spartan7	adiuvo	MCS v	1402	1046	6		4		##	v24.1	1.00	1.0		X					Y	yes	opt	Y	4G	4G	Y	86		32		2023	2024	https://docs.amd.	using their Embedded Sys Dev Board	tightly configured, fixed peripherals
microblaze-v	https://www.amd.com/er		Xilinx	risc-v	32	32	Kintex Ult	Xilinx	uconti	2201	1136	6	4		426	##	v24.1	1.00	1.0	193.5	X		proprietary			Y	yes	opt	Y	4G	4G	Y	86		32		2023	2024	https://docs.amd.	in Vivado at no extra cost	535Mhz max, numbers for 11 diff devices
microcore	http://www.pld	beta	Klaus Schleisiek	forth	12	8	kintex-7-3	James Brakef		513	75	6			336	##	14.7	0.40	1.0	261.9	X		vhdl	30	ucore110	Y	asm	N	Y	512	2K						1999	2023	www.microcore.o	indexing into return stack, auto inc/de	only one block RAM? simplest core
microcore	http://www.pld	beta	Klaus Schleisiek	forth	16	8	kintex-7-3	James Brakef		1101	138	6			171	##	14.7	0.67	1.0	104.1	X		vhdl	17	ucore120	Y	asm	N	Y	4K	4K						1999	2023		indexing into return stack, auto inc/de	no block RAM?, uses tri-state signals
microcore	https://github.c	beta	Klaus Schleisiek	forth	32	8	XP2	Klaus Schleisi		2864		4			33	##	3.12	1.00	1.0	11.5	AILX		vhdl	38	ucore	Y	asm	N	Y	3K	8K	Y	84				1999	2023		**easy to add op-codes, fltg-pt opt., sin**	**12, 16, 27 & 32 bit data sizes**
microcore	https://github.c	beta	Klaus Schleisiek	forth	16	8	XP2	Klaus Schleisi		1976		4			33	##	3.12	0.67	1.0	11.2	AILX		vhdl	38	ucore	Y	asm	N	Y	4K	8K	Y	84				1999	2023		**easy to add op-codes, fltg-pt opt., sin**	**12, 16, 27 & 32 bit data sizes**
microcore	http://www.pld	beta	Klaus Schleisiek	forth	16	8	zu5e	James	find the correct top			6				##	v22.1	1.00	1.0		X		vhdl	38	ucore	Y	asm	N	Y	4K	4K						1999	2023			
microforth	https://github.com/Forth-		Jess Totorica	forth	18	18			altera primitives							##	v23.2	0.67	1.0		A	Y	verilog	34	top	Y		N	Y	64K	64K	N	25				2019	2020	http://mindworks	Arduino-like board/platform based up	AKA F18, educational, loop stack
microwatt	https://git.open	beta	anton blanchard	PPC	64	32	artix7	James	special build instructions							##	v23.2				ALX		vhdl	95	top-arty	Y	yes			4G	4G	Y					2019	2025	https://en.wikiped	open source PPC from IBM	has vivado instructions, supports microPython
milkymist	https://github.c	stable	Sebastien Bourdeaudu	LM32	32	32	spartan-6	James	failed	13531		6	31	78	50	##	14.7	0.80	1.0	3.0	X	Y	verilog	169	system	Y	yes	N	Y	4G	4G	Y			32	6	2007	2014	https://en.wikiped	uses LM32, uses Spartan-6 IO	failed in mapper
mimafpga	https://github.c	stable	Manuel Killinger	accum	24	24	spartan7	James Brakef		275	288	6			125	##	v23.2	0.80	1.0	363.6	X		vhdl	32	mimaproc	Y		N		1M	1M		19				2019	2021		Minimal Machine processor taught at	has testbench
mimafpga	https://github.c	stable	Manuel Killinger	accum	24	24	artix7	James	IP problems			6				##	14.7	0.80	1.0		X	Y	vhdl	32	mimaenvi	Y		N		1M	1M		19				2019	2021		Minimal Machine processor taught at	has testbench
mini16_cpu	https://github.com/miya4		miya	RISC	16	16	kintexus	miya		186		6	1		710			0.67	1.0	2558	X		verilog	13	top											7		2024		Very small and high performance CPU	data width can be expanded
mini16sc_cpu	https://github.com/miya4		miya	RISC	16	16	kria260	miya				6			400	##	v24.1	0.67	1.0		X		verilog	13	top													2024		Very small and high performance CPU	single cycle variant
minicpu	http://www.cs.h	stable	Hirotsugu Nakano	stack	16	5	kintex-7-3	James	lots of	433		6	1	1	128	##	14.7	0.33	1.0	97.7	X		verilog	7	minicpu	Y	yes	N		4K	4K	N	26				2008	2018		same as tiny-cpu	uses Flex, Bison & Perl to create gcc compiler
minicpu_morri	https://github.com/Morri		Michael Morris	6502	8	8	spartan-6	Michael Mor		276		6			104			0.33	2.0	62.2	X		verilog	15	minicpu_c	Y		N		64K	64K	Y	31					2017		simplified 6502, see m65c02a	RE: 8-bit CPU challenge of Arlet Ottens
minicpu-s	https://github.c	stable	Michael Morris	stack	16	8	kintex-7-3	James Brakef		147		6			741	##	14.7	0.67	28.0	120.6	X		verilog	2	both	Y		N					33				2012	2013		separate source for each CPLD chip, u	fits (2) XC9500 CPLD @ 71.4 MHz
minimips	https://opencor	stable	Samuel Hangouet	RISC	32	32	kintex-7-3	James Brakef		2939	1886	6	8		118	##	14.7	1.00	1.0	40.1	X		vhdl	12	minimips	Y	yes	N	N	4G	4G				32	5	2004	2018		based on MIPS I	
minimips_supe	https://opencor	alpha	Miguel Cafruni	RISC	32	32	spartan7	James Brakefield				6	8			##	v23.2	1.00	0.5		X		vhdl	18	minimips	Y	asm	N	N	4G	4G				32	5	2017	2018		based on MIPS I	dual issue to two pipes, 16-bit mulitplier
minirisc	https://opencor	stable	Rudolf Usselmann	PIC16	8	14	spartan-3	Rudolf Usselr		460		4			80			0.33	1.0	57.4	X		verilog	7	risc_core_	Y	yes	N	Y	256	4K	Y					2001	2012			
minsoc	https://opencor	stable	Raul Fajardo etal	OpenRISC	32	32	kintex-7-3	James Brakef		4945		6	4	8	107	##	14.7	1.00	1.0	21.7	ALX	Y	verilog	88	or1200_tc	Y	yes	Y	M	4G	4G	Y			32		2009	2013	https://github.con	minimal OR1200, vendor neutral, has caches	
mips_16	https://opencor	stable	Doyya Doyya	RISC	16	16	kintex-7-3	James	collapsed in compile			6					14.7	1.00	1.0				verilog	12	mips_16_	Y		N		64K	64K		13		8	5	2012	2013		Educational 16-bit MIPS Processor	
mips_fault_tol	https://opencor	stable	Lazaridis Dimitris	MIPS	32	32	kintex-7-3	James Brakef		2017		6	4	6	45	##	14.7	1.00	1.0	22.5	X		vhdl	40	main	Y	yes	N		4G	4G	Y			32	5	2013	2013		arithmetic includes fault detection	no external memory port?
mips_fpga	https://github.c	WIP	Michael Kohn	mips	32	32															L	Y	verilog	6	mipscpu	Y				4G	4G	Y					2024	2025	https://www.mike	derived fromhis risc-v by changing op-	ISA likes and dislikes
mips_harris	http://booksite.	simulatio	David Harris	MIPS	32	32																	system	49	mips_sing	Y	yes	N	Y	4G	4G	Y					2014	2021	https://booksite.e	courseware to go with book	goes with text book exercises
mips_harris	http://booksite.	simulatio	David Harris	MIPS	32	32																	system	49	mips_mul	Y	yes	N	Y	4G	4G	Y					2014	2021	https://www.yout	courseware to go with book	video on Digilent Blog
mips_harris	http://booksite.	simulatio	David Harris	MIPS	32	32																	vhdl	49	mips_sing	Y	yes	N	Y	4G	4G	Y					2014	2021	https://digilent.co	courseware to go with book	complete set of book figures by chapter
mips_harris	http://booksite.	simulatio	David Harris	MIPS	32	32																	vhdl	49	mips_mul	Y	yes	N	Y	4G	4G	Y					2014	2021		courseware to go with book	
mips_linder	https://www.sc	paper	Michael Linder	MIPS	32	32	kintex-7-3	James Brakef		1100		6			238	##	14.7	1.00	1.0	216.5		B	vhdl	39	a_mips	Y	yes	N		4G	4G				32		2007	2007		masters thesis	no LUT RAM, source code in PDF
mips_pipelinec	https://github.c	mature	Mohammad Hossein Y	MIPS	32	32	spartan7	James	macro errors			6				##	v23.2	1.00	1.0				verilog	23	toplevelci	Y	yes	N		4G	4G				32	5	2017	2019		course project, hazard detection as well as forwarding, limited ISA	
mips_sc_rubio	http://www.ece.nmsu.edu		Victor P. Rubio	MIPS	32	32																	vhdl		mips_sc	Y	yes			4G	4G						2004	2004		MIPS RISC Processor for Comp Arch Ed, 2004, single cycle, RTL in PDF	
mips_up_vhdl	https://github.com/cm42		Chandra Mettu	mips	32	32	spartan7	James	no LU	1744	2311	6			250	##	v23.2	1.00	1.0	143.3			vhdl	10	NYU6463	Y	yes	N		4G	4G	Y			32			2020		simple MIPS with comparison to RC5 a	considerable mapping of memory to logic?
mips32	https://opencor	stable	Jin Jifang	MIPS	32	32	kintex-7-3	James Brakef		3696		6		8	192	##	v17.4	1.00	1.0	52.0	X		verilog	17	pipelinem	Y	yes			4G	4G	y	57		32	5	2017	2021		vivado project, ISA at github page	"classic MIPS"
mips32r1	https://opencor	stable	Grant Ayers	MIPS	32	32	arria-2	James Brakef		3716		A	8		79	##	q13.1	1.00	1.0	21.3	AX		verilog	20	processor	Y	yes	N	Y	4G	4G	Y			32	5	2012	2015	https://github.con	Harvard arch	complete software tool chain
mips789	https://opencor	stable	Li Wei	MIPS	32	32	kintex-7-3	James Brakef		1432		6		1	171	##	14.7	1.00	1.0	119.1	AX		verilog	10	mips_core	Y	yes	N		4G	4G	Y			32	5	2007	2014		supports most MIPSI instructions	
mipscpu	https://github.com/mfbso		Matheus Souza	MIPS	32	32	spartan7	James	LPM components												A		system	24	cpu	N		N		4G	4G						2017	2019		MIPS like cpu, course project, VHDL verilog & system verilog	

_uP_all_soft folder	opencores or prmary link	status	author	style / clone	data	inst sz	FPGA	reporter	comments	LUTs ALUT	Dff	LUT?	mults	blk ram	F max	date	tool ver	MIPS /inst	clks/ inst	KIPS /LUT	vendor	SOC	src code	#src files	top file	doc	tool chai	fltg pt	Hav'd	max dat	max inst	byt ead	# inst	adr mo	# reg	pip e	start year	last revis	secondary web link	note worthy	comments
mips-cpu	https://github.c	alpha	Jeremiah Mahler	MIPS	32	32	kintex-7-3	James	added	596		6		1	244	##	14.7	1.00	1.0	409.2	X		verilog	15	cpu	Y	yes	N		4G	4G	Y			32	5	2017	2017		Very early stage project, only impleme	no outputs, missing im_data.txt
mipsfpga	https://www.mi	stable	MIPS Technologies	MIPS	32	32	atrix-7-3	James Brakef		10692		6		47	118	##	14.7	1.00	1.0	11.0	X	Y	verilog	193	mfp_syste	Y	yes	N		4G	4G	Y			32		2014	2018	https://www.yout	M14K core & mipsfpga-plus	DRAM interface, I&D caches. 8789 FF
mips-hls-vivad	https://github.c	stable	Grammatopoulos Vasil	MIPS	32	32																	cpp		cpu	Y	yes	N		4G	4G	Y			32			2019		written in cpp, no inst decode, limited	**xilinx HLS project**
mips-lite	https://github.com/jncrat		Jon Craton	MIPS	32	32	kintex-7-3	James	insufficient memory			6				##	14.7	1.00	1.0				vhdl	65	cpu		asm	N							32		2009	2009			
mips-processor	https://github.com/hcshir		Henry Shires	mips	32	32															A		vhdl	52	mips_proc	Y	yes	N		4G	4G	Y			32	5		2023		course project: single cycle, pipelined	extensive simulation tests
mipsr2000	https://opencor	stable	Lazaridis Dimitris	MIPS	32	32	kintex-7-3	James Brakef		1971		6	4	6	71	##	14.7	1.00	1.0	36.2	X		vhdl	35	Dm	Y	yes	N		4G	4G	Y			32	5	2012	2016		supports almost all instructions of mip	course project
misa-o	https://github.com/Pedro		PedroMagar	risc	16	16																	system	8	misa-o_re	Y		N		64K	64K		16		8			2025	https://www.redc	4 to 16-bit MISC accumulator ISA, mixture of 4 & 16-bit operations, misc register file	
misc16	https://github.com/Steve-		Steve Teal	accum	16	16	**zu-3e**	James	Altera mem			6				##	v21.2	0.22	1.0		A	Y	vhdl	9	misc_fortl	Y	yes	N		64K	64K	N	10					2021		16-bit minimal CPU which only has a single instruction 'mov'	
misc16	https://github.com/Steve-		Steve Teal	accum	16	16	**zu-3e**	James Brakef		197	78	6			500	##	v21.2	0.22	1.0	558.4	X	B	vhdl	1	misc	Y	yes	N		64K	64K	N	10					2021	https://github.con	16-bit minimal CPU, has a single instruction 'mov' & eforth	
misoc	https://github.c	stable	M-Labs	RISC	32	32	arria_2		python source code run thru migen							##	q13.1	0.80	1.0		ALX		V*HDL			Y	yes	N		4G	4G	Y			32		2007	2019	https://m-labs.hk	Video IP for Mist & others	choice of latticemicro32 or mor1kx uP
mist1032	https://github.c	stable	Takahiro Ito	RISC	32	32	arria_2	James	altera	10801		A	4	125	98	##	q18.0	1.00	1.0	9.1			system	50	mist32e1(	Y				4G	4G	Y			64			2014		mist32 uP: embedded version	
mist1032	https://github.c	errors	Takahiro Ito	RISC	32	32	arria_2	James	altera mem			A				##	q18.0	1.00	1.0		A		verilog	87	mist1032sa					4G	4G	Y			64			2014		mist32 uP: out of order version	missing cache_ram_16entry_512bit.v
mist1032	https://github.c	errors	Takahiro Ito	RISC	32	32	cyclone-1	James	altera	33251		4	4	138	32	##	q18.0	1.00	1.0	1.0			verilog	100	mist1032isa					4G	4G	Y			64			2015		mist32 uP: inorder version	high pin count
mitecpu	https://github.com/jbush(		Jeff Bush	accum	8	11															A		verilog	2	tinyproc	Y		N	Y	256		Y	7				2017	2017		only 7 inst, also: RISC-Processsor, **ChiselGPU,** LispMicrocontroller, **PASC** & NyuziProce	
mix-fpga	https://opencor	alpha	Michael Schroeder	accum	31	31	spartan7	James	syntax errors			6				##	v23.2	1.00	1.0				verilog	29	mix	Y		Y		4K	4K	N	49	4	8			2021	https://en.wikiped	binary version of the MIX-Computer as described in "The Art of Computer Programmir	
mocha	https://github.c	stable	Sanjay Gupta	accum	8	8	spartan7	James	missin	390	329	6				##	v23.2	0.33	3.0			Y	vhdl	29	processor	Y	asm	N		64K	64K	Y	31					2018		8-bit microcontroller developed at NIIT University, course materials include full RTL &	
moncky	https://gitlab.com/big-bat		Kris Demuynck	RISC	16	16	artix-7	Kris Demuync		1376		6		33	10	##	v21	0.67	1.0	4.9	X	X	schema	36	top	Y	yes	N		64K	64K	N	32		16		2020	2021	https://hackaday.i	intended as educational, all original	IO: VGA, PS/2, SPI, SD
moncky	https://gitlab.com/big-bat		Kris Demuynck	RISC	16	16	**zu-3e**	James	no me	768	280	6			250	##	v21.1	0.67	1.0	218.1	X	X	schema	36	Moncky3	Y	yes	N		64K	64K	N	32		16		2020	2021	https://hackaday.i	bare CPU	also has verilog
moncky	https://gitlab.com/big-bat		Kris Demuynck	RISC	16	16	**zu-3e**	James	clock (	1196	523	6		33	78	##	v21.1	0.67	1.0	43.8	X	X	schema	36	top	Y	yes	N		64K	64K	N	32		16		2020	2021	https://hackaday.i	from 16x65K to 64KB RAM	two phase clock, ALU & mem have own phase
mor1kx	https://github.c	stable	Julius Baxter	OpenRISC	32	32	kintex-7-3	James Brakef		2718		6	3	3	217	##	14.7	1.00	1.0	80.0	X		verilog	48	mor1kx	Y	yes	N		4G	4G	Y			32		2012	2024	https://www.yout	lots of configuration parameters	considered best openrisc design
moxie	https://github.c	stable	Anthony Green	RISC	32	32	arria-2	James	missing module			A				##	q18.0	1.00	1.0				verilog	16	moxie					4G	4G	Y			16		2009	2017	https://github.com/atgreen/moxie-cores		four read, two write register file missing
moxielite	https://github.c	stable	Anthony Green	RISC	32	32	arria-2	James Brakef		2696		A	4		93	##	q18.0	1.00	1.0	34.6	X		vhdl	11	moxielite					4G	4G	Y			16		2009	2017	https://github.com/atgreen/moxie-cores		
moxielite	https://github.c	stable	Anthony Green	RISC	32	32	kintex-7-3	James Brakef		3159		6	3		152	##	14.7	1.00	1.0	48.0	X		vhdl	11	moxielite_wb					4G	4G	Y			16		2009	2017	https://github.com/atgreen/moxie-cores		
mpdma	https://opencor	beta	quickwayne	uBlaze	32	32	kintex-7-3	James Brakefield				6				##	14.7	1.00	1.0			Y	perl			Y	yes	N		4G	4G	Y			32		2006	2009		Soft MultiProcessor on FPGA	Perl gens *.xmp, mhs, mss & ucf files
mproz	http://www.bitl	stable	K. Lee	stack	16	16	kintex-7-3	James Brakefield				6				##	14.7	1.00	1.0				schematic			Y	asm	N			32K						1999	2007	https://groups.goo	little documentation, CPLD implement	*.1 schematics, also mproz3
mrisc32	https://github.c	alpha	Marcus Geelnard	RISC	32	32	cycloneV	Marcus Geelnard				A			100	##	q13.1	1.00	1.0		A		vhdl	36	mc1	Y	asm	Y		4G	4G	Y	68		32	9	2018	2023	https://www.bitsr	Mostly harmless Reduced Instruction	Cray-1 vector inst, also a1 variant, LLVM suppor
mrisc32	https://github.c	alpha	Marcus Geelnard	RISC	32	32															A	Y	vhdl	36	mc1	Y	asm	Y		4G	4G	Y	68		32		2018	2023	https://www.bitsr	MC1 variant web page	logic that can output a 1920×1080@60 video
mroell_cpu	https://bitbucke	stable	Matthias Roell	accum	8	8	kintex-7-3	James	added	185		6			357	##	14.7	0.33	1.0	637.1	X		vhdl	8	cpu	Y							10				2014	2016		university course project	
msl16		beta	Philip Leong, Tsang, Le	forth	16	4	kintex-7-3	James Brakef		303		6			256	##	14.7	0.67	1.0	566.4	X		vhdl	13	cpu	Y	asm	N		256			16				2001			CPLD prototype	
msp430_fpga	https://github.c	stable	Michael Kohn	msp430	16	16															L	Y	verilog	8	msp430	Y				64K	64K	Y					2024	2025	https://www.mike	derived from riscv_fpga	ISA likes and dislikes
msp430_vhdl	https://opencor	beta	Peter Szabo	MSP430	16	16	kintex-7-3	James Brakef		1735		6			127	##	14.7	0.67	2.0	24.5	AX		vhdl	9	cpu	Y	yes	N		64K	64K	Y			16		2014	2017		Comprehensive verification was not	compiles on cyclone II
multicomp	https://github.com/dougg		Doug Gilliland	accum	8	8															A																	2021	https://hackaday.i	6502, 6800, 6809 & Z80 on Cyclone II;	console available
multicomp	http://searle.hostei.com/g		Grant Searle	accum	8	8															A																	2014	https://blog.gadge	6502, 6800, 6809 & Z80 on Cyclone II;	Basic, CamelForth and CPM; also SD card, UART
multicycle_risc	https://github.c	stable	Yash Sanjay Bhalgat	RISC	16	16	kintex-7-3	James Brakef		1470		6			213	##	14.7	0.67	1.0	97.0	X		verilog	62	risc15	Y		N		64K	64K		15		8		2015	2015		multi-cycle IIT-B-RISC15 ISA	developed on Altera, course project
multi-cycle-cpu	https://github.com/Amrik		Amrik Sadhra	RISC	32	32	spartan6	James Brakef		1134	655	6			83	##	14.7	1.00	1.0	73.0	X		vhdl	48	top_level	Y				4G	4G	Y	21		32		2016	2016	https://www.yout	nicely documented with state diagram	spreadsheet for test programs, ISE project
mx65	https://github.com/Steve-		Steve Teal	6502	8	8	**k7-3**	James	512 LU	1071	195	6		0.5	207	##	14.7	0.33	4.0	15.9	X		vhdl	5	apple1	Y	yes	N		64K	64K	Y					2022	2022		cycle accurate, passes Klaus Dormann 6502 functional tests, has uart	
mx65	https://github.com/Steve-		Steve Teal	6502	8	8	**zu-3e**	James Brakef		485	148	6		1.5	370	##	v21.2	0.33	4.0	63.0	X		vhdl	5	apple1	Y	yes	N		64K	64K	Y					2022	2022		cycle accurate, passes Klaus Dormann 6502 functional tests, has uart	
mxp	http://vectorblo	stable	VectorBlox Computing	vect	8		zynq45-7	vectorblox		39856		6	64	81	175	##	v17.2	1.00	0.1	35.1			proprietary			Y											2012	2017	http://www.ece.u	MXP Matrix Processor is a scalable sof	LUT count for 8 lanes with custom inst
mycpu	http://www.my	mature	Dennis Kuschel	accum	8	8	kintex-7-3	James Brakef		3428		6	1		155	##	14.7	0.33	3.0	5.0	X		vhdl	28	cpu_top	Y		N		64M	64M	Y					2010	2023	http://mynor.org/	originally in TTL, avail. as a kit	my4th: micro-coded, bit serial, runs Forth
myforthproces	https://opencor	stable	Gerhard Hohner	forth	32	8	SP-kintex7	James Brakef		2959		6		6	223	##	14.7	1.00	1.0	75.3	X		vhdl	58	mycpu	Y	yes	N		64M	64M		96				2004	2012		DPANS'94 32-bit Forth, masters thesis	25.15 Whetstones
myfpga_forth	https://github.c	WIP	jemo07	forth	32	8			no top yet														verilog	7		Y		n		4G	4G		16				2023	2023		beginner Forth machine	
myrisc1		stable	Muza Byte	RISC	8	8	arria-2	James Brakef		121		A		2	231	##	q13.1	0.33	1.0	628.7	A		verilog	1	myRISC1	Y		N	Y	256	256	Y	16		4		2011	2011	https://en.wikiped	Verilog source included in PDF file	AKA Mano Machine, LPM macros
myrisc1	https://github.c	stable	Susam Pal	RISC	8	8												0.33	1.0		A		vhdl	5	microproc	Y		N	Y	256	256	Y	16		4		2005	2016	https://en.wikiped	one of several implementations	AKA Mano Machine, LPM macros
nanoblaze	https://opencor	beta	Francois Corthay	picoBlaze	8	18	kintex-7-3	James	punctuation			6				##	14.7	0.33	2.0		X		vhdl	12	nanoblaze		asm			256	2K	Y					2015	2015		nanoBlaze compatable, adjustable data width	
nanoblaze	https://opencor	beta	Francois Corthay	picoBlaze	8	18	kintex-7-3	James Brakef		247		6		1	169	##	14.7	0.33	2.0	113.2	X		vhdl	12	nanoblaze		asm			256	2K	Y					2015	2015		nanoBlaze compatable, adjustable data width	
nanoprocessor	https://github.com/yasan		Yasantha Niroshan	RISC	4	12	artix7	James	no LU	37	50	6			254	##	v24.1	0.10	1.0	687.2	X		vhdl	5	nanoproce	Y		N			8		4					2024		educational: 4 insts MOV, ADD, NEG & JZR	
natalius_8bit_r	https://opencor	beta	Fabio Guzman	RISC	8	16	kintex-7-3	James Brakef		232		6		1	175	##	14.7	0.11	3.0	27.7	X		verilog	12	natalius_p	Y	asm	N	Y	256	2K	Y	29		8		2012	2012		return stack & register file	3 clocks/inst
navre	https://opencor	stable	Sebastien Bourdeaudu	AVR	8	16	kintex-7-3	James Brakef		990		6			207	##	14.7	0.33	1.0	69.0	ALX		verilog	1	softusb_n	Y	yes	N		64K	64K	Y	72		32	2	2010	2013	https://www.milk	AVR clone, part of www.milkymist.org	
nc4016	https://en.wikic	asic	Chuck Moore	forth	16																		proprietary																	chapter in Koopman	
ncore	https://opencor	alpha	Stefan Istvan	accum	16	8	kintex-7-3	James Brakef		223		6			105	##	14.7	0.67	1.0	316.3	X		verilog	3	nCore	Y		N		128K	64K		16		16		2006	2018		This is a little-little processor core	
neo430	https://opencor	alpha	Stephan Nolting	MSP430	16	16	artiix-7	James	chang	947	659	6		2.5	215	##	14.7	0.67	4.0	38.0	ALX	Y	vhdl	19	neo430_t	Y	yes	N		28K	32K	Y			16		2015	2024	https://github.con	edit neo430_sysconfig.vhd to set options	
neo430	https://opencor	alpha	Stephan Nolting	MSP430	16	16	artiix-7	Stephan Nolt		1036	1144	6		2.5	100	##	v19.2	0.67	4.0	16.2	ALX	Y	vhdl	19	neo430_t	Y	yes	N		28K	32K	Y			16		2015	2024	https://github.con	edit neo430_sysconfig.vhd to set optio	default config, includes true RNG
neo430	https://opencor	alpha	Stephan Nolting	MSP430	16	16	artiix-7	Stephan Nolt		576	266	6		1	100	##	v19.2	0.67	4.0	29.1	ALX	Y	vhdl	19	neo430_t	Y	yes	N		28K	32K	Y			16		2015	2024	https://github.con	edit neo430_sysconfig.vhd to set optio	minimal configuration
neo430	https://opencor	alpha	Stephan Nolting	MSP430	16	16	cyclone4	Stephan Nolt		1869	1137	4		8	121	##	q17.1	0.67	4.0	10.8	ALX	Y	vhdl	19	neo430_t	Y	yes	N		28K	32K	Y			16		2015	2024	https://github.con	website has detailed resource unti	default config, includes true RNG
neo430	https://opencor	alpha	Stephan Nolting	MSP430	16	16	cyclone4	Stephan Nolt		590	230	4		6	122	##	q17.1	0.67	4.0	34.6	ALX	Y	vhdl	19	neo430_t	Y	yes	N		28K	32K	Y			16		2015	2024	https://github.con	**website has detailed resource unt**	minimal configuration
neo430	https://opencor	alpha	Stephan Nolting	MSP430	16	16	ice40	Stephan Nolt		3928	1923	4		6	20	##	LR	0.67	4.0	0.9	ALX	Y	vhdl	19	neo430_t	Y	yes	N		28K	32K	Y			16		2015	2024	https://github.con	website has detailed resource unti	default config, includes true RNG
neo430	https://opencor	alpha	Stephan Nolting	MSP430	16	16	ice40	Stephan Nolt		1812	755	4		6	20	##	LR	0.67	4.0	1.9	ALX	Y	vhdl	19	neo430_t	Y	yes	N		28K	32K	Y			16		2015	2024	https://github.con	website has detailed resource unti	minimal configuration
neo430	https://opencor	alpha	Stephan Nolting	MSP430	16	16	virtex-6	Stephan Nolt		402		6		2	204	##	14.7	0.67	4.0	85.0	ALX	Y	vhdl	19	neo430_t	Y	yes	N		28K	32K	Y			16		2015	2021	https://github.con	website has detailed resource untiliza	minimal configuration
neogeo	https://github.com/Mazar		Murray Aickin	68000, z80	16	16															A	Y	verilog															2023	https://en.wikiped	port of Neogeo Core (video arcade	CycloneV, open hardware, retro gaming
next186	https://opencor	stable	Nicolae Dumitrache	x86	16	8	arria-2	James Brakef		1966		A	2		77	##	q13.1	0.67	2.0	13.1	AX		verilog	4	Next186_(	Y	yes	N	N	1M	1M	Y					2012	2013		boots DOS	
next186_soc_p	https://opencor	stable	Nicolae Dumitrache	x86	16	8	kintex-7-3	James	translate errors			6	1			##	14.7	0.67	2.0			Y	verilog	40	ddr_186	Y	yes	N	N	1M	1M	Y					2013	2019		SoC version of next186	boots DOS, does video games & sound
next186mp3	https://opencor	stable	Nicolae Dumitrache	x86	16	8	kintex-7-3					6	1			##	14.7	0.67	2.0			Y	verilog	16	ddr_186	Y	yes	N	N	1M	1M	Y					2013	2014		SoC version of next186	boots DOS, has DSP core, no x86 source
nextz80	https://opencor	stable	Nicolae Dumitrache	Z80	8	8	kintex-7-3	James Brakef		854		6			119	##	14.7	0.33	1.0	46.0	X	B	verilog	3	NextZ80C	Y	yes	N	N	64K	64K	Y					2011	2019			claim of 700 LUTs in Spartan-3 probably wrong
nibblercpu	https://github.com/bchan		Bryan Chan	accum	4	8	spartan7	James	combi	9066	84	6				##	v23.2	0.16	1.0				system	24	nibbler	Y		N	Y	4K	4K							2017	http://www.raysl	originally a TTL project	https://www.bigmessowires.com/nibbler/
nibblercpu	https://gist.github.com/er		erin candescent	accum	4	8	spartan7	James Brakef		39	27	6			200	##	v23.2	0.16	2.0	410.3			vhdl	1	nibblercpu	Y		N	Y	4K	4K							2014	https://www.bigm	4-bit CPU in VHDL	seondary web link has documentation
nige_machine	https://github.c	stable	Andrew Read	forth	32	8	kintex-7-3	James Brakef		5033		6	8	33	123	##	14.7	1.00	1.0	24.5	X		vhdl	29	Board	Y	yes	N		16M	16M		512		512			2014		standalone Forth system	https://www.youtube.com/watch?v=PRltE8q62
niloofar1	http://ce.sharif.	errors	Mahdi Amiri	RISC	16	16	kintex-7-3	James	ran out of memory			6				##	14.7	0.67	1.0				verilog	3	nf1	Y												2001		derived from risc-16	ASIC, uses Leonardo for synthesis
nios2		proprietar	Altera	Nios II	32	32	stratix-3	Altera	consis	1020		A			290	##	q13.1	0.90	1.0	255.9	A		proprietary			Y	yes	opt		4G	4G	Y			32		2004			fltg-pt, caches & MMU options	Nios II/f: fastest version, DMIPS adj, 2.15 CoreM
nios2		proprietar	Altera	Nios II	32	32	stratix-5	Altera	consis	584		A			420	##	q16.0	1.00	1.0	719.2	A		proprietary			Y	yes	opt		4G	4G	Y			32		2004			fltg-pt, caches & MMU options	Nios II/e: min LUTs version, DMIPS adj, 1.68 Cor
niosprocessor	https://github.com/Julien		Julien Malka	Nios II	32	32															A		vhdl	25	cpu	Y	yes	N		4G	4G	Y			32		2019	2019		Project for Computer Architecture cou	uses much Altera source code
nios-v	https://www.intel.com/cc		Intel	risc-v	32	32	Agilex 7	Altera	compa	421		A			442	##	q24.2	1.00	1.0	1050	A		proprietary			Y	yes	opt		4G	4G	Y			32		2021	2024	https://www.intel	requires Quartus Prime Pro? Ashling F	ALM mystery: off by 2X? No FF counts
nnarm	ftp://ftp.gwdg.d	untested	Sheng Shen	ARM	32	16															A		verilog	40	syn	Y	yes	N		4G	4G	Y			16			2001		mentioned at https://en.wikipedia.org/wiki/Amber_(processor_core), ran afoul of ARI	
nocpu	https://github.c	beta	John Tzonevrakis	RISC	8	8	kintex-7-3	James Brakef		175		6			243	##	14.7	0.33	1.5	306.1	X		verilog	5	cpu	N	no	N		256	256	Y			4			2017		minimal & complete	8 ALU inst, 3 port reg file
non-von-1	https://www.ch	stable	Christopher Fenton	accum	8	8	kintex-7-3	James Brakef		230	89	6			556	##	14.7	0.33	1.0	797.1			verilog	1	nonvontop		no	N		64		Y	30					2018		SIMID in tree structure	A & B regs, instructions broadcast
nop	https://github.com/NOP-F		Mingdao Liu	RISC	32	32																Y	scala	83	main	Y	yes	Y		4G	4G	Y			32			2023	https://en.wikiped	full size uP, doc in Chinese	see wikipedia link
nova1bach	https://github.com/jadels		Jan Adelsbach	nova	16	16	spartan7	James	multiply driven nets			6				##	v23.2	0.67	1.0			Y	verilog	10	nova_cpu	Y	yes	N		64K	64K				7			2016		implementation of a DataGeneral NOVA in Verilog	
nova-soc	https://github.com/scottll		Scott Baker	nova	16	16	**zu-3e**	James	no mem init file			6				##	v21.2	0.67	2.0			Y	vhdl	14	soc	Y	yes	N		64K	64K				7		2016	2020		Nova CPU + RAM + UART + Timer + I/O Ports, Sierra Circuit Dsgn, missing hex file	
npower	https://github.c	WIP	Robert Finch	ppc	32	32	artix-7	James	cntlz32.sv missing												X	Y	system	21	rfpower	Y	yes			4G	4G	Y			32		2020	2024		powerPC subset, "can run most comm	has artix project file
nybbleForth	https://github.c	errors	Lars Brinkhoff	forth	16	4	kintex-7-3	James	missing init file			6				##	14.7	0.67	1.0				verilog	1	cpu	Y	yes			4K	4K	Y	11				2017	2017		empty design, no init file	tiny
nyuzi_gpu	https://github.c	stable	Jeff Bush	GPGPU	32	32	arria-2	James	syntax errors			A				##	q18.0	1.00	1.0		AX	Y	system	70	nyuzi	Y	yes	Y		4G	4G	Y	80		64		2015	2024	https://github.con	32 scalar & 32 vector reg	should run on either altera or xilinx
nyuzi_gpu	https://github.c	stable	Jeff Bush	GPGPU	32	32	cyclone-4	Jeff Bush		74000		4			54		q18.0	16.00	1.0	11.7	AX	Y	system	70	nyuzi	Y	yes	Y		4G	4G	Y	80		64		2015	2024	https://github.con	32 scalar & 32 vector reg	
nyuzi_gpu	https://github.c	stable	Jeff Bush	GPGPU	32	32	artix7	James	missin	82767	#####	6	64	17	50	##	v23.2	1.00	1.0	0.6	AX	Y	system	70	nyuzi	Y	yes	Y		4G	4G	Y	80		64		2015	2024	https://github.con	32 scalar & 32 vector reg	should run on either altera or xilinx
oberon_sdram	http://projectob	beta	Nicolae Dumitrache	RISC	32	32	kintex-7-3	James Brakef		2103		6		1	104	##	14.7	1.00	1.0	49.5	X		verilog	16	risc5	Y	yes	Y		4G	4G				16		2013	2017		minimalist Wirth, part of Project Ober	modified to use DRAM, serial mult
oc54x	https://opencor	beta	Richard Herveille	DSP	16	16	kintex-7-3	James Brakef		2225		6	1		180	##	14.7	0.67	1.0	54.1	X		verilog	10	oc54_cpu	Y	yes	N	Y	64K	64K						2002	2009		40-bit accumulator, barrel shifter	C54x clone
octagon	https://opencor	beta	Jon Pry	MIPS	32	32	kintex-7-3	James Brakef		3021		6	4	9	333	##	14.7	1.00	1.0	110.2	X		vhdl	46	octagon		asm			4G	4G	Y			32		2015	2015	https://github.con	8 thread barrel processor, largely MIPS compatible	
octavo	http://fpgacpu.c	beta	Charles LaForest	reg	16	16	stratix-4	Charles LaFor		500		A	1		550			0.67	1.0	737.0	A		verilog	18	Octavo	Y	asm	N					14		16	10	2012	2019	https://github.con	8 core barrel, adjustable data width	~= performance across word sizes, no call/rtn ir
odess	https://opencor	stable	Dmytro Senyakin	RISC	##	16	cyclone-5	James	too bi	130160		A	##	462		##	q18.0	4.00	0.3		A		system	27	CoreQuad	Y	asm	Y		4G	4G				16		2017	2017	https://opencores	Altera proj, Multicore, P&R results at	37-bit adr, quad issue, caches, 32-64-128 fltg-p
odess	https://opencor	stable	Dmytro Senyakin	RISC	##	16	cyclone-5	James	reduce	35984		A	72	112	103	##	q18.0	4.00	1.0	11.4	A		system	27	CoreOneV	Y	asm	Y		4G	4G				16		2017	2017	https://opencores	Altera proj, Multicore, P&R results at	37-bit adr, quad issue, caches, 32-64-128 fltg-p
odess	https://opencor	stable	Dmytro Senyakin	RISC	##	16	cyclone-5	James	slow t	50135		A	72	112	90	##	q18.0	4.00	1.0	7.2	A		system	27	CoreOneV	Y	asm	Y		4G	4G				16		2017	2017	https://opencores	Altera proj, Multicore, P&R results at	37-bit adr, quad issue, caches, 32-64-128 fltg-p
odess	https://opencor	stable	Dmytro Senyakin	RISC	##	16	stratix-5	Dmytro Senya		32978		A	72	112	192	##	q17.1	4.00	1.0	23.3	A		system	27	CoreOneV	Y	asm	Y		4G	4G				16		2017	2017	https://opencores	Altera proj, Multicore, P&R results at	37-bit adr, quad issue, caches, 32-64-128 fltg-p
odess	https://opencor	stable	Dmytro Senyakin	RISC	##	16	stratix-5	Dmytro Senya		148078		A	72	122	184	##	q17.1	4.00	0.3	19.9	A		system	27	CoreQuad	Y	asm	Y		4G	4G				16		2017	2017	https://opencores	Altera proj, Multicore, P&R results at	37-bit adr, quad issue, caches, 32-64-128 fltg-p
odess	https://opencor	stable	Dmytro Senyakin	RISC	##	16	stratix-5	Dmytro Senya		50814		A	72	112	180	##	q17.1	4.00	1.0	14.1	A		system	27	CoreOneV	Y	asm	Y		4G	4G				16		2017	2017	https://opencores	Altera proj, Multicore, P&R results at	37-bit adr, quad issue, caches, 32-64-128 fltg-p
oldland-cpu	http://jamieiles.	errors	Jamie Iles	RISC	32	32	arria-2	James	syntax errors			A				##	q18.0	1.00	1.0		A		verilog	22	oldland_c	Y		N	N	4G	4G	Y			16	5	2015	2017	https://github.con	has caches & MMU	runs on Cyclone V
oldland-cpu	http://jamieiles.	errors	Jamie Iles	RISC	32	32	arria-2	James	syntax errors			A				##	q18.0	1.00	1.0		A	Y	verilog	32	keynsham	Y		N	N	4G	4G	Y			16	5	2015	2017	https://github.con	has caches & MMU	runs on Cyclone V
oms8051mini	https://opencor	alpha	Simon Teran, Dinesh A	8051	8	8	kintex-7-3	James Brakef		1991		6	1	32	133	##	14.7	0.33	5.0	4.4	X	Y	verilog	66	digital_co	Y	yes	N		64K	64K	Y					2000	2018			
one-der	http://www.drdobbs.com/		Al Williams	CISC	32		spartan3	James	missimg file			4				##	14.7	1.00	1.0				verilog	23	topbox												2009	2009		The One Instruction Wonder	TTA
opa	https://github.c	stable	Wesley W. Terpstra	RISC	32	32	cyclone-5	Wesle	larges	8540		A			125		q15.0	1.00	0.5	29.3	A		vhdl												32		2013	2016		An Out-of-Order Superscalar Soft CPU	tested, incomplete
opc.opc2cpu	https://github.c	stable	revaldinho	accum	8	16	kintex-7-3	James	reduce	117		6			556	##	14.7	0.15	4.0	178.1	X		verilog	2	opc2cpu	Y	asm	N	N	256	1K	Y	12	3			2017	2021	https://revaldinho	OPC2 revised OPC1, for XC9572 CPLD	see hackaday One Page Computing Challenge

_uP_all_soft folder	opencores or prmary link	status	author	style / clone	data	inst sz	FPGA	repor ter	com ents	LUTs ALUT	Dff	LUT?	mults	blk ram	F max	date	tool ver	MIPS /inst	clks/ inst	KIPS /LUT	ven dor	SOC	src code	#src files	top file	doc	tool chai	fltg pt	Hav'd	max dat	max inst	byt ead	# inst	adr mo	# reg	pip e	start year	last revis	secondary web link	note worthy	comments
opc.opc3cpu	https://github.c	stable	revaldinho	accum	16	16	kintex-7-3	James	reduce	174		6			526	##	14.7	0.30	4.0	226.9	X		verilog	2	opc3cpu	Y	asm	N	N	64K	64K	N	13	3			2017	2021	https://revaldinh	OPC3 16-bit OPC1, for XC95144 CPLD	see hackaday One Page Computing Challenge
opc.opc5cpu	https://github.c	stable	revaldinho	RISC	16	16	kintex-7-3	James	reduce	273		6			294	##	14.7	0.40	3.0	143.6	X		verilog	7	opc5cpu	Y	asm	N	N	64K	64K	N	15	4	16		2017	2021	https://revaldinh	OPC5 RR inst, ISA similar to OPC1	see hackaday One Page Computing Challenge
opc.opc5lscpu	https://github.c	stable	revaldinho	RISC	16	16	kintex-7-3	James	Brakef	383		6			247	##	14.7	0.67	3.0	144.0	X		verilog	2	opc5lscpu	Y	asm	N	N	64K	64K	N	18	4	16		2017	2021	https://revaldinh	OPC5LS OPC5 with predicate inst	see hackaday One Page Computing Challenge
opc.opc6cpu	https://github.c	stable	revaldinho	RISC	16	16	kintex-7-3	James	Brakef	450		6			222	##	14.7	0.67	2.0	165.4	X		verilog	2	opc6cpu	Y	asm	N	N	64K	64K	N	27	4	16		2017	2021	https://revaldinh	OPC6 based on OPC5LS, more inst	see hackaday One Page Computing Challenge
opc.opc7cpu	https://github.c	stable	revaldinho	RISC	32	16	kintex-7-3	James	Brakef	624		6			303	##	14.7	1.00	2.0	242.8	X		verilog	2	opc7cpu	Y	asm	N	N	1M	1M	N	32	5	16		2017	2021	https://revaldinh	OPC7 32bit, based on OPC5LS, more i	see hackaday One Page Computing Challenge
opc.opc8cpu	https://github.c	beta	revaldinho	RISC	24	24	kintex-7-3	James	no tes	516		6			323	##	14.7	0.80	2.0	250.1	X		verilog	1	opc8cpu	Y	asm	N	N	16M	16M	N	32	4	16		2017	2021	https://revaldinh	OPC8 24bit, based on OPC5LS, more i	see hackaday One Page Computing Challenge
opc.opccpu	https://github.c	stable	revaldinho	accum	8	16	kintex-7-3	James	reduce	101		6			526	##	14.7	0.15	4.0	195.4	X		verilog	2	opccpu	Y	asm	N	N	256	2K	Y	13	3			2017	2021	https://revaldinh	OPC1 one page computer for CPLD	**see hackaday One Page Computing Challe**
opc_spinal	https://justanotherelectronicsblog.com/?p=543			accum	8	16																	scala	1	opc			N		1K	8K	Y	16				2019	2019		just the single web page	
open8_urisc	https://opencor	stable	Kirk Hays, Jshamlet	RISC	8	8	kintex-7-3	James	Brakef	691		6	1		263	##	14.7	0.33	1.0	125.6	X		vhdl	9	Open8	Y	yes	N		64K	64K	Y			8		2006	2023		accum & 8 regs, clone of Vautomation uRISC processor, in use	
openc	https://github.com/T-hea		T-Head Semiconductor	risc-v	32	32																	verilog			Y	yes	N		4G	4G	Y			32			2021	https://www.cnx-	Alibaba ASIC RISC-V uP: e902-e906-c906-and-c910, docs in Chinese, many many large	
openfire_core	https://opencor	alpha	Alex Marschner, Steph	uBlaze	32	32	kintex-7-3	James	empty project file			6					14.7	0.33	1.0				verilog	12	openfire_	Y	yes	N	N	4G	4G	Y			32		2007	2009		OpenFire Processor Core	"FPGA Proven"
openfire2	https://opencor	beta	Antonio Anton	uBlaze	32	32	kintex-7-3	James	Brakef	1201		6	3	2	105	##	14.7	1.00	1.0	87.4	X	Y	verilog	27	openfire_	Y	yes	N	N	4G	4G	Y			32		2007	2012		"FPGA Proven"	derived from Stephen Craven's OpenFire
opengateware	https://github.com/openg		opengateware	z80	8	8															A	Y	vhdl, verilog			Y	yes	N		64K	64K	Y						2022	https://github.con	compatible Congo Bongo/Tip Top arca	several others at opengateware
openmsp430	https://opencor	stable	Oliver Girard	MSP430	16	16	stratix-3-	Oliver Girard		1147		A	1		98			0.67	2.0	28.5	AX		verilog	30	openMSP4	Y	yes	N	N	64K	64K	Y			16		2009	2018		near cycle accurate	performance spreadsheet
openpiton	https://github.c	difficult	mmckeown	SPARC	32	32	kintex-7-3	James	too many files			6				##	14.7	1.00	1.0				verilog			Y	yes	Y	N	4G	4G	Y			64		2015	2019	http://parallel.pri	Princeton Un.	both FPGA & ASIC, very many source files
openscale	http://www.lirn	stable	Lyonel Barthe	uBlaze	32	32	spartan-3	Lyonel Barthe		1563		4			91		i12.1	1.00	1.0	58.2	X	Y	vhdl	26	sb_core		yes			4G	4G	Y	86		32	5	2010	2012	www.lirmm.fr/AD	NoC secretblaze	data is for single secretblaze
openxlr8	https://github.com/Aloriu		alorium technology	AVR	8	16															A	Y	verilog	24	xlr8_alori	Y	yes	N		64K	64K	Y			32			2019	https://www.alori	AVR clone, Snō and Hinj Arduino comp	https://www.youtube.com/watch?v=Drr1M9z1
or1200	https://github.c	stable	Damjan Lampret	OpenRISC	32	32	kintex-7-3	James	Brakef	5231		6	4	8	118	##	14.7	1.00	1.0	22.5	X		verilog	78	or1200_tc	Y	yes	Y	M	4G	4G	Y			32		2010	2015	https://openrisc.i	best older openrisc implementation	no LUT RAM for reg file
or1200_hp	https://opencor	stable	Strauch Tobias	OpenRISC	32	32	virtex-5	Strauc	3 slot	5602		6			185	##		1.00	1.0	33.1	X		verilog	39	or1200_ic	Y	yes	Y	M	4G	4G	Y			32		2010	2013	https://openrisc.i	3 slot barrel version of OR1200	numbers from published paper
or1200_soc	https://opencor	beta	gaz	OpenRISC	32	32	cyclone-2	James	missing files			4				##	q11.1	0.67	2.0			Y	verilog	39	top	Y	yes	Y	M	4G	4G	Y			32			2011	https://openrisc.i	OpenRISC on Terasic DE1 board	
or1200mp	https://github.c	stable	Stefan Wallentowitz	OpenRISC	32	32	kintex-7-3	James	Brakef	4960		6	4	8	111	##	14.7	1.00	1.0	22.4	X		verilog	104	or1200_tc	Y	yes	Y	M	4G	4G	Y			32		2012	2012	https://openrisc.i	multiprocessor variant, single core	
or1k	https://opencor	stable	Julius Baxter, Stefan Kr	OpenRISC	32	32	kintex-7-3	James	Brakef	3299		6	3	3	189	##	14.7	1.00	1.0	57.3	AX		verilog	39	mor1kx	Y	yes	N	M	4G	4G	Y			32		2001	2018	https://opencores	no longer supported, see mor1kx	cappuccino ALU
or1k_marocchi	https://github.c	stable	Andrey Bacherov	RISC	32	32	spartan7	James Brakefield				6				##	v23.2	0.67	1.0				verilog	39	or1k_mar	Y	yes	Y		4G	4G	Y			32		2012	2022	https://github.con	continous regression tests	Implements a variant of Tomasulo algorithm
or1k_soc	https://opencor	mature	Xianfeng Zeng	OpenRISC	32	32	arria-2	James	syntax errors			6				##	q18.0	1.00	1.0		A	Y	verilog	194	or1k_soc_	Y	yes			4G	4G	Y			32		2009	2010	https://openrisc.i	SoC using OpenRISC 1200	huge tar file
or1k-cf	https://opencor	alpha	Kenr	OpenRISC	32	32																	confluence														2004	2009			
osu8	https://www.pj	alpha	Paul Stoffregen	accum	8	8																	schematic			Y	asm	N	N	64K	64K	Y	24				1994	2005	https://github.con	OSU8 Microprocessor Project "instruc	*.1 schematics, doc at web page, currently acti
p16	http://www.ultratechnolc		Don Golding	forth	16	5	kintex-7-3	James	bad syntax			6					14.7	0.67	1.0				vhdl	1	p16			N		64K	64K						2000		http://ftp.forth.org/svfig/kk/11-2021-Golding.pdf		
p16b		beta	C.H. Ting	forth	16	5	kintex-7-3	James	case c	367		6			355	##	14.7	0.67	1.0	648.1	X		vhdl	1	cpu16	Y	asm	N		64K	64K		28				2000			part of eForth?	data width can be expanded
p16c5x	https://opencor	mature	Michael Morris	PIC16	8	14	kintex-7-3	James	Brakef	378		6			252	##	14.7	0.33	1.0	220.2	AX		verilog	3	P16C5x	Y	yes	N	Y	256	4K	Y					2013	2014			
p24e		beta	C.H. Ting	forth	24	6	spartan_3	James	Brakef	1175		4		16	51	##	14.7	0.83	1.0	36.0	X		vhdl	1	p24c	Y	asm	N		2K	2K		28				2000			part of eForth?	data width can be expanded
pacoBlaze	www.bleyer.org	mature	Pablo Kocik	picoBlaze	8	18	spartan-3	Pablo Kocik		177		4		1	117			0.33	2.0	109.1	X		verilog	18	pacoblaze	Y	asm	N		256	2K	Y	57			2		2006		3 versions, behavioral coding	
pancake	https://people.e	stable	Bruce Land	stack	16	5	kintex-7-3	James	bypas	441		6	1	1	128	##	14.7	0.67	1.0	194.8	X		verilog	7	de2_minic	Y	yes	N		4K	4K		31				2010	2014	http://www.cs.hir	The Pancake Stack Machine dervied fr	Cornell ECE5760
parwan		stable	Zainalabedin Navabi	accum	8	8	kintex-7-3	James	Brakef	157		6			435	##	14.7	0.33	4.0	228.5	X		verilog	16	par_beh	Y	yes	N	N	4K	4K	Y					1995	1997	2nd uP in director	from VHDL: Analysis and Modeling of	AKA cpu8, both vhdl & verilog versions
parwan		stable	Zainalabedin Navabi	accum	8	8	kintex-7-3	James	Brakef	161		6			76	##	14.7	0.33	4.0	38.8	X		vhdl	2	parwan	Y	yes	N	N	4K	4K	Y					1995	1997	2nd uP in director	from VHDL: Analysis and Modeling of	AKA cpu8, both vhdl & verilog versions
pasc	https://github.com/jbush(		Jeff Bush	RISC	16	16															AX		verilog			Y		N		64K	64K	N	20	2	8		2017	2019	https://github.con	16 RISC cores	quartus project
patmos	https://github.c	stable	Martin Schoeberl	RISC	32	32																	scala														2015	2023	http://patmos.cor	university project, ASIC tapeout	http://www.t-crest.org/
pauloblaze	https://github.c	mature	Paul Genssler	picoBlaze	8	18	kintex-7-3	James	Brakef	314	80	6			118	##	v25,1	0.33	1.0	124.0	X		vhdl	7	pauloBlaz	Y	asm	N		256	2K	Y					2015	2021		LUT6 req'd, course project, slower more LUTs than original claims easier to modify an	
pavr	https://opencor	alpha	Doru Cuturela	AVR	8	16	kintex-7-3	James	Brakef	2630		6		1	132	##	14.7	0.33	1.0	16.5	X		vhdl	18	pavr_cont	Y	yes	N	Y	4K	4M	Y	72		32	6	2003	2009		superset of AVR	
pcycle	https://github.com/domir		Dominik Salvet	accum	4	8	spartan7	James	altera primitives			6				##	v25,1	0.33	1.0		A		vhdl	5	pcycle	Y		N	Y	16	128		12				2015	2021		inspired by redstone processor in Minecraft, 1st custom VHDL design by author	
pdp1	https://opencor	alpha	Yann Vernier	PDP1	18	18	spartan-3	James	Brakef	1390		4		6	138	##	14.7	0.50	10.0	5.0	X		vhdl	15	top	Y	yes	N	N	4K	4K	Y	28				2011	2017	http://pdp-1.comp	PDP-1 descended from MIT TX-0	uses Minimal UART from opencores
pdp11_fpga	https://github.c	WIP	Michael Kohn	pdp11	16	16															L	Y	verilog	6	pdp11	Y				64K	64K	Y					2024	2025	https://www.mike	~same performance as msp430_fpga	ISA likes and dislikes
pdp11_reduce	https://github.c	simulatio	Mohamed Omran	PDP11	16	16	spartan7	James	no outputs			6				##	v25,1	0.67	2.0		X		vhdl	9	system	Y	yes	N	N	64K	64K		24	10	8			2021		simplified pdp11, 24 inst	no byte data size, ucode, 2-12 clocks/inst
pdp11-34verilc	www.heeltoe.cc	stable	Brad Parker	PDP11	16	16	arria-2	James	Brakef	2532		A			126	##	q13.1	0.67	2.0	16.7	AX	Y	verilog	24	pdp11	Y	yes	N	N	64K	64K		70	13	8		2009			boots & runs RT-11, EIS inst & MMU	
pdp11-soc	https://github.com/scottll		Scott Baker	pdp11	16	16	**zu-3e**	James	no mem init file			6				##	v21.2	0.67	3.0			Y	vhdl	15	soc	Y	yes	N	N	64K	64K		70	13	8		2016	2020		PDP-11/20 CPU + RAM + UART + Timer + I/O Ports, Sierra Circuit Design now open sou	
pdp1bach	https://github.c	simulatio	Jan Adelsbach	PDP1	18	18	spartan7	James	PDP-1	635	128	6			109	##	v25,1	0.67	2.0	57.5	X	Y	verilog	16	pdp1_cpu	Y	yes	N	N	4K	4K	Y	28					2015			
pdp2011	http://pdp2011.	stable	Sytse van Slooten	PDP11	16	16	kintex-7-3	James	Brakef	5060		6	1		205	##	14.7	0.67	2.0	13.6	AX	Y	vhdl	3	cpu	Y	yes	Y	N	64K	64K		70	13	8		2008	2019	http://pdp2011.sy	SoC, build files for A&X boards	complete impl including orig IO devices
pdp6	https://github.com/Morri		Michael Morris	PDP6	36	36	spartan7	James	Altera primitives			6					v25,1				A	Y	verilog	16	pdp6	Y				256K	256K							2018	https://en.wikipe	ISA identical to PDP-10	PDP-10 was much more successful
pdp8	https://opencor	alpha	Joe Manojlovick, Rob D	PDP8	12	12	kintex-7-3	James	Brakef	1219		6	1		183	##	14.7	0.50	2.0	37.5	X	Y	vhdl	55	cpu	Y	yes	N	N	32K	32K				8		2012	2016		PDP-8 Processor Core and System	Boots OS/8, runs apps, several variants
pdp8l	https://opencor	beta	Ian Schofield	PDP8	12	12	cyclone-3	James	Brakef	1088		4		48	63	##	q13.1	0.50	2.0	14.4	A		vhdl	11	top	Y	yes	N	N	4K	4K						2013	2013		Minimal PDP8/L implementation with 4K disk monitor system	
pdp8-soc	https://github.com/scottll		Scott Baker	PDP8	12	12	**zu-3e**	James	no mem init file			6				##	v21.2	0.40	2.0			Y	vhdl	15	soc	Y	yes	N	N	4K	4K						2016	2020		implemented for the Lattice iCE40-hx8	PDP-8 CPU + RAM + UART + Timer + I/O Ports
pdp8verilog	www.heeltoe.cc	stable	Brad Parker	PDP8	12	12	kintex-7-3	James	Brakef	505		6			366	##	14.7	0.50	2.0	181.3	X		verilog	18	pdp8	Y	yes	N	N	32K	32K				8		2005	2010		boots & runs TSS/8 & Basic	
pdp-8x	https://github.com/meng		Mats Engstrom	PDP8	12	12																	schematic			Y	yes	N	N	4K	4K							2019		Digital schematic, TTL	
pet_fpga	https://github.c	stable	Thomas Skibo	6502	8	8	kintex-7-3	James	Brakef	1052		6			242	##	14.7	0.33	4.0	19.0	X		verilog	1	cpu6502	Y	yes	N	N	64K	64K	Y					2007	2011	https://github.con	for Commodore PET	
pet-on-a-chip	https://github.com/ept22		Ezra Thomas	RISC	8	16	spartan7	James	verilog errors			6				##	v25.1	0.67	2.0			Y	verilog	19	top	Y	asm	N	Y	64K	64K	Y	40	5	8	2	2021	2024	https://ezrasrobot	robot controller, senior design project	cust pcb & uP, derivative of tiny_soc
phinix_cpu_de	https://github.c	WIP	Martin Andronikos	risc	16	16															A		schem	12	processor	Y	yes	N		64K	64K		40		16			2023	https://github.con	several cores? culminating in phinix+	docs separate folder, smolproc has system veril
pic_coonan		alpha	Tom Coonan	PIC16	8	14	kintex-7-3	James	Brakef	328		6		1	165	##	14.7	0.33	1.0	166.1	X		verilog	7	piccpu	Y	yes	N	Y	256	4K	Y					1999				risc8 by Tom Coonan also a PIC uP
pic-16c5x	https://tams-wv	errors	Ernesto Romani	PIC16	8	12	kintex-7-3	James	std library problems			6				##	14.7	0.33	2.0				vhdl	16	pic_core	Y	yes	N	Y	256	4K	Y					1998	2002			as part of thesis?
picoblaze	https://www.xil	stable	Ken Chapman	picoBlaze	8	18	kintex-7-3	James	Brakef	110		6		2	217	##	14.7	0.33	2.0	325.5	X		vhdl	1	kcspm6	Y	asm	N		256	2K	Y					2003		https://en.wikipe	2 clocks/inst, no prog ROM	this is the original picoBlaze author
picoblaze	https://www.xil	stable	Ken Chapman	picoBlaze	8	18	kintex-7-3	James	Brakef	317		6		2	195	##	14.7	0.33	2.0	101.6	X	Y	vhdl	19	kc705_kcp	Y	asm	N		256	2K	Y					2003		https://en.wikipe	2 clocks/inst	this is the original picoBlaze author
picoblaze	https://www.xil	stable	Ken Chapman	picoBlaze	8	18	spartan-3	James	Brakef	178		4		1	182	##	14.7	0.33	2.0	168.9	X		vhdl	1	kcspm3	Y	asm	N		256	2K	Y					2003		https://en.wikipe	2 clocks/inst, no prog ROM	this is the original picoBlaze author
piped_mips	https://github.com/AmirT		Maryam Hilmy Awad	mips	24	24	spartan7	James	empty ASIC design			6				##	v24.1	0.80	1.0		X		vhdl	19	maincode	Y	yes			16M	16M	Y			32	5		2022		top has component declarations & ma	LUT ram & blkRAM on both clock edges; MIPS w
piropiro	https://github.c	stable	pandora2000	RISC	32	32	kintex-7-3	James	port m	7491		6	11	1	118	##	14.7	1.00	1.0	15.7	X		vhdl	42	top			Y	N	64K	64K	Y			32		2010	2011		five variants	no doc, xilinx constraint file
plasma	https://opencor	stable	Steve Rhoads	MIPS	32	32	kintex-7-3	James	Brakef	2462		6		3	97	##	14.7	1.00	1.0	39.5	X		vhdl	22	plasma	Y	yes	N		4G	4G	Y			32		2001	2016	http://plasmacpu.	wide outside use, opencores page has list of related publications	
plasma_cortex	https://github.com/Nucle		Dylan Brophy	RISC	32	16	spartan6	James	Brakef	1566	626	6	4		105	##	14.7	1.00	1.0	67.0	X		vhdl	4	cpu	Y	yes	N		4G	4G	Y			8			2018	https://hackaday.io/project/160180-plasma-cortex-open-source-cpu-in-vhdl		
plasma_fpu	https://opencor	stable	Maximilian Reuter	MIPS	32	32	kintex-7-3	James	errors			6				##	14.7	1.00	1.0				vhdl	20	plasma	Y	yes	Y		4G	4G	Y			32		2015	2019		plasma with FPU	based on Plasma by Steve Rhoads
pmd85	https://github.com/PetrM		PetrM1	8080	8	8															A	Y	system	28	sys_top	Y	yes	N		64K	64K	Y					2021	2025	https://www.yout	Czechoslovakian PC using Intel 8080 c	https://github.com/MiSTer-devel/PMD85_MiS
pomegranate	https://github.c	WIP	Zachary Pearce	RISC	32	16+			vhdl style very modern, no top, opcodes via enumeration														vhdl	20		Y	asm	N		4G	4G		24		32		2023	2024		easy to configure uP core for embedd	senior thesis, see images subdirectory
popcorn	http://www.fpg	stable	Jeung Joon Lee	accum	8	8x	kintex-7-3	James	Brakef	267		6			347	##	14.7	0.33	1.0	428.4	X		verilog	4	pc	Y		N		64K	64K	Y	43				1998	2000		small 8 bit uP	
power_a2	https://github.c	stable	IBM (open PPC)	PPC	64	32	vu3p-2		TCL fil	380000													vhdl	285		Y	yes	Y		16E	16E	Y			32		2019	2020		PPC RTL, asic gate RTL	Virtex VU3P-2 FPGA implementation (380K luts
power_a2i	https://github.c	stable	IBM (open PPC)	PPC	64	32	vu3p-2		TCL files														vhdl	285		Y	yes	Y		16E	16E	Y			32		2019	2022	https://git.openp	PPC RTL, asic gate RTL	a2i and a2o variants
power_a2o	https://github.c	stable	IBM (open PPC)	PPC	64	32	vu3p-2		TCL files														vhdl	285		Y	yes	Y		16E	16E	Y			32		2019	2022	https://git.openp	PPC RTL, asic gate RTL	a2i and a2o variants
power_a2p	https://github.c	stable	IBM (open PPC)	PPC	32	32	artix-7																vhdl	285		Y	yes	Y		16E	16E	Y			32		2019	2022	https://git.openp	A2P is a mashup of OpenPOWER archi	using Cmod A7-35T board
powerpc_fpga	https://github.c	stable	Michael Kohn	ppc	32	32															L	Y	verilog	7	powerpc	Y						Y						2024	https://www.mike	minimal powerpc	ISA likes and dislikes
ppcpu	https://github.c	WIP	Piotr Węgrzyn	RISC	16	32	spartan7	James	ASIC design			6				##	v25.1						verilog	31	cpu	Y	yes			64K	64K	Y	43		8		2019	2023		8 regs, 16-bit imm, LLVM compiler	LLVM & OS, all inst have 16-bit imm/adr
ppx16	https://opencor	stable	Daniel Wallner	PIC16	8	14	kintex-7-3	James	missin	409		6			238	##	14.7	0.33	1.0	192.1	X		vhdl	10	P16C55	Y	yes	N	Y	256	4K	Y					2002	2009		both 16C55 & 16F84	with fake instruction ROM
prawn		errors	Tadatoshi Ishii	accum	8	8	spartan_	James	missing files			6				##	14.7	0.33	3.0				vhdl	2	prawn	Y	yes	N	N	4K	4K	Y					1992			reduced version of parwan from VHDL: Analysis and Modeling of Digital Systems, 1993	
processor-core	https://github.com/Hanxi		Steven Hua	RISC	32	32															A		vhdl			Y		N	N	4G	4G		24		32		2018	2018		clean, simple, prob classwork	Quartus proj, basic RISC instructions
propeller	https://propelle	proprietar	Chip Gracey	RISC	32	32																	verilog							4G	4G				512	5	2014	2020	https://github.con	original propeller has verilog (FPGA) s	ISA: op/ddd/sss format with predication
propeller_p8x3	https://www.pa	stable	Chip Gracey	RISC	32	32	kintex-7-3	James	Brakef	9498		6		20	160	##	14.7	1.00	0.1	134.8	X		verilog	9	top	Y	yes										2014			eight propellers, clocking from ucf file	several FPGA card build files
PSX_MiSTer	https://github.c	beta	MiSTer-devel	mips	32	32															A		vhdl	120	sys_top	Y	yes			4G	4G	Y			32		2021	2022	https://en.wikipe	MiSTer version of original Playstation	VHDL, verilog & system verilog RTL
pt13	http://www.sin	stable	Daniel Ogilvie	accum	8	8	kintex-7-3	James	Brakef	301		6			357	##	14.7	0.33	3.0	130.5			verilog	1	pt13	Y	asm	N	Y	64K	8K	Y	40	3			2011	2018	https://www.edn.	PT13 is optimized to be completely en	micro-code & register updates, minimal ISA
pulserain	https://github.c	errors	PulseRain Tech LLC	8051	8	8	arria-2	James	missing files			A				##	q18.0	0.33	3.0		A		system veril		PulseRain_	Y	yes	N	Y	64K	64K	Y					2017	2018	https://www.puls	intended for Max10	
pulserain	https://github.c	stable	PulseRain Tech LLC	8051	8	8	arria-2	James	some	2376		A	2	41	130	##	q18.0	0.33	3.0	6.0	A		system	25	FP51_fast	Y	yes	N	Y	64K	64K	Y					2017	2018	https://www.puls	1 clk/inst, intended for Max10	
pumpkin	https://github.com/Steve-		Steve Teal	accum	16	16	**zu-3e**	James	Brakef	166	67	6			625	##	v21.2	0.67	2.0	1261			vhdl	6	hello_wor	Y	asm	N		4K	4K		14					2020		scalable, 16-bit, 16 instruction soft CP	LUT RAM inferred (small size)
pumpkin	https://github.com/Steve-		Steve Teal	accum	16	16	**zu-3e**	James	Brakef	230	131	6		1	450	##	v21.2	0.67	2.0	656.1			vhdl	6	myco	Y	asm	N		4K	4K		14					2020		scalable, 16-bit, 16 instruction soft CP	emulates Myco, forced block RAM
p-vex	https://github.com/tvana		Thijs van As	VLIW	32	##	kintex-7-3	James	bypas	1660		6		1	233	##	14.7	1.00	1.0	140.1			vhdl	26	system	Y	yes	N					73		32	4	2005	2015	http://www.vliw.	1, 2 or 4 issue VLIW, uses HP VEX tool	probable degeneracy, LUT RAM for program me
pycpu	https://pycpu.w	myhdl	Norbert Feurle			8																	myhdl														2013		https://pycpu.wor	python hardware processor	
qnice-fpga	https://qnice-fp	stable	Bernd Ulmann	RISC	16	16	kintex-7-3	James	1432 l	3071	255	6			76	##	v25.1	0.67	1.0	17	X	Y	vhdl	40	quince_cp	Y	yes	N	N	64K	64K	N	18	4	16		2020	2024	https://github.con	derived from NICE: http://www.vaxm	PDP11-like, no byte operations
qnice-fpga	https://qnice-fp	stable	Bernd Ulmann	RISC	16	16	kintex-7-3	James	config parms not work			6				##	v25.1	0.67	1.0		X	Y	vhdl	40	MEGA65	Y	yes	N	N	64K	64K	N	18	4	16		2020	2024	https://github.con	derived from NICE: http://www.vaxm	PDP11-like, no byte operations
qrisc32	https://opencor	alpha	Viacheslav	RISC	32	32	arria-2	James	Brakef	3075		A	4		144	##	q13.1	1.00	1.0	46.9	A		system	8	qrisc32	Y	yes	N		4G	4G	Y			32	4	2010	2011		qrisc32 wishbone compatible risc core	for PhD thesis
qs5-rible	http://www.san	stable	John Rible	RISC	8	16	kintex-7-3	James	Brakef	468		6			135	##	14.7	0.33	1.0	95.3	X		verilog	1	qs5_mix			N		256	32K	Y					1998	1999		used in his class, also uses eP32	
qupls	https://github.c	WIP	Robert Finch	RISC	64	40		Robert Finch		160K											X		system	98	qupls	Y	asm	Y							64		2023	2024	http://www.finitr	Qupls (Q+): 2024 version of the Thor u	variety of three operand & u-coded instructions
r32v2020	https://github.com/dougg		Doug Gilliland	RISC																																	2021	2022			huge download, canceled
r4000		errors	Michael Povlin	MIPS	32	32	kintex-7-3	James	lots of problems			6				##	14.7	1.00	1.0				verilog														1994	1995		does not implement 64-bit data	only a few insts implemented, test vehicle
r8051	https://github.c	stable	Li Xinbing	8051	8	8	kintex-7-3	James	Brakef	1031		6	1		139	##	14.7	0.33	4.0	11.1	X		verilog	2	r8051	Y	yes	N	N	64K	64K	Y					2015	2019			
r8-core	https://github.com/vctrop		Victor O. Costa	RISC	16	16	spartan7	James	large r	1660	658	6	1	17	21	##	v25.1	0.67	1.0	8		Y	vhdl	14	r8_uc	Y	asm	N		64K	64K	N	35		16			2019		university project, doc in portuguese (	expanded R8 ISA
raptor16	www.spacewire	stable	Steve Haywood	CISC	16	16	kintex-7-3	James	Brakef	590	512	6			319	##	14.7	1.40	2.7	280.2	X		vhdl	1	raptor16	Y	yes	N	N	64K	64K	N					2004			8 data & 8 adr regs	no multiply, 8 adr modes
raptor64	https://opencor	alpha	Robert Finch	RISC	64	32																	verilog	63	raptor64	Y		Y	Y	4G	4G	Y	105	2	96	9	2005	2013		16 register sets, inst & data cache, me	ISA not finished, core runs
rca110	https://github.com/jadels		Jan Adelsbach	rca110	24	24																Y	verilog	2	rca110_cp	Y												2015	http://www.bitsavers.org/pdf/rca/110/TP1134_RCA110_PgmrRef_Aug62.pdf		

_uP_all_soft folder	opencores or prmary link	status	author	style / clone	data	inst sz	FPGA	repor ter	com ents	LUTs ALUT	Dff	LUT?	mults	blk ram	F max	date	tool ver	MIPS /inst	clks/ inst	KIPS /LUT	ven dor	SOC	src code	#src files	top file	doc	tool chai	fltg pt	Hav'd	max dat	max inst	byt ead	# inst	adr mo	# reg	pip e	start year	last revis	secondary web link	note worthy	comments
rcpu	https://github.com/redfas		redfast00	RISC	8	16															L		verilog	5	rcpu	Y	yes	N		4K	4K	Y			6			2019	https://github.co	verilog implementation of Python emulator, six 16-bit registers	
recon	https://github.com/jefflie		jeff lieu	Nios II	32	32																	verilog			Y	yes	opt		4G	4G	Y			32			2019	https://hackaday.	NIOS helper files	software helper files also
recore54		beta	Hans Tiggeler	PIC16	8	14	kintex-7-3	James	Cannot find <rcore_pk			6					14.7	0.33	1.0				vhdl	20	rcore54_s	Y	yes	N	Y	256	4K	Y					1999			not available at ht-lab website	www.ht-lab.com
reduceron	https://www.cs.	stable	Matthew Naylor/Tommy Thorm													##					AX				Reduceron												2008	2018	https://github.co	hardware for functional programming	red-lava generates the RTL
reflet	https://github.com/Arkae		Maxime Bouillot	accum	8	16	spartan7	James Brakef		688	411	6			128	##	v24.1	0.67	1.0	124.6	X		verilog	9	reflet_cpu		asm	N		64K	64K				16		2020	2024	https://github.co	original design, data size adj	most ops between accumulator & register, risc
ReLM-PoC	https://github.com/Tanun		Tanuma Hideki	accum	32	5															A		verilog	5	relmv	Y				4G	4G		32					2024		unusual accumulator ISA: ways to change meaning of 32-bit operand	
reonv	https://github.c	difficult	Lucas Castro	risc-v	32	32	kintex-7-3	James	many files			6				##	14.7	1.00	1.0				vhdl			Y	yes	N		4G	4G	Y			32		2017	2018	https://strijar.live	uses Leon infrastructure with risc-v ISA	
retro-compute	https://github.com/dougg		Doug Gilliland	misc	8	16																															2019	2024	https://github.co	Gilliland's builds of various 8 & 16-bit uPs, huge, several builds each	
reverse-u16	https://github.c	stable	A.T.	Z80	8	8	cylcone-4	James Brakef		11224		4		60		##	14.7	0.33	4.0		X	Y	vhdl	29	zxpoly	Y	yes	N	N	64K	64K	Y					2015			SOC project using T80, HDMI generato	retro Z80 based on T80 by Daniel Wallner
rf68000	https://opencor	alpha	Robert Finch	68000	32	16	zu5e	James	missing IP														system	7	rf68000	Y	yes	N	N	4G	4G	Y			16		2008	2024		mc68000 similar core, BCD instructions have variances	
rf6809	https://opencores.org/pr		Robert Finch	6809	12	12	artix-7	James Brakefield				6		5		##	v21.2	0.50	4.0		X	Y	system	21	rf6809	Y	asm	N		64G	64G	Y	44	13	8		2022	2022	http://www.finitr	Different from rtf6809: 36-bit adrs, op	12-bit version, has inst. Cache
rf6809	https://opencores.org/pr		Robert Finch	6809	8	8	artix-7	Robert Finch		4200		6		4	120	##	v21.2	0.33	4.0	2.4	X	Y	system	21	rf6809	Y	yes	N		16M	16M	Y	44	13	8		2022	2024	http://www.finitr	Different from rtf6809: 24-bit adrs, op	8-bit version, has inst. Cache
rf6809	https://opencores.org/pr		Robert Finch	6809	12	12	artix-7	Robert Finch		6500		6		5	120	##	v21.2	0.50	4.0	2.3	X	Y	system	21	rf6809	Y	asm	N		64G	64G	Y	44	13	8		2022	2024	http://www.finitr	Different from rtf6809: 36-bit adrs, op	12-bit version, has inst. Cache
rf80386	https://github.c	WIP	Robert Finch	x86	32	8															X		system	65	rf80386	Y	yes			4G	4G	Y						2024		has both 8088 and 80386, testing unfinished, work on rf8088 discontinued	
rfPhoenix	https://github.c	alpha	Robert Finch	GPGPU	32	40																	system	83						4G	4G	Y						2022		gpgpu Under Construction, derived from Nyuzi core by Jeff Bush	
risc_core_i	https://opencor	planning	Manuel Imhof	RISC	16	16	kintex-7-3	James Brakef		349		6	1		526	##	14.7	0.67	3.0	336.8	X	B	vhdl	13	CPU	Y	asm	N		1K	1K				8	4	2001	2009		Havard arch, thesis project	derived clocks: estimated derating
risc_cpu	https://electron	untested		accum	8	8																	vhdl					N		32	32	Y	8					2017			
risc_uw_dnn	https://github.com/Shiche		Justin Qiao	RISC	32	32															A	Y	system	98	cpu	Y	asm	Y		4G	4G	Y	28		32	5	2022	2023	https://github.co	real-time device 4 recognizing handwr	senior project at UW, MIPS derivative (WISC-SP
risc0	https://sourcefo	beta	Niklaus Wirth	RISC	32	32	kintex-7-3	James Brakef		1186		6	4	6	110	##	14.7	0.67	1.0	61.9	X		verilog	8	RISC0	Y	yes	N		4G	4G						2011	2018	https://people.inf.	minimalist Wirth, education tool	Lola: https://people.inf.ethz.ch/wirth/Lola/inde
risc-16	https://user.eng	stable	Bruce Jacob	RISC	16	16												0.67					vhdl	12	soc	Y	yes	N		64K	64K	N	9		8		2000	2015	https://user.eng.u	single cycle, pipeline & OO variants	Little Computer (LC-896) derivative
risc16_verilog	https://github.com/musta		Mustafa Cataltas	risc	16	16	artix-7	James	empty design							##	14.7				X		verilog	1	CORG_Pro	Y		N		256	128	N	17		8			2024	https://github.co	educational, 16-bit MIPS	MuSe & DoMe archs, Python simulation
risc16f84	https://opencor	stable	John Clayton	PIC16	8	14	kintex-7-3	James Brakef		375		6			392	##	14.7	0.33	2.0	172.5	AX		verilog	1	risc16f84_	Y	yes	N	Y	256	4K	Y					2002	2018		derived from CQPIC by Sumio Morioka	other variants with RTL
risc5	http://www.pro	beta	Niklaus Wirth	RISC	32	32	atrix7-35	James Brakef		2913		6		48	50	##	v20.1	1.00	1.0	17.2	ALX	Y	verilog	8	RISC5Top	Y	yes	Y		4G	4G				16		2013	2018	https://people.inf.	minimalist Wirth, part of Project Ober	32x32 multiplier, wikipedia entry
risc5	http://www.pro	beta	Niklaus Wirth	RISC	32	32	kintex-7-3	James Brakef		2441		6	4	1	92	##	14.7	1.00	1.0	37.8	ALX	Y	verilog	8	RISC5	Y	yes	Y		4G	4G				16		2013	2017	http://www.astro	minimalist Wirth, part of Project Ober	32x32 multiplier, wikipedia entry
risc5	http://www.pro	beta	Niklaus Wirth	RISC	32	32	zu-2e	James Brakef		2001	392	6	4		177	##	v20.1	1.00	1.0	88.3	ALX	Y	verilog	8	RISC5	Y	yes	Y		4G	4G				16		2013	2017	http://www.astro	minimalist Wirth, part of Project Ober	32x32 multiplier, wikipedia entry
risc5	http://www.pro	beta	Niklaus Wirth	RISC	32	32	zu-3e	James	IBUF clocking			6	4		213	##	v21.1	1.00	1.0		ALX	Y	verilog	8	RISC5Top	Y	yes	Y		4G	4G				16		2013	2017	http://www.astro	minimalist Wirth, part of Project Ober	32x32 multiplier, wikipedia entry
risc5	http://www.pro	beta	Niklaus Wirth	RISC	32	32	zu-3e	James Brakef		1936	392	6	4		213	##	v21.1	1.00	1.0	109.9	ALX	Y	verilog	8	RISC5	Y	yes	Y		4G	4G				16		2013	2017	http://www.astro	minimalist Wirth, part of Project Ober	32x32 multiplier, wikipedia entry
risc5a	http://www.pro	beta	Niklaus Wirth	RISC	32	32											v21.1	1.00	1.0		ALX	Y	verilog	8	RISC5	Y	yes	N		4G	4G				16		2013	2017	http://www.astro	minimalist Wirth, part of Project Ober	no floating-point
risc5x	https://opencor	stable	MikeJ	PIC16	8	14	kintex-7-3	James	RLOC constraint errors			6					14.7	0.33	1.0				vhdl	15	cpu	Y	yes	N	Y	256	4K	Y					2002	2011		makes extensive use of xilinx primitives	
risc63	https://github.com/domir		Dominik Salvet	RISC	64	16	kintex7	James Brakef		2103	1080	6			240	##	14.7	2.00	1.0	227.8	X		vhdl	16	risc63	Y		N		256K	256K	Y	39		16		2020	2024		tightly packed 16-bit ISA, no mult, no d	BS thesis in Chech
risc8	https://web.arcl	stable	Tom Coonan	PIC16	8	12	kintex-7-3	James Brakef		355		6			154	##	14.7	0.33	2.0	71.5	X		verilog	8	cpu	Y	yes	N	Y	256	2K	Y					1999	1999	https://github.co	excellent HTML doc	directory contains derivative design by another
risc8softcore	https://github.com/osrese		Trammell Hudson	AVR	8	16															L		verilog	6	risc8-soc	Y	yes	N	Y	64K	64K	Y					2020	2021		mostly compatible with the AVR instruction set	
riscff		proprietar	ExpressIf	RISC	16	16																	proprietary														2004			now produce ESP8266 & ESP32	
risc-fuggit	https://github.com/itsShn		Nikhil Shah	RISC	32	32																	verilog	33	riscmain	y		N		4G	4G				32			2019		non-standard set of conditional branches, schematic conflicts with documentation on	
riscmcu	https://opencor	stable	Yap Zi He	AVR	8	16	arria-2	James	LPM parameter errors			4				##	q18.0	0.33	1.0		A		vhdl	15	v_riscmcu	Y	yes	N	Y	128	512	Y	92		16	3	2002	2009		thesis	added 5 inst to AVR
riscompatible	https://opencor	beta	Andre Soares	RISC	32	32	kintex-7-3	James	set IO	2167		6		1	145	##	14.7	1.00	3.0	22.3	X		vhdl	12	riscompat	Y	yes	N	Y	4G	4G	Y			16		2014			based on RISCO processor by Junqueira & Suzim 1993	
risc-processor	https://github.c	stable	Jeff Bush	RISC	32	32	kintex-7-3	James Brakef		1445		6		6	161	##	14.7	1.00	1.0	111.6	X		verilog	22	fpga_top	Y	yes	N		4G	4G	Y	21		32		2008	2019	https://github.co	two designs with same name	MIT course work
riscuva1	https://www.sc	stable	S. de Pablo	picoBlaze	8	14	kintex-7-3	James Brakef		109		6			370	##	14.7	0.33	2.0	560.7	X		verilog	1	riscuva1	ome		N	Y	256	1K	Y	35				2006	2006	https://github.co	Verilog source included in PDF file	also VHDL version by Bikash Gogoi with identica
riscv_3wide_o	https://github.c	simulatio	aritramanna	risc-v	32	32																	system	23	design	Y	yes	N		4G	4G				32			2025	https://www.redd	3-wide superscalar, out-of-order RISC-	good schematic diagrams
riscv_biriscv	https://opencores.org/pr		Ultra Embedded	risc-v	32	32																	verilog			Y	yes	N		4G	4G	Y	32		32			2021	https://github.co	dual issue	also single issue version
riscv_black-par	https://github.com/black-		Daniel Petrisko	risc-v	64	32																	system verilog			Y	yes	Y		16E	16E	Y			32			2021		cache-coherent, RV64GC multicore	
riscv_bonfire	https://github.c	vado proje	Thomas Hornschuh	risc-v	32	32	kintex-7	James Brakefield				6				##	14.7	1.00	1.0				vhdl		bonfire_cp	Y	yes			4G	4G	Y			32			2018	http://bonfirecpu.	vivado project, based on lxp32	comingled lxp32 & RISCv; poorly organized gith
riscv_boom	https://github.com/riscv-b		UC Berkeley	risc-v	32	32																	scala			Y	yes	Y		4G	4G	Y	45		32				https://boom-core	Berkeley Out-of-Order RISC-V Processor	
riscv_briscv	https://ascslab.org/resear		various	risc-v	32	32																				Y	yes	Y		4G	4G	Y	45		32		2018	2020	https://opencores	six implementiations of risc-v	Boston Un. Course work
riscv_clarinet	https://github.com/HPC-L		Riya Jain etal	risc-v	32	32																	bluespec verilog			Y	yes	Y		4G	4G	Y	45		32	5		2020	https://github.co	RISC-V with posit arithmetic, bluespec	verilog for riscv flute & (3) posit sizes
riscv_clarvi	https://github.c	stable	Robert Eady	risc-v	32	32	arria-2	James	Altera	2616		A			178	##	q18.0	1.00	1.0	68.2	A	B	system	7	clarvi	Y	yes	N		4G	4G	Y			32	6	2016	2017	https://www.cl.ca	educational simple RISC-V implementa	doesn't make use of block RAM RTL
riscv_cpu	https://github.com/nobot		misha kevlishvili	risc-v	32	32												1.00	1.0				verilog			Y	yes	N		4G	4G	Y	45		32		2019	2019	https://www.yout	simple and easy to understand design	
riscv_croyde	https://github.com/ben-m		Ben Marshall	risc-v	64	32																Y	system	35	core_top	Y	yes	N		16Q	16Q	Y			32	3	2021	2021		64-bit rv64imck ISA	small, simple yet SOC, see also his tim & vanilla
riscv_cva6	https://github.com/pulp-p		openhwgroup	risc-v	64	32												1.00	1.0							Y	yes	Y		4G	4G	Y			32	6	2018	2022	https://github.co	single issue, in-order CPU which imple	**was riscv_ariane**
riscv_cva6	https://github.com/openh		openhwgroup	risc-v	64	32												1.00	1.0							Y	yes	Y		4G	4G	Y			32	6	2018	2022	https://github.co	32 and 64-bit RISC-V cores CVxxx: AKA	ariane, PULP/rocket & Ibex, directory name was
riscv_dark	https://github.c	beta	Marcelo Samsoniuk	risc-v	32	32	kintex-7-3	Marcelo Sam		1000		6			220	##	v20.1	1.00	1.0	220.0	XL		verilog	4	darkriscv	Y	yes	N		4G	4G	Y	45		32	2	2018	2024	https://opencores	written in one night, low line count	ku040 overclock 400MHz, builds for 18 fpga bo
riscv_dark	https://github.c	alpha	Marcelo Samsoniuk	risc-v	32	32	kintex-7-3	James Brakef		1422		6		1	167	##	14.7	1.00	1.0	117.2	XL		verilog	2	darksocv	Y	yes	N		4G	4G	Y	45		32	2	2018	2024	https://blog.hacks	written in one night, low line count	readme is descriptive, uses cache
riscv_drim-s	https://github.c	alpha	Integrated Circuits Lab	risc-v	32	32															A		system	107	module_to	Y	yes	N		4G	4G	Y			32	6	2021	2024		6-stage core, 2-Wide Superscalar, implementing the RiscV ISA (RV32IM)	
riscv_engine-v	https://github.com/micro-		Antti Lukats	risc-v	32	32				306		4						1.00	6.7		AL		verilog	11		Y	yes	N		4G	4G	Y	45		32		2018	2018	https://riscv.org/2	RISC-V contest 2nd place, 8-bit ALU	no source for xilinx, no implementation docs
riscv_fazyrv	https://github.com/meinil		Meinhard Kissich	risc-v	32	32															X		system	23	fazyrv_top	Y	yes			4G	4G	Y			32			2024	https://dl.acm.org	minimal-area RISC-V core with a scalable data path to 1, 2, 4, or 8 bits	
riscv_femtoRV	https://github.c	stable	Bruno Levy	risc-v	32	32																	verilog	45	femtosoc	Y	yes	N		4G	4G	Y	45		32		2020	2025	https://brunolevy.	eight riscv uP, teaches FPGAs to unive	100MB of images deleted
riscv_fpga	https://github.c	stable	Michael Kohn	risc-v	32	32															L	Y	verilog	13	riscv	Y				4G	4G	Y						2025	https://www.mike	comparisons with his 68K & x86	complaints about RISC-V
riscv_fwrisc	https://github.com/mballa		Matthew Balance	risc-v	32	32	ice40	Matthew Bala		1653		4				##		1.00	6.7		AL		system	8	fwrisc_fpg	Y	yes	N		4G	4G	Y	45		32		2018	2018	https://opencores	featherweight entry 2018 RISC-V cont	0.15 DMIPS/MHz
riscv_fwrisc	https://github.com/mballa		Matthew Balance	risc-v	32	32	igloo2	Matthew Bala		1060		4			20	##		1.00	6.7	2.8	AL		system	8	fwrisc_fpg	Y	yes	N		4G	4G	Y	45		32		2018	2018	https://opencores	featherweight entry 2018 RISC-V cont	0.15 DMIPS/MHz
riscv_glacial	https://github.com/brouh		Eric Smith	risc-v	32	32												1.00	6.0				schema	4	glacial	Y	yes	N		4G	4G	Y	45		32		2018	2019		designed for 2018 RISC-V SoftCPU Contest, for "smallest implementation" category, 8-	
riscv_GRVI-pha	http://fpga.org/	beta	Jan Gray	risc-v	32	32	virtex-u-2	Jan Gray		320		6		1	375	##	v16.4	1.00	1.0	1172	X		proprietary			Y	yes	N		4G	4G	Y	45		32	3	2015	2018	https://www.yout	hand fitted & placed	"Hoplite" router, 1680 cores in XCVU9P
riscv_harris	http://pages.hmc.edu/har		Dave Harris	risc-v	32	32																	system	53		Y	yes	N		4G	4G	Y	45		32		2019	2021		courseware to go with book	no top?
riscv_harris	http://pages.hmc.edu/har		Dave Harris	risc-v	32	32																	vhdl	46		Y	yes	N		4G	4G	Y	45		32		2019	2021		courseware to go with book	no top?
riscv_harzad5	https://github.com/Wren		Luke Wren	risc-v	32	32															L		verilog	18	hazard5_c	Y	yes	N		4G	4G	Y			32	5	2019	2023	https://github.co	RISC-V processor designed for the RISCBoy games console	
riscv_harzad5	https://github.com/Wren		Luke Wren	risc-v	32	32															L		verilog	28	hazard3_c	Y	yes	N		4G	4G	Y			32	3	2019	2024	https://github.co	RISC-V processor designed for the RIS	supports ASIC sysnthesis
riscv_hl5	https://github.c	stable	Paolo Mantovani	risc-v	32	32																	system	12	hl5	Y	yes	N		4G	4G	Y	45		32		2017	2020		32-bit RISC-V processor designed with HLS, coded in SystemC	
riscv_humming	https://github.c	stable		risc-v	32	32	kintex-7-3	James	too many Ios			6				##	14.7	1.00	1.0			Y	verilog	141	e203_cpu	Y	yes	N		4G	4G	Y			32		2016	2022	https://github.co	e200 has opensource	also have a chip
riscv_humming	https://github.c	stable		risc-v	32	32	kintex-7-3	James Brakef		14119		6		32	62	##	14.7	1.00	1.0	4.4	X	Y	verilog	141	e203_soc	Y	yes	N		4G	4G	Y			32		2016	2022	https://github.co	e200 has opensource	also have a chip
riscv_humming	https://github.c	untested		risc-v	32	32																Y	verilog			Y	yes	N		4G	4G	Y			32		2017	2022	https://github.co	AKA e200, Chinese	software tools take 80MB
riscv_ibex_dem	https://github.com/lowrisc/ibex-demo-system			risc-v	32	32	artix7	James Brakefield				6					v24.1	1.00	1.0		X	Y	system	16	top_artya	Y	yes	N		4G	4G	Y			32			2024		RISC-V SoC targeting the Arty-A7 FPG	xdc files
riscv_ibex_low	https://github.c	stable	Philipp Wagner	risc-v	32	32																	system	27	ibex_core	Y	yes	N		4G	4G	Y			32		2020	2026	https://www.lowr	AKA zero-riscy, also see pulp	four performance levels, several tapeouts
riscv_jive	https://github.com/fredre		Frédéric REQUIN	risc-v	32	32												1.00	20.0				verilog	19	jive_cpu_t	Y	yes	N		4G	4G	Y			32			2018		Size-Optimized Microcoded RISC-V CP	16-bit ALU
riscv_kian	https://github.com/spline		splinedrive	risc-v	32	32																	verilog	17	kianv	Y	yes	N		4G	4G	Y			32			2021		very simple riscv cpu/soc one single file implementation	
riscv_lattice	https://www.lat	stable	Lattice Semi	risc-v	32	32	machXO3	Lattice Semic		1507		4		4	60	##		1.00	1.0	39.8	L	Y	verilog			Y	yes	N		4G	4G	Y			32	5		2021		RV32I ISA, 5 stage pipeline, configured & generated using Lattice Propel	
riscv_lowrisc	https://github.com/lowRIS		Alex Bradbury	risc-v	32	32																Y	verilog			Y	yes	N		4G	4G	Y			32			2017	http://www.lowri	version 0.4-lowRISC with tagged memory and minion core	
riscv_microrv3	https://github.com/agra-u		agra-uni-bremen	risc-v	32	32																	scala	50	MicroRV3	Y	yes	N		4G	4G	Y			32		2021	2024	https://agra.infor	multi-cycle risc-v	
riscv_microsen	https://github.c	stable	Microsemi	risc-v	32	32	polarfire	microsemi		8614		4	2	10	122		L11.8	1.00	1.0	14.2			proprietary			Y	yes	N		4G	4G	Y			32		2016	2018	https://www.micr	is encrypted IP	has caches
riscv_minerva	https://github.com/lambd		lambdaconcept	risc-v	32	32																	nmigen			Y	yes	N		4G	4G	Y			32	6		2020		microarchitecture of Minerva is largely inspired by the LatticeMico32 processor	
riscv_minimax	https://github.com/gsmec		Graeme Smecher	risc-v	32	16	KU060	Graeme Sme		423	61	6			200	##	v22.2	1.00	4.0	118.2	X		verilog	2	minimax	Y	yes	N		4G	4G	Y			32		2022	2023		LUT count comparisons with other risc	most 32-bit insts microcoded, limited 16-bit ISA
riscv_myth	https://github.com/kuby1		Kubiran Karakaran	risc-v	32	32																																	https://tl-x.org		
riscv_n_chip8	https://github.com/nobot		misha kevlishvili	risc-v	32	32																	verilog	2	riscv	Y	yes	N		4G	4G	Y			32			2023	https://www.yout	simple RV32I on Tang Nano 9K	video: Tang Nano & LCD doing Chip-8 games
riscv_naxriscv	https://github.com/Spinal		Charles Papon?	risc-v	32	32	artix7	Charle	AKA sp	13300	#####	6	4	12	155			1.00	0.4	29.1			scala			Y	yes	N		4G	4G	Y			32			2024	https://spinalhdl.g	OoO execution w/reg renaming, Superscalar(2 decode, 3 execution units, 2 retire), 2.9	
riscv_neorv32	https://github.c	stable	Stephan Nolting	risc-v	32	32	cyclone-IV	Stepha	rtl fpg	1223	607	4			130	##	q19.1	1.00	2.0	53.1	ALX	Y	vhdl	25	neorv32_t	Y	yes	N		4G	4G	Y			32	2	2020	2025	https://opencores	**very well documented, customizal**	minimal configuration, minimal riscv
riscv_neuro_m	https://github.com/cepdn		Un Peradeniya, Sri Lan	risc-v	32	32																	verilog			Y	yes	N		4G	4G	Y			32			2026	https://cepdnaclk.	full Spiking Neural Network (SNN) training and inference pipeline	
riscv_niosv	https://www.int	proprietar	Intel	risc-v	32	32	agilex	intel	fastes	1509		A		2	566	##	q21.3	1.00	1.0	375.2	A		proprietary			Y	yes	N		4G	4G	Y			32	5		2021		free license, small inst & data mem	RV32IA spec, M20K for reg file, interrupts
riscv_niosv	https://www.int	proprietar	Intel	risc-v	32	32	arria-10	intel	fastes	1375		A		2	306	##	q21.3	1.00	1.0	222.3	A		proprietary			Y	yes	N		4G	4G	Y			32	5		2021		free license, small inst & data mem	RV32IA spec, M20K for reg file, interrupts
riscv_niosv	https://www.int	proprietar	Intel	risc-v	32	32	stratix-10	intel	fastes	1580		A		2	362	##	q21.3	1.00	1.0	229.1	A		proprietary			Y	yes	N		4G	4G	Y			32	5		2021		free license, small inst & data mem	RV32IA spec, M20K for reg file, interrupts
riscv_noel	https://www.gaisler.com/		gaisler	risc-v	32	32															AX		vhdl	40		Y	yes	N		4G	4G	Y			32			2022	https://www.gaisl	many config options	32 & 64-bit, software tools, bit files
riscv_nox	https://github.com/aignac		Anderson Ignacio	risc-v	32	32															X	Y	system	37	nox_soc	Y	yes	N		4G	4G	Y			32	4		2024	https://arxiv.org/	asic & fpga, multiple cores/chip, 4	xdc's, performance compares with several
riscv_orca	https://github.c	beta	VectorBlox	risc-v	32	32	stratix-5	vectorblox		1082		A		?	244	##	14.7	0.98	1.0	221.0	A		vhdl	13	orca	Y	yes	N		4G	4G	Y			32		2016			*, /, fltg-pt all optional	RV32IM
riscv_paranut	https://github.com/hsa-ee		Alexander Bahle	risc-v	32	32																	vhdl	~100	paranut	Y	yes	N		4G	4G	Y			32			2021	https://ees.hs-aug	SIMD vect & simul multi-threading in	Effic embed Sys group Un of Applied Sciences A
riscv_pequeno	https://github.com/iammi		Mitu Raj	risc-v	32	32	artix7	Mitu R	16 cus	2084	1564	6			100	##		1.00	1.0	48.0	X		system	31	pqr5_core	Y	yes	N		4G	4G	Y	53		32	5	2022	2025	https://chipmunkl	multi-page tutorial on uP design, pequ	https://github.com/iammituraj/iammituraj
riscv_percival	https://github.com/artecs		ArTeCS (Un Madrid)	risc-v	64	32	kintex7	ArTeC	larges	57129	#####	6			50		v20.2	1.00	2.0	0.4	X		system	~60		Y	yes	N		16E	16E	Y			32		2017	2022	https://github.co	Open-Source Posit RISC-V Core with Quire Capability, cav6(AKA Ariane) derivative	
riscv_neuro_m	https://github.com/cepdn		Un Peradeniya, Sri Lan	riscv	32	32																																		spiking neuron accerator	
riscv_piccolo	https://github.com/bluesp		BlueSpec	risc-v	32	32																	bluespec verilog			Y	yes	N		4G	4G	Y			32	3	2018	2018		RISC-V CPU, simple 3-stage pipeline, for low-end applications (e.g., embedded, IoT), wi	
riscv_picorv32	https://github.c	beta	Clifford Wolf	risc-v	32	32	GW1NR-9	Jean-L	small	2764	1833	4		8	27	##		1.00	3.0	3.3	X		verilog	1	picorv32	Y	yes	N		4G	4G	Y			32		2016	2022	https://www.cnx-	mimimal features, soc options	https://github.com/sipeed/TangNano-9K-exam
riscv_picorv32	https://github.c	beta	Clifford Wolf	risc-v	32	32	GW1NR-9	Jean-L	large	8594	5278	4	2	32	27	##		1.00	3.0	1.0	X	Y	verilog	1	picorv32	Y	yes	N		4G	4G	Y			32		2016	2022	https://www.cnx-	mimimal features, soc options	inclueds all peripherals
riscv_picorv32	https://github.c	beta	Clifford Wolf	risc-v	32	32	kintex-U-3	Cliffor	small	761	442	6			454	##	v16.2	1.00	3.0	198.9	X		verilog	1	picorv32	Y	yes	N		4G	4G	Y			32		2016	2022		mimimal features, soc options	LUTs & Fmax for Kintex, Virtex & Ultrascale+
riscv_picorv32	https://github.c	beta	Clifford Wolf	risc-v	32	32	xcku3p-3	Cliffor	small	761	442	6			769	##	v16.2	1.00	3.0	336.8	X		verilog	1	picorv32	Y	yes	N		4G	4G	Y			32		2016	2022	https://github.co	mimimal features, soc options	designed for minimum LUTs

_uP_all_soft folder	opencores or prmary link	status	author	style / clone	data	inst sz	FPGA	repor ter	com ents	LUTs ALUT	Dff	LUT?	mults	blk ram	F max	date	tool ver	MIPS /inst	clks/ inst	KIPS /LUT	ven dor	SOC	src code	#src files	top file	doc	tool chai	fltg pt	Hav'd	max dat	max inst	byt ead	# inst	adr mo	# reg	pip e	start year	last revis	secondary web link	note worthy	comments
riscv_picorv32	https://github.c	beta	Clifford Wolf	risc-v	32	32	xcku3p-3	Cliffor	large	2019	1085	6			769	##	v16.2	1.00	3.0	127.0	X	Y	verilog	1	picorv32	Y	yes	N		4G	4G	Y			32		2016	2022	https://github.con	mimimal features, soc options	designed for minimum LUTs
riscv_pito	https://github.com/hossei		Hossein Askari	risc-v	32	32	ZCU102	Hosse	includ	201079		6	##	####	250						X		system	31	rv32_core	Y	yes	N		4G	4G	Y			32	8	2020	2022	https://barvinn.re	RISC-V Barrel Processor for Deep Neu	has NN accelerator
riscv_potato	https://github.c	beta	Kristian Skordal	risc-v	32	32	kintex-7-3	James Brakef		2467		6			116	##	14.7	1.00	1.0	47.1	X	B	vhdl	24	pp_core	Y	yes	N	N	4G	4G	Y	30		32		2014	2020		risc-V interger only, no mult	"rocket-core" version at risc.org
riscv_pulpino	https://github.com/pulp-p		Andreas Kurth	risc-v	32	32	arria-2	James	missing files			A				##	q18.0						system	9		Y	yes	N		4G	4G	Y			32		2015	2020	http://www.pulp-	pulpissimo is single core "pulp" with interest in non-riscv ISA expansion	
riscv_qerv	https://github.com/olofk/		Olof Kindgren	risc-v	32	32																	verilog	8	serv_top	Y	yes	N		4G	4G	Y	45		32		2023	2025	https://www.gam	risc-v with 4-bit serial ALU	
riscv_reindeer	https://github.com/olofk/		pulserain.com	risc-v	32	32															AL		verilog			Y	yes	N		4G	4G	Y	45		32	4	2018	2018	https://riscv.org/2	RISC-V contest prize	
riscv_reonv	https://github.com/lcbcFc		Lucas Castro	risc-v	32	32	spartan6	Wajih Yousse		3370		6			133			1.00	1.0	39.4						Y	yes	N		4G	4G	Y	45		32			2018	https://www.hind	Lightweight Cryptographic Instruction	risc-v version on Leon3 tools
riscv_riscboy	https://github.com/Wren		Luke Wren	risc-v	32	32																	verilog	54	riscboy_fp	Y	yes	N		4G	4G	Y	45		32		2018	2021		portable games console desgn, PCB dsgn, see riscv_hazard3&5	
riscv_rocket	https://github.c	chisel	Andrew Waterman	risc-v	32	32																Y	chisel			Y	yes	N		4G	4G	Y			32		2016	2018			
riscv_rp32	https://github.c	alpha	Iztok Jeras	risc-v	32	32																	system	28	r5p-mous	Y	yes	N		4G	4G	Y			32			2022		four variants including single cycle, m	synthesis collapse
riscv_rpu	https://github.c	untested	Colin Riley	risc-v	32	32	artix-7	Colin Riley		3291	1156	6	12	1	200	##	14.7	1.00	1.0	60.8	X	Y	vhdl	14	core	Y	yes	N		4G	4G	Y			32		2015	2020	http://labs.domipl	Series of 16 tutorials on uP design, wo	RPU uP, TPU now discarded
riscv_rsd	https://github.com/rsd-de		Susumu Mashimo	risc-v	32	32	zynq	Susumu Mash		28166		6			90			1.00	1.0	3.2			system verilog			Y	yes	N		4G	4G	Y			32			2020	https://www.rsg.c	RISC-V out-of-order superscalar proce	can be synthesized for small FPGAs
riscv_rtg4	https://github.c	mature	microsemi	risc-v	32	32																				Y	yes	N		4G	4G	Y			32		2018	2020	https://github.con	risc-v for actel FPGAs, tcl files only	based on rocket chip
riscv_rudolv	https://github.com/bobbl/		Jörg Mische	risc-v	32	32	kintex-7-3	Jörg Mische		545		6			200	##		1.00	1.0	367.0	ALMX		verilog	4	pipeline	Y	yes	N		4G	4G	Y			32	5		2021		RISC-V processor for real-time system	34 clock mult & divide
riscv_rv01_cor	https://opencor	stable	Stefano Tonello	risc-v	32	32	kintex-7-3	James Brakef		13997		6	4	62	130	##	14.7	1.00	1.0	9.3	X		vhdl	65	rv01_selft	Y	yes	N		4G	4G	Y			32		2015	2017		all files in one directory	two self test tops
riscv_rv12	https://github.com/roalog		Roa Logic BV	risc-v	32	32	arria-2	James Brakefield				A				##	q18.0						system verilog			Y	yes	N		4G	4G	Y			32				https://roalogic.com		
riscv_rv16poc	https://github.com/Anton		Anton Mause	risc-v	16	32															A		vhdl	16	rv16poc			N		64K	4K	Y	33		32		2019	2023		small 16 bit CPU based on RISC-V RV3	reduced version of Actel RISC-V?
riscv_rv32soc	https://github.com/tomve		tom verbeure	risc-v	32	32	spartan3	James Brakef		1787	843	4	4	6	50	##	14.7	1.00	1.0	28.0	AX		verilog	18	top	Y	yes	N		4G	4G	Y			32			2018	https://tomverbeu	vexriscv in verilog, VexRiscV CPU - A N	near infinite amount of configuration options
riscv_rv3n	https://github.com/risclit		Li Xinbing	risc-v	32	32																	verilog	17		Y	yes	N		4G	4G	Y			32			2020		RV32IMC processor core, which has a new pipeline with "3+N" stages	
riscv_rvbs	https://github.com/CTSRD		Alexandre Joannou	risc-v	32	32																	bluesp	33		Y	yes	N		4G	4G	Y			32			2020		descript of the RISC-V instruction set in Bluespec, requires bluespec, no verilog code	
riscv_scarv-cpu	https://github.com/scarv/		Daniel Page	risc-v	32	32																Y	verilog	31	frv_core	Y	yes	N		4G	4G	Y			32		2019	2020	https://www.ukris	side channel hardened, no cache, branch prediction or virtual memory, research proje	
riscv_scr1	https://github.com/syntac		Syntacore	risc-v	32	32	arria-2	James Brakefield				A				##	q18.0						system	47	scr1_top_	Y	yes	N		4G	4G	Y			32		2017	2018	http://syntacore.com		
riscv_scr1	https://github.com/syntac		Syntacore	risc-v	32	32																	system	47	scr1_core	Y	yes	N		4G	4G	Y			32		2017	2021	http://syntacore.com		
riscv_serv	https://github.com/olofk/		Olof Kindgren	risc-v	32	32	cyclone1	Olof Kindgre		239	164	4		0.5	80	##		1.00	32.0	10.5	A		verilog	63	serv_top	Y	yes	N		4G	4G	Y	45		32		2018	2023	https://riscv.org/2	**smallest risc-v core, many boards**	https://github.com/olofk/corescore
riscv_serv	https://github.com/olofk/		Olof Kindgren	risc-v	32	32	ice40	Olof Kindgre		198	164	4			32	##		1.00	32.0	5.1	L		verilog	63	serv_top	Y	yes	N		4G	4G	Y	45		32		2018	2023	https://riscv.org/2	RISC-V contest prize, 1-bit ALU	https://github.com/olofk/corescore
riscv_serv	https://github.com/olofk/		Olof Kindgren	risc-v	32	32	vu37p	Olof Kindgre		125	164	6		0.5	125	##		1.00	32.0	31.3	X		verilog	63	serv_top	Y	yes	N		4G	4G	Y	45		32		2018	2023	https://riscv.org/2	6K cores in vu37p, reg-file in blk-RAM	https://github.com/olofk/corescore
riscv_shakti	https://github.com/anmol		IIT Madras	risc-v	32	32												1.00	1.0				bluesp	25		Y	yes	N		4G	4G	Y			32	3	2014	2021	https://shakti.org.	~8 different riscv cores, Madras India	several web sites & datings
riscv_sifive	https://www.sif	asic		risc-v	32	32																	proprietary			Y	yes	N		4G	4G	Y			32				https://www.sifiv	ASIC IP house, 32-bit "freedom" core	free Artix-7 bitstream
riscv_sifive	https://www.sif	asic		risc-v	64	32																	proprietary			Y	yes	N		4G	4G	Y			32				https://www.sifiv	ASIC IP house, 64-bit "freedom" core	free Artix-7 bitstream
riscv_snitch	https://github.c	WIP	Florian Zaruba	risc-v	32	32																	system	87	snitch	Y	yes	N		4G	4G	Y			32			2023	https://www.pulp	single-stage, single-issue, in-order RISC-V core (RV32I or RV32E), 32-bit integer and 64	
riscv_sodor	https://github.c	scala	UC Berkeley	risc-v	32	32																	scala			Y	yes	N		4G	4G	Y			32					1, 2, 3 and 5 stage pipe versions	
riscv_soomrv_	https://github.com/mathi		Mathis Salmen	risc-v	32	32	arty-7	Mathis Salmen				6			42								system	78	top	Y	yes	Y		4G	4G	Y			32		2023	2025	https://github.con	SoomRV for Arty A7 devbd & run Linu	4-wide superscalar OoO Execution
riscv_spu32	https://github.com/maikn		Merten Maik	risc-v	32	32																Y	verilog		top	Y	yes	N		4G	4G	Y			32		2019	2021	https://giters.com	actively being developed	
riscv_steel	https://opencores.org/pro		Rafael Calcada	risc-v	32	32	atrix-7-3	James Brakef		1784		6			116	##	v19.2	1.00	1.0	65.0			verilog	21	steel_top	Y	yes	N		4G	4G	Y			32	3	2020	2024	https://github.con	github version has vivado proj	under grad thesis, several web locations
riscv_steel	https://github.com/riscv-s		Rafael Calcada	risc-v	32	32	**zu-2e**	James Brakef		1775		6			208	##	v19.2	1.00	1.0	117.4			verilog	21	steel_top	Y	yes	N		4G	4G	Y			32	3	2020	2024	https://github.con	github version has vivado proj	under grad thesis
riscv_swerv	https://github.com/chipsa		david harris & sarah ha	risc-v	32	32																	verilog			Y	yes	N		4G	4G	Y			32			2024	https://github.con	rvfpga, swervolf, also _books/Digital_	digikey courseware, three variants
riscv_swerv	https://github.com/chipsa		Western Digital	risc-v	32	32	ZCU102	Weste	high L	30128		6	4	62	100			1.00	0.5	6.6			system	45	veer_wra	Y	yes	N		4G	4G	Y			32	9	2019	2022	https://blog.weste	9 stage pipe, dual issue	risc-v SoC for fpga, riscv_swerv_eh1_fpga now
riscv_taiga	https://gitlab.cc	stable	Eric Matthews	risc-v	32	32	zynq			1551				1	123			1.00	1.0	79.3	AX		system	46		Y	yes	N		4G	4G	Y			32		2017	2022		TAIGA: A new RISC-V soft-processor fr	33% smaller & 39% faster than LEON3
riscv_tinsel	https://github.com/POETS		Ghaith Tarawneh	risc-v	32	32																	bluespec verilog																https://poets-proj	message-passing architecture designed for FPGA clusters	
riscv_tl-verilog	https://github.com/RISCV-		Steve Hoover	risc-v	32	32																	system verilog			Y	yes	N		4G	4G	Y			32		2020	2023	https://github.con	TL-verilog implementation of RISC-V	65 particpants (sub-directories), 5 day course
riscv_tordboya	https://github.com/Bruno		Bruno Levy	risc-v	32	32	aritx-7	Bruno	RV32I	1000	600	6			120	21/	v23.1	1.00	1.0	120.0			verilog	15	soc	Y	yes	N		4G	4G	Y			32		2022	2023	https://brunolevy.	pipelined RISC-V uP & courseware	several configuration resource stats
riscv_ucoded	https://github.com/andmi		andmiele	risc-v	32	32								4				1.00	3.0				system	14	systemTo	Y	yes	N		4G	4G	Y			32			2022		micro-coded, 3-4 clocks/inst, base integer ISA	
riscv_uriscv	https://github.com/ultrae		ultra_embedded	risc-v	32	32												1.00	2.0				verilog	7	riscv_core	Y	yes	N		4G	4G	Y			32			2021	https://opencores	Simple, small, multi-cycle 32-bit RISC-V CPU implementation	
riscv_urv-core	https://github.c	error	Tomasz Włostowski	risc-v	32	32	kintex-7-3	James	missing files							##	14.7	1.00	1.0				verilog			Y	yes	N		4G	4G	Y			32		2015	2015			
riscv_vanilla	https://github.c	verified	Ben Marshall	risc-v	32	32	artix-7	Ben Marshall		2422		6			150			1.00	2.0	31.0			verilog	26	frv_cpu_a	Y	yes	N		4G	4G	Y			32	5		2019		"toy" 5 stage RISC-V CPU, implementing the rv32imc	
riscv_vanilla	https://github.c	verified	Ben Marshall	risc-v	32	32	zu-5e	James	IO lim	2422		6				##	v21.1	1.00	2.0				verilog	26	frv_cpu_a	Y	yes	N		4G	4G	Y			32	5		2019		"toy" 5 stage RISC-V CPU, implementing the rv32imc	
riscv_vexiiriscv	https://github.com/Spinal		Andreas Wallner	risc-v	32	32						6						2.50	0.3				scala			Y	yes	N		4G	4G	Y			32			2024		improved VexiRiscv: single/dual issue,	quad core
riscv_vexriscv	https://github.c	beta	Charles Papon	risc-v	32	32	artix-7	Charles Papo		481		6			346			0.52	1.0	374.1	X		scala		smallest	Y	yes			4M	4M	Y						2023	https://riscv.org/2	preformance #s for 8 configurations o	"Briey" is SOC variant
riscv_vexriscv	https://github.c	beta	Charles Papon	risc-v	32	32	artix-7	Charles Papon?				6						0.52	1.0		X		verilog			Y	yes			4M	4M	Y						2018	https://nlnet.nl/ev	verilog source: see riscv_rv32soc	scala not needed
riscv_vexriscv	https://github.c	scala	Charles Papon	risc-v	32	32	atrix-7-3	Charles Papo		1399		6			295			1.00	1.0	210.9	X	Y	scala	54	full no cac	Y	yes	N		4G	4G	Y			32			2023		preformance #s for 9 configurations o	"Briey" is SOC variant
riscv_vexriscv	https://github.c	scala	Charles Papon	risc-v	32	32	atrix-7-3	Charles Papo		504	505	6			243			0.52	1.0	250.7	X	Y	scala	54	small, no i	Y	yes	N		4G	4G	Y			32	2-5		2025	https://github.con	preformance #s for 9 configurations o	DMIPS/MHz
riscv_vexriscv	https://github.c	scala	Charles Papon	risc-v	32	32	atrix-7-3	Charles Papo		556	566	6			240			0.52	1.0	224.5	X	Y	scala	54	small no b	Y	yes	N		4G	4G	Y			32	2-5		2025	https://github.con	preformance #s for 9 configurations o	DMIPS/MHz
riscv_vexriscv	https://github.c	scala	Charles Papon	risc-v	32	32	atrix-7-3	Charles Papo		816	534	6			232			0.82	1.0	233.1	X	Y	scala	54	small, pro	Y	yes	N		4G	4G	Y			32	2-5		2025	https://github.con	preformance #s for 9 configurations o	DMIPS/MHz
riscv_vexriscv	https://github.c	scala	Charles Papon	risc-v	32	32	atrix-7-3	Charles Papo		730	570	6			220			0.70	1.0	211.0	X	Y	scala	54	small & ic	Y	yes	N		4G	4G	Y			32	2-5		2025	https://github.con	preformance #s for 9 configurations o	DMIPS/MHz
riscv_vexriscv	https://github.c	scala	Charles Papon	risc-v	32	32	atrix-7-3	Charles Papo		1418	949	6			216			1.21	1.0	184.3	X	Y	scala	54	full, single	Y	yes	N		4G	4G	Y			32	2-5		2025	https://github.con	preformance #s for 9 configurations o	DMIPS/MHz, 2.30 Coremark/MHz
riscv_vexriscv	https://github.c	scala	Charles Papon	risc-v	32	32	atrix-7-3	Charles Papo		1840	1158	6			199			1.21	1.0	130.9	X	Y	scala	54	full, cache	Y	yes	N		4G	4G	Y			32	2-5		2025	https://github.con	preformance #s for 9 configurations o	DMIPS/MHz, 2.30 Coremark/MHz
riscv_vexriscv	https://github.c	scala	Charles Papon	risc-v	32	32	atrix-7-3	Charles Papo		1935	1216	6			200			1.38	1.0	142.6	X	Y	scala	54	full max p	Y	yes	N		4G	4G	Y			32	2-5		2025	https://github.con	preformance #s for 9 configurations o	DMIPS/MHz, 2.57 Coremark/MHz
riscv_vexriscv	https://github.c	scala	Charles Papon	risc-v	32	32	atrix-7-3	Charles Papo		2021	1541	6			151			1.24	1.0	92.6	X	Y	scala	54	full MMU	Y	yes	N		4G	4G	Y			32	2-5		2025	https://github.con	preformance #s for 9 configurations o	DMIPS/MHz, 2.35 Coremark/MHz
riscv_vexriscv	https://github.c	scala	Charles Papon	risc-v	32	32	atrix-7-3	Charles Papo		2883	2130	6			180			1.21	1.0	75.5	X	Y	scala	54	linux bala	Y	yes	N		4G	4G	Y			32	2-5		2025	https://github.con	preformance #s for 9 configurations o	DMIPS/MHz, 2.27 Coremark/MHz
riscv_vhdl	https://opencor	errors	Sergey Khabarov	risc-v	64	32	kintex-7-3	James	many files, missing typ			6				##	14.7	1.00	1.0			Y	vhdl & verilog			Y	yes	N		4G	4G	Y			32		2016	2018	https://github.con	System-On-Chip based on bare Rocket	both rocket & river cores
riscv_vroom	https://github.com/Moon		Paul Campbell	risc-v	32	32	zu9p	Paul Campbell				6			25		v22.2	4.00	1.0				system	51	cpu	Y	yes	N		4G	4G	Y			32		2019	2023	https://hackaday.	high-end RISC-V implementation	8 IPC (instructions per clock) peak, goal ~4 aver
riscv_wildcat	https://github.com/schoe		Martin Schoeberl	risc-v	32	32	artix7	Martin Schoe		993	442	6			111			1.00	1.0	112.0	X		scala	32	singlecycl	Y	yes	N		4G	4G	Y			32	4		2025	https://arxiv.org/	comparison of 3, 4 & 5 stage pipeline,	book: Digital Design with Chisel
riscv_wildcat	https://github.com/schoe		Martin Schoeberl	risc-v	32	32	cyclone4	Martin Schoe		1727	452	4			85			1.00	1.0	48.9	A		scala	32	singlecycl	Y	yes	N		4G	4G	Y			32	4		2025	https://arxiv.org/	comparison of 3, 4 & 5 stage pipeline,	papers show 3 & 5 stage pipelines
riscv_wolv-z7	https://github.com/taner		Taner Öksüz	risc-v	32	32															AX		system	46	cpu	Y	yes	Y		4G	4G	Y			32			2023	https://github.con	SP & DP fltg-pt in VHDL & Sys Verilog	branch target address cache with bimodal bran
riscv_zscale	https://github.c	scala	UC Berkeley	risc-v	32	32																	scala			Y	yes	N		4G	4G	Y			32		2015	2017		not maintained & not conformant	
rise	https://opencor	beta	Jlechner etal	RISC	16	16	kintex-7-3	James	missing black boxes			6	1				14.7	0.67	1.0		X		vhdl	26	rise	Y	asm	N		64K	64K				16	5	2006	2010	en.wikiversity.org,	ARM style register usage	
rj32	https://github.c	alpha	Ryan Sanche	RISC	16	16																	verilog	8	top	Y	asm	N		64K	64K	Y	32		16		2013	2022	https://github.con	Digital schematic editer	nanogo compiler, **youtube videos**
rjsc5	https://github.com/rj45/r		Ryan Sanche	risc-v	32	32																	schem	6		Y	yes	N		4G	4G	Y			32			2022		Digital schematic, 16-bit data paths, micro-coded, multi-cycle	
rjstack	https://github.com/rj45/r		Ryan Sanche	stack	16	8																	schem	3	rjstack	Y	asm	N		64K	64K		8					2025	https://github.con	only eight instructions	uses customasm as assembler
robin	https://github.com/varke		Michel Anders	RISC	16	16																Y	verilog	12	robin	Y	yes	N		64K	64K	Y	20		8		2020	2020	https://github.con	**C compiler**, SoC design targeted at th	docs at **https://varkenvarken.github.io/rob**
rois	https://opencor	alpha	James Brakefield	RISC	24	24	spartan7	James	area e	402	82	6		1	103	##	v25.1	0.83	1.0	213.4	X		vhdl	2	rois24_24	Y		N		16M	16M	N	30		64	1	2016	2025		single pipe stage, passes simulation	cvtd to numeric.std & unsigned
rois	https://opencor	alpha	James Brakefield	RISC	24	24	kintex-7-3	James Brakef		384		6		1	170	##	14.7	0.83	1.0	368.8	X		vhdl	2	rois24_24min			N		16M	16M	N	30		64	1	2016	2017		single pipe stage, passes simulation	24-bit word operations only
rois	https://opencor	alpha	James Brakefield	RISC	24	24	kintex-7-3	James Brakef		382		6		1	120	##	14.7	0.83	1.0	261.7	X		vhdl	2	rois24_24up			N		16M	16M	Y	55		64	1	2016	2017		single pipe stage, pre simulation stage	8, 16 & 24-bit load/store
rois	https://opencor	alpha	James Brakefield	RISC	24	24	zu-2e	James	no blk	627		6			382	##	v19.2	0.83	1.0	507.1	X		vhdl	2	rois24_24min			N		16M	16M	N	30		64	1	2016	2017		single pipe stage, passes simulation	24-bit word operations only
rois	https://opencor	alpha	James Brakefield	RISC	24	24	xczu-1	James	area e	485		6		1	192	##	v25.1	0.83	1.0	329.8	X		vhdl	2	rois24_24up			N		16M	16M	Y	55		64	1	2016	2017		single pipe stage, passes simulation	cvtd to numeric.std & unsigned
rrisc	https://github.com/rener		Rene Schallner	RISC	8	8	spartan-7	James	uses L	330	154	6			125	##	v25.1	0.33	8.0	15.6	X		vhdl	8	top	Y	asm	N		64K	64K	Y			8		2020	2022	https://git.sr.ht/~	originally TTL/schematic, beginner's p	doc PDF file huge
rtf64	https://github.c	alpha	Robert Finch	RISC	64	8																	system	3	rtf64	Y	yes	Y				Y			32		2020	2021		variable length instructions	Posit support, glossary & references
rtf65002	https://opencor	alpha	Robert Finch	accum	32	8	kintex-7-3	James Brakef		11216		6	4	6	123	##	v14.1	0.67	2.0	3.7	X		verilog	10	rtf65002d	Y		N		4G	4G	Y			16		2013	2013	https://github.con	32-bit 6502 + 6502 emulation	"proven"
rtf6809	https://github.c	alpha	Robert Finch	6809	8	8	kintex-7-3	James	many	7506		6	1	2	106	##	14.7	0.33	4.0	1.2	X		verilog	4	rtf6809	Y	yes	N	N	64K	64K	Y	44	13	8		2012	2015	http://www.finitr	6809 with 32-bit "FAR" addressing	see also rf6809 variant
rtf68ksys	https://opencor	alpha	Robert Finch	68000	16	16	spartan-3	James	need t	13639		4	12	17		##	14.7	0.67	4.0		X	Y	verilog	49	rtf68kSys	Y	yes	N	N	4G	4G	Y			16		2011	2011	https://github.con	based on Tobias Gubener's TG68	
rtf8088	https://opencor	planning	Robert Finch	x86	16	8	kintex-7-3	James Brakef		4514		6	4		174	##	14.7	0.67	3.0	8.6	X		verilog	57	rtf8088	Y	yes	N	N	1M	1M	Y					2012	2013	https://github.con	8-bit memory data, e.g. 8088	
rtx2000	http://www.mp	asic	Tom Hand	forth	16	16																	proprietary																	Harris Corp., FPGA version at MPEfort	descendent of the Novix NC4016
s1_core	https://opencor	stable	Fabrizio Fazzino etal	SPARC	64	32	kintex-7-3	James Brakef		52845		6	8	59	56	##	v14.1	2.00	1.0	2.1	AX		verilog	136	s1_top	Y	yes	Y	N	4G	4G	Y			32		2007	2012	https://en.wikipec	reduced version of OpenSPARC T1	Vivado run
s16x4a	https://github.c	stable	Samuel Falvo II	forth	16	4	kintex-7-3	James Brakef		514		6			476	##	14.7	0.67	1.0	620.7	X	B	verilog	1	s16x4a	Y		N	N	64K	64K	Y	12				2012	2017		kestrel #2, byte & word data	derived from Myron Plichota's design (streame
s430	https://www.p-	stable	Paul Taylor	MSP430	16	16	artix-7	Paul Taylor		449		6			100			0.67	9.0	16.6			vhdl	1	s430					64K	64K	Y					2019	2019		msp430 subset with 8-bit alu	coded for size & not for speed
s4pu	https://baioc.github.io/po		Gabriel de Sant'Anna	forth	16	16	cyclone2	Gabriel de Sa		3306	1622	4		86	50	##	q13.1	0.67	1.0	10.1	A		vhdl	17	s4pu	Y	asm	N		64K	64K		32				2017	2020	https://gitlab.com/baioc/s4pu		in Portuguese
s64x7	https://github.c	stable	Samuel Falvo II	forth	64	8																	verilog	4	s64x7					16E	16E	Y	56					2017		64-bit simple Forth engine	very little doc
s6soc	https://opencor	stable	Dan Gisselquist	RISC	32	32	spartan-6	James	sparta	2820		6	1	10	133	##	14.7	1.00	1.0	47.3	X	Y	verilog	31	toplevel			N	N	4G	4G	N	20		16	5	2015				uses ZIP CPU
s80186	https://github.c	stable	Jamie Iles	x86	16	8	cyclone-V	Jamie Iles		1750		A			60			0.67	2.0	11.5	A	Y	system	50	core	Y		N		1M	1M	Y					2017	2021	https://www.jami	80186 binary compatible core	implementing the full 80186 ISA
sap	https://opencor	stable	Ahmed Shahein	accum	8	8	kintex-7-3	James	no LU	48		6			200	##	14.7	0.10	4.0	104.2	X		vhdl	15	mp_struct			N		16	16	Y	5				2012	2022	https://shirishkoir	Simple as Possible Computer from Ma	https://www.youtube.com/watch?v=prpyEFxZ
sap	https://code.austinmorlar		austin morlan	accum	8	8																	verilog	4	top			N		16	16	Y	5					2021	https://austinmor	another sap-1 implementation	good looking writeup
sap	https://github.c	stable	Federico Zotti	accum	8	8	kintex-7-3	James	no LU	48		6			200	##	14.7	0.10	4.0	104.2	G		vhdl	9	sap-1-TOP			N		16	16	Y	5					2024		Simple as Possible Computer	Gowin 9K project
sardmips	https://opencor	systemC	Igor Loi	MIPS	32	32																	systemC			Y	yes	N		4G	4G	Y			32		2006	2009		synthesizable parametric IP core supporting full MIPS R2000 ISA	
sayeh_cpu	https://github.com/armin		Armin Kazemi	RISC	16	16												0.67	1.0				vhdl		Sayeh	Y	asm	N		64K	64K				64			2017		16-bit MIPS, data flow schematic	64 word reg file?
sayeh_process	https://opencor	stable	Alireza Haghdoost, Arr	RISC	16	8	kintex-7-3	James Brakef		479		6	1		164	##	14.7	0.67	1.0	229.7	X		verilog	13	Sayeh	Y		N		64K	64K				32		2008	2009	haghdoost.persiangig.com		simple RISC
sayuri_cpu	http://www.mo	stable	Toyoaki Sagawa	RISC	32	32	kintex-7-3	James Brakef		1604		6			208	##	14.7	1.00	1.0	129.9	X		vhdl	13	cpu01				Y	4G	4G				32		2000	2000		dead weblink	high number of DFF
sbc6502	https://github.com/daven		Dave Nardella	6502	8	8															AGX	Y	verilog	26	top	Y	yes	N	Y	64K	64K	Y						2024	https://www.linke	projects for cmod-a7, de10, tang9K, ta	written from scratch, uses 8-digit display board
sbc6502	https://github.com/daven		Dave Nardella	6502	8	8	artix7	James	was m	1074	382	6		12	42	##	v23.2	0.80	1.0	31.3	AGX	Y	verilog	19	top	Y	yes	N		256	256	Y						2024	https://www.linke	linked in page has full description	web page also has soft 6502 for Gowin, Xilinx &
sbc6502	https://github.c	simulatio	Dave Nardella	accum	8	8			no outputs												GIX		verilog	2	minicpu	Y		N		256	256	Y	9					2024	https://www.linke	was his initial project, now included ir	linkedin doc for both minicpu & 6502

_uP_all_soft folder	opencores or prmary link	status	author	style / clone	data	inst sz	FPGA	repor ter	com ents	LUTs ALUT	Dff	LUT?	mults	blk ram	F max	date	tool ver	MIPS /inst	clks/ inst	KIPS /LUT	ven dor	SOC	src code	#src files	top file	doc	tool chai	fltg pt	Hav'd	max dat	max inst	byt ead	# inst	adr mo	# reg	pip e	start year	last revis	secondary web link	note worthy	comments
sc20	http://www.fort	proprietar	Brad Eckert	forth	32	8	virtex-6	Brad Eckert		1977		6			150			1.00	1.0	75.9	X		proprietary															2010		PDF file, Forth Inc.	
scamp-cpu	https://github.c	beta	James Stanley	accum	16	16																	verilog	76	fpga-cpuir	Y	asm	N		64K	64K							2022	https://hackaday.(	TTL & Verilog home built, has OS	pictures of TTL version
scarts	https://opencor	beta	Jlechner, Martin Walte	RISC	16	16	kintex-7-3	James	missing signal declarat			6					14.7	0.67	1.0				vhdl	18	scarts		yes	N		64K	64K		122		16	4	2011	2012		Scarts Processor	GCC compiler
schoolmips	https://github.com/MIPSf		Andrea Guerrieri	RISC	32	32																					yes			4G	4G								https://github.com	small MIPS CPU core originally based (	schoolMIPS has several versions
scpu	https://github.com/mohd		mohdfahad12328	cisc	8	8																	verilog	8	cpu	Y	yes	N		64K	64K	Y	21	6	8			2023		8-bit CPU which is inspired by RISC and CISC Architectures and designs, eight accumula	
SediCiPU2	https://github.com/alexfr		Alexey Frunze	risc	16	16	tang20k														G	Y	system	53	top	Y	asm	N		64K	64K	Y		6	8			2026	https://www.hack	16-bit PC based on the SediCiPUv2 CP	tang 20K, VGA display, PC & SP regs
secretblaze	http://www.lirn	beta	Lyonel Barthe	uBlaze	32	32	spartan-3	Lyonel Barthe		1563		4			91		i12.1	1.00	1.0	58.2	X		vhdl	26	sb_core		yes			4G	4G	Y	86		32	5	2010	2012	www.lirmm.fr/ADAC		
sifp	https://github.c	WIP	Zoltan Pekic	misc	16	16	spartan3-	James Brakef		3157	563	4		3	123	##	14.7	1.33	2.0	25.8	X		vhdl	36	mercury	Y		N		64K	64K		30		4		2023	2024	https://hackaday.i	"Single Instruction Format Processor"	five micro-operations per inst
simple_ttl_cpu	https://github.com/mons(		Ken Boak	accum	8	8																	schema	10	Nybble													2021		Digital schematic, very minimal	designed for manual operation
simplecpu	https://www-us	simulatioı	Michael Freeman	RISC	32	32	kintex-7-3	James	way to many los														vhdl	17	top								8				2018	2019	https://www-user	Educational, also a version 2 with VHD	both mips & riscv RTL
simplecpu_v1a	https://www.simplecpude		Michael Freeman	accum	16	16	artix-7	James	many	92	83	6		1		##	v24.2	0.67	3.0		X		vhdl	17	mcp_top_	Y		N		256	256		8				2018	2025	https://www.simp	courseware, vivado project, incomplet	micro coded, no store instruction
simplecpu_v3	https://www.sir	WIP	Michael Freeman	risc	16	16																	vhdl	17	top	Y				64K	64K				4		2018	2025		courseware, vivado project, incomplet	no RTL
simplecpu_v4	https://www.sir	WIP	Michael Freeman	risc	32	32																	vhdl	17	top	Y							20				2018	2025		courseware, vivado project, incomplet	MIPS/DLX rerivative, no RTL
simplecpu12	https://github.com/jasom		Jan Sommer	accum	12	12	spartan6	Jan Sommer		498	212	6		6	180	##	14.7	0.50	2.0	90.4	X		vhdl	9	top_level	Y		N	Y	4K	4K	N	32	3				2020		educational, has stack pointer	looks like an accumulator dsgn
simplecpucore	https://github.com/Karan		Karang	arm	32	32	kintex-7-3	James	bare c	946	309	6			100	##	v24.1	1.00	1.0	105.7	AX		vhdl	11	arm_core	Y	yes	N		4G	4G	Y			16	5		2017		CPU core for ARMv3, educational	no RTL comments, shows ASIC layout
simple-v	https://libre-soc.org/docs		Luke Leighton	PPC	64	32																	python			Y		Y				Y			32		2018	2022	https://libre-soc.o	Scalable Vectors for Power ISA	has the respect of Mitch Alsup
single-cyc-cpu	https://github.c	mature	Victor A Pajaro	MIPS	32	32															A		vhdl	30	AlvarezPajaro_singl			N		4G	4G	Y			32			2019		nice schematic and clear description, course work	
slurm	https://github.com/james		James Sharp	RISC	16	16															L	Y	verilog	54	slurm16_s	Y	asm	N		64K	64K	Y	20		16			2022		SLURM16 SoC - SLightly Useful RISC M	Video console system-on-chip made for the iCE4
SNESTang	https://github.com/nand2		nand2mario	6502	8	8																	verilog	109	nano20k	Y	yes	N		16M	16M	Y					2024	2025		SNES for Sipeed Tang FPGA Boards	65C816, 20K or larger tang boards
sngl-cycl-mips-	https://github.com/Diablo		DiabloBlood	MIPS	32	32	cycloneV	James	missing file			6									A		verilog	16	mipscpu	Y	yes	N		4G	4G	Y			32			2021		course project, single cycle, **digrams for R2R, memory read/write, branch, etc**	
socdp8	https://github.c	beta	Folke Will	PDP8	12	12	spartan7	James Brakef		583	268	6			106	##	v24.1	0.50	3.0	30.4	X	Y	vhdl	34	pdp8	Y	yes	N	N	32K	32K				8		2019	2019		SoC implementation of a PDP-8/I for t	includes extended ALU
socz80	http://sowerbut	stable	Will Sowerbutts	Z80	8	8	spartan-6	James	constr	2568		6		15	93	##	14.7	0.33	3.0	4.0	X		vhdl	25	top_level	Y	yes	N	N	64K	64K	Y					2013	2014		based on Daniel Wallner's T80, for Papillio Pro board	
softavrcore	https://opencores.org/prc		Andras Pal	AVR	8	16	artix7-3	James	empty design												LX	Y	verilog	14	top	Y	yes	N		64K	64K	Y					2019	2023	https://szofi.net/p	full implementation of AVR 2-stage pi	variants: VR2, AVR2.5, AVR3, AVR4 & AVR5
softcore-cpu	https://github.com/Ayme		Aymen Sekhri	RISC	32	16															A		vhdl	15	control_u	Y	asm	N		4G	4G	Y	32		7		2019	2021		course project, seven "x86" registers, 32-bit immediates, multi-cycle design	
softpc	https://github.com/alreac		Michael S	Nios II	32	32	cyclone-1	Micha	block	613		4		1	180		q17.1	1.00	5.0	58.9			vhdl	13	nios2ee	Y	yes	opt		4G	4G	Y			32			2019		nine variations in attempt to improve	16-bit ALU
spam-1	https://github.com/Johnlo		John Lonergan	vliw	8	48																	verilog		cpu	Y	yes	N		64K	64K	Y					2019	2023	https://hackaday.i	8 Bit CPU Hardware Implementation i	TTL modules with verilog RTL
sparc64soc	https://opencor	alpha	Dmitry Rozhdestvenski	SPARC	64	32	kintex-7-3	James	errors			6				##	14.7	2.00	1.0			Y	verilog	263	W1	N		Y									2009	2010		huge source file count	work in progress with no progress
spartanmc	http://www.spa	stable	Falk Hassler	RISC	18	18	kintex-7-3	James Brakef		853		6	1	2	120	##	14.7	0.67	1.0	94.6	X	Y	verilog	38	spartanm	Y	asm										2012	2014		SPARC like register windows	
sp-i586	https://github.c	stable	Lini Mestar	x86	32	8	kintex-7-3	James Brakef		32144		6	4	28	73	##	14.7	1.00	2.0	1.1	X		verilog	37	top_sys	Y	yes	Y		4G	4G	Y					2016	2016	http://lmeshoo.ne	gate level dsgn, vivado project also	http://img.youtube.com/vi/2W1guyhCJuE/0.jp
spu-lite	https://github.com/danny		Dongyun Lee	risc	32	32																	verilog	113	top-level	Y	yes	Y		4G	4G				128	11		2025		Sony Cell SPU uP: Dual-Issue Piped Mu	128-bit SIMD register operations
spu-mark-ii	https://github.c	WIP	Felix Queißner	stack	16	16																	vhdl	37	soc	Y		N		64K	64K	Y	34				2020	2023	https://ashet.com	micro-code ISA stack machine	ISA at doc/specs/spu-mark-ii.md
src	https://github.com/KyleLa		Heuring & Jordan	RISC	32	32																	verilog															2018	http://www.zeepe	book by Heuring & Jordan	also Kilts cpt17 Adv FPGA dsgn
srm	https://github.c	WIP	Santiago Licudis	RISC	32	32																																2024	https://www.yout	RISC-V like ISA specification, no RTL	distinct signed and unsigned instructions
ssbcc	https://opencor	stable	Rodney Sinclair	forth	8	9	kintex-7	Rodney Sincla		196		6			474		14.7	0.33	1.0	797.9	ALX		verilog	3	core	Y	asm	N	Y	1K	8K	Y	41		3		2012	2020	https://github.com	Python program generates the Verilog	inst after branch/call/rtn always execs
ssppu	https://github.com/redost		redoste	8085	8	16															X	Y	vhdl	20	board		asm	N		64K	64K	Y	5					2022	https://archive.or	SAP-1 (Simple-As-Possible) architectu	small subset of 8085
stack_machine	http://people.ec	stable	Bruce R. Land	forth	16	5	cyclone10	James Brakef		5101		4	6	29	66	##	q18.0	0.67	0.3	25.9	X		verilog	9	VGA_sram	Y	asm	N	N	64K	4K	N					2009	2011	https://people.ece	(3) uP cores, Cornell course material	VGA output, uses Nakano's tiny_cpu
stack-cpu	https://github.com/Arlet/		Arlet Ottens	stack	16	16															X		verilog	2	cpu					64K	64K	N	23					2017		3 or 4 stacks, load/store with stack de	xilinx block RAM
stacks-16-bit	https://github.com/rcrist/		rcrist	RISC	16	16																	schema	36														2022	https://www.instr	Digital schematic, TTL & 3 layer bread	pictures of 3 layer breadboard
storm_core	https://opencor	beta	Stephan Nolting	ARM7	32	32	kintex-7-3	James Brakef		2312		6	3		179	##	14.7	1.00	1.0	77.4	AX		vhdl	16	core	Y	yes	N		4G	4G	Y			32	8	2011	2014		Storm Core (ARM7 compatible)	I & D caches not compiled
storm_soc	https://opencor	beta	Stephan Nolting	ARM7	32	32	kintex-7-3	James Brakef		3514		6	3	4	159	##	14.7	1.00	1.0	45.2	X	Y	vhdl	40	storm_top	Y	yes	N		4G	4G	Y			32	8	2012	2015		STORM SoC	cache & no peripherals
streamer16	http://www.ultr	stable	Myron Plichota	forth	16	3	kintex-7-3	James Brakef		143		6			417	##	14.7	0.20	1.2	485.6	X		vhdl	8	streamer	Y	yes	N	N	64K	64K	N	8	2			2001	2001	http://www3.sym	MIPS/inst reduced	2nd web adr non-functional
sub86	https://opencor	alpha	Jose Rissetto	x86	16	8	kintex-7-3	James Brakef		1916		6			172	##	14.7	0.67	3.0	20.1	X		verilog	1	sub86	Y	yes	N	N	64K	64K	Y			7		2012	2013		very small x86 subset core	no segment registers, limited op-codes
suite-16	https://github.com/mons(		Ken Boak	accum	16	8																	schema	7														2020		Digital schematic, version of sweet-16	
superscalar_sri	https://github.com/awedi		Aditya Sriram	RISC	16	16																	vhdl	27	datapath	Y	asm	N		64K	64K	N			8		2022	2023	https://anubhavbl	Superscalar RISC, CS 683, derived fron	2-way out of order, GHDL
SuperscalarOo	https://github.com/aman		Amanda Matthes	risc	64	32																	system	10	top	Y						N	2		32		2019	2023		BS thesis: SystemVerilog implementat	six 64-bit ports, simulation?
superscaler-ris	https://github.com/risclite		Li Xinbing	risc-v	32	32																	verilog	15	ssrv_top	Y	yes	N		4G	4G	Y			32		2019	2020		Super-scalar out-of-order RV32IMC	performance: 6.4 CoreMark/MHz
supersmall	http://www.eec	stable	Michael Ritchie	RISC	32	32	stratix_3	Michael Ritch		207		A		2+8	126	##	q9.0	1.00	16.0	38.1	A		verilog														2005	2009		2-bit serial, Mostly MIPS-I compliant	Copyright 2005,2006,2009 Jonathan Rose, and
suska-III	http://www.exp	beta	Wolfgang Forster	68000	16	16	arria-2	James Brakef		7388		A			55	##	q13.1	0.67	4.0	1.3	A		vhdl	11	wf68k00ip	Y	yes	N	N	4G	4G	Y			16		2003	2013		for use as an Atari ST	
suslik	https://opencor	alpha	Goran Dakov	RISC	32	32	kintex-7-3	James	missing file(s)			6				##	14.7	1.00	1.0				verilog	4	cpu	om	asm										2015	2016		"arithmetic core"	has testbench & caches
svc16	https://github.c	emulatior	Jan Neuendorf	mem	16	16																	rust	5	main	Y		N		64K	64K	N					2024	2025	https://github.com	Three address memory to memory CP	Rust source files, see rust-hdl, gens verilog
sweet32	https://opencor	alpha	Valentin Angelovski	MIPS	32	16	kintex-7-3	James Brakef		1050		6	1		142	##	14.7	1.00	1.0	135.1	X	B	vhdl	2	Sweet32_	Y	yes	N	N	4G	4G	Y	26		16		2014	2015		targets MACHXO2, no RAM	
sweet32	https://opencor	alpha	Valentin Angelovski	MIPS	32	16	kintex-7-3	James Brakef		1797		6	1	2	185	##	14.7	1.00	1.0	103.1	X	Y	vhdl	28	sweet32_	Y	yes	N	N	4G	4G	Y	26		16		2014	2015		targets MACHXO2, DDR RAM	clock divider to Sweet32_v1_core
sweet32	https://opencor	alpha	Valentin Angelovski	MIPS	32	16	kintex-7-3	James Brakef		1177		6	1		116	##	14.7	1.00	1.0	98.8	X	B	vhdl	2	Sweet32_	Y	yes	N	N	4G	4G	Y	26		16		2014	2015		targets MACHXO2, no RAM	
swssp	https://www.ipo	patented	Othman Ahmad	RISC	8+	8+																	schematic			Y			Y						8+		2014	2021	https://groups.goo	patent, "simplest scalable" data/inst s	a template for dsgn configuration of uP
swt16	https://github.com/captai		captaindane	RISC	16	16																	verilog	10	swt16-top	Y	asm	N	Y	64K	64K	Y	31		16	5		2020		16-bit, 5-stage RISC uP. RTL description in Verilog. Includes assembler, simulator, and (	
symphony	http://www.ece	alpha	Jason Yu	vect	32	32																	verilog	47	vpu_top												2007	2008		vector addon to NIOS	
synpic12		stable	Miguel Angel Ajo Pelay	PIC12	8	12	kintex-7-3	James Brakef		474		6		1	197	##	14.7	0.33	1.0	136.8	AX		vhdl	7	synpic12	Y	yes	N	N	256	2K	Y					2011	2011	http://projects.nb	CHDL to verilog	bad weblink
SYNtzulu	https://github.c	e risc-v/se	Gianluca Leone	risc-v	32	32	ice40	Gianlu	4 large	3473		4	2	21	24			2.00	32.0	0.4	L	Y	verilog	56	Syntzulu	Y	yes	N		4G	4G	Y	45		32			2025	https://ieeexplore	Tiny RISC-V(serv)-crtld **SNN**: processing 4 synapses & 1 neuron per clock cycle **(spiki**	
sys_180x	https://github.com/zpekic		Zoltan Pekic	1802	8	8																Y	vhdl	65	CDP180X	Y	yes	N		64K	64K	Y	100		16			2020	https://hackaday.i	ucoded 1802 using mcc ucode compile	https://github.com/zpekic/MicroCodeCompiler
sys_emz1001	https://github.com/zpekic		Zoltan Pekic	S2000	4	8	spartan3	Zoltan Pekic		1022	344	4				##	14.7	0.16			X	Y	vhdl	26	EMZ1001A	Y	asm	N	Y	128	4K		59					2022	https://hackaday.i	recreation of Iskra EMZ1001 4-bit mic	no block ram? Picture of original chip
sys_microbasic	https://hackaday.io/proje		Zoltan Pekic	cisc	8	16											14.7					Y	vhdl	36	microbasi	Y	ISE			64K	64K	Y					2025	2026	https://hackaday.i	TinyBasic interpreted by custom micro	updated to 32-bits from 16, uses am9511
sys0800	https://github.c	stable	Zoltan Pekic	TMS0800	4	12																	vhdl	26	sys0800	Y	yes	N	Y	12	512						2019	2020	https://hackaday.i	calculator chip, both TI Datamath and	256x52 micro code
sys9080	https://github.c	stable	Zoltan Pekic	8080	8	8																	vhdl	15	sys9080	Y	yes	N	N	64K	64K	Y					2017	2023	https://opencores	8-bit 8080 CPU based on 29XX bit-slice series of devices AMD 1978 51 pge ap note	
system01	http://members	beta	John Kent, David Burne	6801	8	8	kintex-7-3	James Brakefield				6					14.7	0.33	4.0				vhdl			Y	yes	N	N	64K	64K	Y					2003	2009			
system05	https://opencor	beta	John Kent, David Burne	6805	8	8	kintex-7-3	James Brakef		834		6			204	##	14.7	0.33	4.0	20.2	X	Y	vhdl	10	System05	Y	yes	N	N	64K	64K	Y					2003	2009	http://members.optushome.com.au/jekent/		
system09	https://opencor	stable	John Kent, David Burne	6809	8	8	kintex-7-3	James Brakef		1631		6		41	88	##	14.7	0.33	3.0	6.0	AX	Y	vhdl	40	cpu09l	Y	yes	N	N	64K	64K	Y	44	13	8		2003	2021	http://members.o	from John Kent web page	opencores download URL incorrect, use col E
system11	https://opencor	alpha	John Kent, David Burne	68HC11	8	8	kintex-7-3	James Brakef		1218		6			153	##	14.7	0.33	4.0	10.3	X	Y	vhdl	17	cpu11	Y	yes	N	N	64K	64K	Y					2003	2009	http://members.o	known bugs & untested instructions	
system68	https://opencor	stable	John Kent, David Burne	6801	8	8	spartan-3	James Brakef		2235		4		4	46	##	14.7	0.33	4.0	1.7	X	Y	vhdl	21	cpu68	Y	yes	N	N	64K	64K	Y					2003	2009	http://members.optushome.com.au/jekent/		
system6801	https://opencor	stable	Michael L. Hasenfratz	6801	8	8	cyclone-3	James Brakef		1507		4		3	73	##	14.7	0.33	4.0	4.0	A		vhdl	15	wb_cyclor	Y	yes	N	N	64K	64K	Y					2003	2009	http://members.o	based on John Kent's 6801	tested on Apex20K, Cyclone & Straix boards
t180-cpu		stable	Leonard Brandwein	accum	16	8	kintex-7-3	James	bypas	709		6			83	##	14.7	0.67	3.0	26.2	X		vhdl	23	cpu	Y		N	N	64K	64K	Y	182				2016	2016	https://www.vtto	8-bitter with pc, sp, a, b, c & d regs	based on Viktor Toth's 4 bit microcontroller
t400	https://opencor	stable	Arnim Laeuger	COP400	4	8	spartan-2	Arnim Laeuge		643		3		2	60			0.16	4.0	3.7	AX		vhdl	36	t400_core	Y	yes	N	Y	64	1K	Y					2006	2009		implementation of National's 4-bit COP400 microcontroller	
t48	https://opencor	stable	Arnim Laeuger	MCS-48	8	8	cyclone-1	Arnim Laeuge		738		4		1	59			0.33	4.0	6.6	AX		vhdl	70	t48_core	Y	asm	N		256	1K						2004	2022		T48 uController	used in several projects
t51	https://opencor	stable	Andreas Voggeneder	8051	8	8	kintex-7-3	James Brakef		1942		6	1		147	##	14.7	0.33	4.0	6.2	AX		vhdl	17	T8032	Y	yes	N	N	64K	64K	Y					2002	2010		8052 & 8032	8032 SoC
t65	https://opencor	stable	Daniel Wallner	6502	8	8	kintex-7-3	James Brakef		575		6			291	##	14.7	0.33	4.0	41.7	AX		vhdl	7	T65	Y	yes	N	N	64K	64K	Y					2002	2010		6502, 65C02 & 65C816; wide use	
t6507lp	https://opencor	beta	Gabriel Oshiro, Samue	6502	8	8	spartan-6	James	errors								14.7		4.0				verilog	22	t6507lp	Y	yes	N	N	64K	64K	Y					2009	2010		for use in ATARI 2600	
t80	https://opencor	stable	Daniel Wallner	Z80	8	8	kintex-7-3	James	Z80 m	1389		6			163	##	14.7	0.33	3.0	12.9	X		vhdl	5	T80a	Y	yes	N	N	64K	64K	Y					2002	2018		Z80, 8080 & gameboy inst sets, several usages	
table887	https://github.c	alpha	Robert Finch	RISC	16	16	kintex-7-3	James Brakef		643		6		2	208	##	14.7	0.67	1.0	217.1	X		verilog	2	table887_	Y		N	N	64K	64K		28		8		2014	2016			included with Table888 source code
table888	https://github.c	alpha	Robert Finch	RISC	32	16	kintex-7-3	James Brakef		5756		6	9	6	137	##	14.7	2.00	1.0	47.6	X		verilog	3	table888	ome				4G	4G	Y	130		8		2014	2016		2016 version gives same reults as 201	code for cache & mmu incomplete
tarhi	https://github.c	alpha	Dagvadorj Galbadrakh	RISC	32	32	kintex-7-3	James	everyt	396		6		1	123	##	14.7	1.00	4.0	77.9	X		verilog	4	tarhi_controller			N		16M	16M	N	11		4		2013	2013		no doc, extremely small RISC	difficulty with timing, try 7.0ns
td4	https://github.c	stable	cielo_ee	accum	8	8	spartan-3	James Brakef		102					200	##	14.7	0.20	1.0	392.2	X		verilog	5	td4_top						16	Y					2012	2015			very small uP
temlib	http://temlib.or	stable		SPARC	32	32	kintex-7-3	James Brakef		2579		6		32	111	##	14.7	1.00	1.0	43.1	X		vhdl	48	mcu_simple			Y	N	4G	4G	Y			64		2013	2015		copywrite: experimental use	has caches
temlib	http://temlib.or	stable		SPARC	32	32	kintex-7-3	James Brakef		3730		6	5		111	##	14.7	1.00	1.0	29.8	X		vhdl	48	fpu_simple			Y	N	4G	4G	Y			64		2013	2015		copywrite: experimental use	options for fltg-pt, pipeline, mul & div configura
terracresta	https://github.c	beta	Darren Olafson	68000	16	16															A		verilog	50		Y	yes	N		4G	4G	Y			16		2018	2022		FPGA compatible core of Nichibutsu M	fx86k & t80 cores
tg68	https://opencor	stable	Tobias Gubener	68000	16	16	kintex-7-3	James Brakef		2331		6			44	##	14.7	0.67	4.0	3.2	X		vhdl	2	TG68_fast	Y	yes	N	N	4G	4G	Y			16		2007	2012		TG68 - execute 68000 Code	for use with Minimig
tg68kc	https://opencor	stable	Tobias Gubener	68000	16	16	kintex-7-3	James Brakefield										0.67	4.0		X		vhdl	3	TG68docC	Y	yes	N	N	4G	4G	Y			16		2013	2021		68020 ISA (68000, 68010 & 68020 choice)	
the12X_12uP		alpha	James Brakefield	stack/acc	12	12	kintex-7-3	James Brakef		972		6	1	1	123	##	14.7	0.50	1.0	63.3	X		vhdl	2	the12x_12	Y		Y	N	4K	4K	N	54		64	1	2015			combo stack/accumulater design	load/store arch, not optimized
theia_gpu	https://opencor	beta	Diego Valverde	RISC	96	64	kintex-7-3	James	huge a	934049		6				##	14.7	0.40	1.0		GP		verilog	32	theia												2009	2012		Ray Cast Programable graphic Process	four cores, huge LUT count, 2/3rds LUT RAM
thm-oberon	https://github.com/hgeiss		Hellwig Geisse	risc5	32	32															A	Y	verilog	18	risc5	Y	yes	Y		4G	4G	Y			16			2023	https://github.com	port of Niklaus Wirth's Oberon system	use v1 RTL, from Andreas Pirklbauer's
thor	https://opencor	mature	Robert Finch	RISC	64	16	**zu-5e**	James	WIP							##	v21.1	2.00	1.0				system	27	thor2021	Y	asm	Y		16E	16E	Y			64		2015	2021	https://github.com	Thor-5: L1 & L2 caches, GP float & vec	plans for more features, eventually 2M LUTs
thor	https://opencor	mature	Robert Finch	RISC	32	32		Robert Finch		90000				306									verilog		thor	Y	asm	Y		4G	4G	Y			64		2015	2023	https://github.com	Thor 2015, 2021-3 docs	variable length instructions
thor	https://opencor	mature	Robert Finch	RISC	64	32		Robert Finch		210000				306									verilog		thor2	Y	asm	Y		4G	4G	Y			64		2015	2023	https://github.com	Thor-2: L1 & L2 caches, GP float & vec	96-bit registers
thor	https://opencor	mature	Robert Finch	RISC	64	16		Robert Finch		210000				306									verilog		thor5	Y	asm	Y		4G	4G	Y			64		2015	2023	https://github.com	Thor-5: L1 & L2 caches, GP float & vec	plans for more features, eventually 2M LUTs
tigli_cpu		stable	Cleiton Juffo	RISC	16	16	kintex-7-3	James Brakef		636		6			455	##	14.7	0.67	4.0	119.7	X		verilog	24	cpu	Y		N	Y	64K	64K		16		16		2013	2013		course project, not pipelined	no LUT RAM for reg file
tim	https://github.com/ben-m		Ben Marshall	RISC	32	8	**zu-3e**	James	degenerate synthesis			6				##	v21.1	0.33	3.0				vhdl	15	top	Y		Y		4G	4G	Y	50				2014	2015		TIM: Tiny Instruction Machine, variable length inst	
tiny_cpu	http://www.cs.h	errors	K. Nakano	stack	16		kintex-7-3	James	multiple assignments (			6				##	14.7	0.66	3.0		AX		verilog	11	DE2_TINY	Y	yes	N		4K	4K						2007	2009	http://www.cs.hir	different from tinycpu	**uses Flex, Bison & Perl to create gcc compi**
tiny_soc	https://github.com/ept22		Ezra Thomas	RISC	8	16																Y	verilog	16	top	Y	asm	N	Y	64K	64K	Y	44		16			2020	https://ezrasrobot	small cpu with VGA	includes GPU (char gen)
tiny_zuse	https://github.com/fstolz(		Florian Stolz	zuse z3	22	8																	verilog	4	project	Y		Y	Y	64		N	9					2024	https://tinytapeou	partial implement of Zuse Z3 ALU	design from tinytapeout run tt06
tiny64	https://opencor	stable	Ulrich Riedel	RISC	32	32	kintex-7-3	James Brakef		874		6			189	##	14.7	1.00	2.0	107.9	X		vhdl	6	tinyx					64K	64K		14		8		2004	2007		data size from 32 to 64 bits	micro-coded sub-ops
tiny8	https://opencor	altera dsg	Ulrich Riedel	accum	8	8	arria-2	James	needs async ROM			A				##	q18.0	0.33	3.0		A		ahdl							256	64K	Y			256		2002	2009		Altera megafunctions	

_uP_all_soft folder	opencores or prmary link	status	author	style / clone	data	inst sz	FPGA	repor ter	com ents	LUTs ALUT	Dff	LUT?	mults	blk ram	F max	date	tool ver	MIPS /inst	clks/ inst	KIPS /LUT	ven dor	SOC	src code	#src files	top file	doc	tool chai	fltg pt	Hav'd	max dat	max inst	byt ead	# inst	adr mo	# reg	pip e	start year	last revis	secondary web link	note worthy	comments
tinyavr	https://github.com/Kamp		Daniel Kampert	AVR	8	16															X		vhdl	17	top	Y	yes	N		64K	64K	Y						2022		AVR compatible CPU	
tinycomputer	https://github.com/zpekic		Zoltan Pekic	accum	4	8	spartan3	James Brakef		643	286	4			100	##	14.7	0.17	1.0	26.0	X	Y	vhdl	29	tinycompu	Y		N			256		20		16			2017		4-bit Up via 2901 slice & micro code	no data RAM memory
tinycpu	https://opencor	alpha	Jordan Earls	RISC	8	8	arria-2	James Brakef		136		A			384	##	q13.1	0.17	2.0	235.5	AX		vhdl	2	tinycpu		asm	N	N	1K	1K		12		4		2012	2012	directory contains	subset of 6502	MIPS/inst reduced due to few inst
tinyfpga	https://github.c	stable	Ken Jordan	accum	8	8	kintex-7-3	James Brakef		185		6		1	175	##	14.7	0.33	3.6	86.9	X		vhdl	12	system			N	N	16	16	Y	10				2017	2017		educational 8-bitter with 4-bit addres	why use block RAM?
tiny-gpu	https://github.c	WIP	Adam Majmudar	risc	8	16																	system	12	gpu	Y		N		256	256	Y	11		16			2024	https://news.ycon	multiple compute cores	no overall controller at this time
tinyisa	https://github.com/dillonl		Dillon Huff	RISC	32	32																	verilog	49	cpu	Y		N		4G	4G	N	13		32			2019		very small ISA with multi-cycle, pipelined & with forwarding implementations	
tiny-riscv	https://github.com/husho		Hyounguk Shon	RISC	32	32																	verilog	35	riscv_top	Y		N		4G	4G	Y	24		32			2019		course work, reduced risc-v, 24 inst, four variations: cache, multi-cycle, pipeline & sing	
tinyvliw8	https://opencor	alpha	Oliver Stecklina	VLIW	8	32	kintex-7-3	James	hacke	895		6			149	##	14.7	0.33	1.0	55.0	X		vhdl	19	sysarch			N	Y	256	1K	Y					2013	2020		tinyVLIW8 soft-core processor	bare core, Altera LPM for RAMs
tis-100	https://github.com/Maste		Felix Queißner	accum	8	8																	vhdl	2	tis100	Y	asm	N		256	256	Y	13				2015	2016	https://en.wikiped	programming/puzzle video game by Zachtronics Industries	
tisc	https://opencor	beta	Vincent Crabtree	accum	8	8	kintex-7-3	James Brakef		195		6			87	##	14.7	0.33	1.0	147.1	X		vhdl	1	TISC			N		256	1K	Y			2		2009	2009		Tiny Instruction Set Computer	minimal accumulator machine
tms1000	https://opencores.org/prc		Nand Gates	TMS1000	4	8																	verilog	4	tms1000	Y		N		64	1K		54				2021	2021		Four function BCD calculator chip	used in several TI products
tms1000_fpga	https://github.c	WIP	Michael Kohn	tms1000	4	8													3.0		L		verilog	3	tms1000	Y				1K	64							2024	https://www.mike	TMS1000 and TMS1100	
tms9900	https://github.com/dnotq		Matthew Hagerty	TMS9900	8	8																	vhdl	14	f18a_top	Y		N		64K	64K	Y			16			2019	https://github.con	F18A is a gaming box, conflicts with Cl	Tang Nano 9K F18a Clone
TongjiWork_C	https://github.com/Zhefei		ZhefeiGong	mips	32	32																	vhdl	58	cpu	Y	yes	N		4G	4G		54		32			2024		single cycle and multi-cycle MIPS	31 & 54 inst.
totalcpu	https://opencor	alpha		RISC	12+	12	kintex-7-3	James Brakef		229	64	6	1		149	##	14.7	0.33	3.0	71.7	X		verilog	10	cpu			N							16		2007	2009		data width 12 bits and up, no data memory	
tproc	https://forum.digilent.con		Ted Rossin	maoc	16	16																	verilog	11	tproc1	Y	tproc	N	Y	64K	64K	Y	29	7	16		2025	2026		tProc is a synthesizable 16 bit uP	irregular ISA, CmodA7 xdc files
tpu	https://github.c	untested	Colin Riley	RISC	16	16	spartan-6	James Brakef		1514	984	6	2	11	118	##	14.7	0.67	1.0	52.0	X		vhdl	20	tpu_top	Y		N		64K	64K	Y			8		2016	2016	https://domipheu	Test Processing Unit. Or Terrible Processing Unit. A simple 16-bit CPU in VHDL for edu	
tridora-cpu	https://gitlab.com/sledere		Sebastian Lederer	stack	32	16	artix-7	James	incom	1019	362				100	##	v24.2	1.00	4.0	24.5	X	Y	verilog	20	top	Y	yes	N		4G	4G						2024	2026	https://hackaday.	32-bit stack machine, Wirth pascal	3-bit to 16-bit instructions, some with enables
trivialmips	https://github.com/trivialmips/TrivialMIPS			mips	32	32															X		system	53	trivialMIP	Y		Y		4G	4G	Y			32			2019	https://en.wikiped	MIPS clone, superscaler, done as a prelude to Loongson	
troc16_16	https://github.c	WIP	James Brakefield	RISC	16	16	spartan7	James Brakef		232	50	6	1	1	74	##	v24.2	0.67	1.0	212.8	X		vhdl	5	troc16_16	Y		N		64K	64K	N	25	2	32			2025	https://events.vto	adequate 16-bit uP with op-code space for full TROC ISA; no shift, extract or divide at t	
troc16_16	https://github.c	WIP	James Brakefield	RISC	16	16	spartan7	James	area o	285	50	6	1	1	71	##	v24.2	0.67	1.0	167.9	X		vhdl	5	troc16_16	Y		N		64K	64K	N	25	2	32			2025	https://events.vto	adequate 16-bit uP with op-code spac	half word aligned, 4 tag bits, signed mult
tt06-MiniCPU	https://github.com/jacqu		Jacqueline Gislai	accum	4	8																	verilog	2	tt_um_4b	Y		N		16	NA		11		4			2024	https://app.tinyta	4-bit uP, 16 memory locations, four registers, 11 instructions; tiny tapeout project	
tt-cpu	https://github.com/Moon		Paul Campbell	accum	4	4																	verilog	3	cpu	Y		N		128	128		25		3			2022	https://tinytapeou	4-bit accum, 7-bit PC, 2 7-bit index regs and a carry bit, 8 & 12-bit instructions	
tugman	https://tildegit.org/potatc		tugman?	stack	18	18	tang20k					4			111						G	Y	verilog	8	top	Y	asm	N		256K	4K	N						2025	https://anycpu.or	18-bit stack machine VLIW style	Based on James Bowman's J1 CPU
turbo8051	https://opencor	beta	Dinesh Annayya	8051	8	8	kintex-7-3	James Brakef		1985		6	1		127	##	14.7	0.33	4.0	5.3	AX		verilog	74	oc8051_t	Y	yes	N	N	64K	64K	Y					2011	2016		includes perpherials	
turbo9	https://github.c	WIP	Kevin Phillipson	6809	8	8	artix-7	James	no tim	1428	530	6		8	112	##	v23.2	0.33	3.0	8.6	X	Y	verilog	96	soc_top_g	Y	yes	N		64K	64K	Y	44	13	8	6		2024	https://hackaday.	Compact & Efficient Pipe'd 6809 uP IP	masters thesis, full testbench, ucoded
turbo9	https://github.c	WIP	Kevin Phillipson	6809	8	8	artix-7	Kevin Phillips		1464	505	6		8	112	##	v22.2	0.33	3.0	8.4	X	Y	verilog	96	soc_top_g	Y	yes	N		64K	64K	Y	44	13	8	6		2024	https://www.yout	competes well against other 8-bitters	four videos, see github page
tv80	https://opencor	mature	Guy Hutchison, Howar	Z80	8	8	kintex-7-3	James Brakef		1207		6			182	##	14.7	0.33	3.0	16.6	AX		verilog	6	tv80n	Y	yes	N	N	64K	64K	Y					2004	2020	https://github.con	derived from Daniel Wallner's T80, ASIC implementations	
ucode_cpu	http://minnie.tu	stable	Warren Toomey	RISC	16	16	atrix-7-3	James	4K LU	6748		6	1	1		##	14.7	0.67	2.0		A		vhdl	16	cpu			N	N	64K	64K	N			16		2012	2015			originally schematic based (Logisim)
ucore	https://opencor	stable	Whitewill	MIPS	32	32	kintex-7-3	James Brakef		2469		6		1	231	##	14.7	1.00	1.0	93.5	X		verilog	25	ucore	Y	yes	N		4G	4G	Y			32	6	2005	2010		MMU & caches	
ucpuvhdl	https://github.c	stable	Reed Foster	RISC	8	16	kintex-7-3	James	512 LU	933		6			118	##	14.7	0.33	2.0	20.8	X		vhdl	29	core	Y	asm	N		256	64K	Y	12	2	7		2016	2017	https://github.con	six tutorials on uCPUvhdl	using muCPUv2_1 of 3 upwards compatible des
uos	https://opencor	mature	Daniel Roggen	accum	8	16	kintex-7-3	James Brakef		441		6			270	##	14.7	0.33	3.0	67.4	X		vhdl	14	cpu	Y								3	4		2014	2017		UoS Educational Processor	inspired by x86 ISA
up1232	http://www.dte	stable	Santiago de Pablo	RISC	8	16	kintex-7-3	James Brakef		220		6			244	##	14.7	0.33	3.0	122.0	X		vhdl	3	up1232a			N		64K	64K	Y	33	2	32		2000	2000		bare core, prog size 4K to 64K	description in source files
up3	https://people.e	stable	Bruce Land	accum			cyclone2	Bruce Land		186		4		1		##	q8.0						verilog	1	de2_top															Cornell ECE576	basic core is scomp, used by up3 & de2_top'
urisc		errors	Farhad Mavaddat	RISC	16	16	kintex-7-3	James	missing module			6				##	14.7	0.67	4.0				vhdl	31	urisc	Y				64K	64K	N	1				1987	2012	https://cs.uwaterl	Ultimate Reduced Inst Set Computer Un. Of Waterloo	
usimplez	https://opencor	stable	Pablo Salvadeo etal	accum	12	12	stratix-2	Pablo Salvade		48		4			134		q9.1	0.17	2.0	237.9	A		vhdl	3	usimplez_cpu			N		512	512		8				2011		http://www-gti.de	part of university course, simplez+i4 has an index register	
uTTA		stable	Hans Tiggeler	TTA	16	16	kintex-7-3	James Brakef		810		6	1		57	##	14.7	0.67	1.0	47.4	X		vhdl	23	utta_struc	N	asm	N											http://www.ht-lat	time triggered arch	bad weblink
v1_coldfire	https://www.sil	proprietar	IPextreme	68000	16	16	cyclone-3	freescale		5000		4			80			0.89	1.0	14.2	A		verilog			Y	yes	N	N	4G	4G	Y			16		2008		https://www.silva	free for Altera	3500 LUTs on Stratix-III
v586	https://opencor	beta	Jose Rissetto	x86	32	8	kintex-7-3	James Brakef		22282		6	12	16	102	##	14.7	1.00	2.0	2.3	X		verilog	22	v586	Y	yes	N		1M	1M	Y					2014	2016	https://github.con	MMU & caches, branch cache	www.youtube.com/channel/UCNbm8Bah54cw
v586	https://opencor	beta	Jose Rissetto	x86	32	8	**zu-3e**	James Brakefield				6	12	16	102	##	v21.1	1.00	2.0		X		verilog	22	core	Y	yes	N		1M	1M	Y					2014	2016	https://github.con	MMU & caches, branch cache	www.youtube.com/channel/UCNbm8Bah54cw
v6502	https://github.com/RyuKc		Ryu Kojiro	6502	8	8	**zu-3e**	James	bare c	868	131	6			250	##	v21.1	0.33	3.0	31.7	X		vhdl	23	v6502	Y	yes	N	N	64K	64K	Y					2019	2020	https://opencores	6502 with extras: 16-bit stack pointer	www.youtube.com/watch?v=K3jH-f_r80E
v65c816	https://github.com/RyuKc		Valerio Venturi	6502	8	8	cyclone-IV	Valerio Ventu		1693		4			25			0.33	3.0	1.6	A		vhdl	26	v6502	Y	yes	N	N	64K	64K	Y					2011	2023	https://opencores	6502 with extras: 16-bit stack pointer	https://www.youtube.com/watch?v=K3jH-
v65c816	https://github.com/RyuKc		Valerio Venturi	6502	8	8	cyclone-IV					4			25			0.33	3.0		A		vhdl	29	v65c816	Y	yes	N	N	64K	64K	Y					2011	2023	https://opencores	renamed v6502WS to v65c816, softco	https://www.youtube.com/watch?v=K3jH-
v8cpu	https://github.com/vserge		Vanya Sergeev	RISC	8	16															X		verilog	3	v8cpu	Y	asm	N	Y	64K	64K	Y	15		16			2018		simple(educational) multi-cycle von Neumann architecture 8-bit CPU written in ~440 l	
vax	https://github.c	WIP	Glenn West	vax	32	8															X	Y	vhdl	27	vax_top	Y	yew			4G	4G	Y	75	16	16			2025		VAX-11/780 uP 4 FPGA & OpenVMS	75+ instructions implemented (37% of VAX ISA)
verilog1802	https://github.c	errors	James Bowman	1802	8	8	kintex-7-3	James	errors			6				##	14.7	0.33	4.0				verilog	3	cdp1802	Y	yes	N	N	64K	64K	Y					2015	2020		runs CamelForth	all except RAM in one source file
verilog-6502	https://github.c	stable	Arlet Ottens	6502	8	8	kintex-7-3	James Brakef		407		6			200	##	14.7	0.33	4.0	40.6	X		verilog	2	cpu		yes	N	N	64K	64K	Y					2007	2018	http://ladybug.xs4all.nl/arlet/fpga/6502/		
verilog-6502	https://github.c	stable	Arlet Ottens	6502	8	8	**zu-3e**	James Brakef		475	112	6			333	##	v21.1	0.33	3.0	77.2	X		verilog	2	cpu		yes	N	N	64K	64K	Y					2007	2018	http://ladybug.xs4all.nl/arlet/fpga/6502/		sync memory, e.g. use block RAM
verilog-65C02	https://github.c	alpha	Arlet Ottens	6502	16	8	kintex-7-3	James	remov	599		6		2	204	##	14.7	0.67	4.0	57.1			verilog	5	gop16		yes	N	N	4G	4G						2011	2018	http://forum.6502	16-bit data RAM "bytes"	boot ROM mapped to LUTs?
verilog-65C02	https://github.c	alpha	Arlet Ottens	6502	16	8	**zu-3e**	James Brakef		327	98	6			370	##	v21.1	0.33	3.0	124.6	X		verilog	26	cpu		yes	N	N	64K	64K	Y					2011	2021	https://github.con	used in 100MHZ 6502 DIP module	rewritten for 6LUTs, spartan6 version has black
verilogboy	https://hackada	alpha	Wenting Zhang	risc-v	8	8	**zu-3e**	James Brakef		872	608	6			313	##	v21.1	1.00	3.0	119.5	X		verilog	36	vbh	Y	yes	N	N	64K	64K	Y						2019	https://github.con	Game Boy in Verilog, both CPU (SM83	uses riscv_picorv32 core
verilogboy	https://hackada	alpha	Wenting Zhang	SM83	8	8	**zu-3e**	James Brakef		2415	1601	6		4	238	##	v21.1	0.33	3.0	10.8	X	Y	verilog	22	boy	Y	yes	N	N	64K	64K	Y						2019	https://github.con	Game Boy in Verilog, both CPU (SM83	also https://github.com/neildryan/GBA
verilog-harvard	https://github.com/jaywo		Jae-Won Chung	RISC	16	16	**zu-3e**	James	multi-	165	96	6			250	##	v21.1	0.67	1.0	1015	X		verilog	7	cpu02	Y		N	N	64K	64K	N	23		4		2019	2019		multi-driven nets	multi cycle CPU that has an IPC of 1
verilog-harvard	https://github.com/jaywo		Jae-Won Chung	RISC	16	16	**zu-3e**	James	multi-driven net			6				##	v21.1	0.67	1.0		X		verilog	7	cpu03	Y		N	Y	64K	64K	N	23		4	5	2019	2019		multi-driven nets	5-stage pipelined CPU, same for cpu4 thru cpu1
verilog-harvard	https://github.com/jaywo		Jae-Won Chung	RISC	16	16	**zu-3e**	James	multi-driven net			6				##	v21.1	0.67	1.0		X		verilog	7	cpu04	Y		N	Y	64K	64K	N	23		4	5	2019	2019		multi-driven nets	Data forwarding from the ALU
verilog-harvard	https://github.com/jaywo		Jae-Won Chung	RISC	16	16	**zu-3e**	James	multi-driven net			6				##	v21.1	0.67	1.0		X		verilog	7	cpu05	Y		N	Y	64K	64K	N	23		4	5	2019	2019		multi-driven nets	Branch prediction with a BTB with 2-bit saturat
verilog-harvard	https://github.com/jaywo		Jae-Won Chung	RISC	16	16	**zu-3e**	James	multi-driven net			6				##	v21.1	0.67	1.0		X		verilog	7	cpu06	Y		N	Y	64K	64K	N	23		4	5	2019	2019		multi-driven nets	tournament branch predictor
verilog-harvard	https://github.com/jaywo		Jae-Won Chung	RISC	16	16	**zu-3e**	James	multi-driven net			6				##	v21.1	0.67	1.0		X		verilog	7	cpu07	Y		N	Y	64K	64K	N	23		4	5	2019	2019		multi-driven nets	Memory latency parameter
verilog-harvard	https://github.com/jaywo		Jae-Won Chung	RISC	16	16	**zu-3e**	James	multi-driven net			6				##	v21.1	0.67	1.0		X		verilog	8	cpu08	Y		N	Y	64K	64K	N	23		4	5	2019	2019		multi-driven nets	instruction cache and data cache
verilog-harvard	https://github.com/jaywo		Jae-Won Chung	RISC	16	16	**zu-3e**	James	multi-driven net			6				##	v21.1	0.67	1.0		X		verilog	9	cpu09	Y		N	Y	64K	64K	N	23		4	5	2019	2019		multi-driven nets	DMA module and its interrupt mechanism
verilog-harvard	https://github.com/jaywo		Jae-Won Chung	RISC	16	16	**zu-3e**	James	multi-driven net			6				##	v21.1	0.67	1.0		X		verilog	10	cpu10	Y		N	Y	64K	64K	N	23		4	5	2019	2019		multi-driven nets	DMA interleaved with instructions that access t
verilog-harvard	https://github.com/jaywo		Jae-Won Chung	RISC	16	16	**zu-3e**	James	multi-	171		6			357	##	v21.1	0.67	1.0	1399	X		verilog	5	cpu01	Y		N	N	64K	64K	N	23		4		2019	2019		multi-driven nets	single cycle CPU that has an IPC of 1
verilog-harvard	https://github.com/jaywo		Jae-Won Chung	RISC	16	16																	verilog	74	cpu	Y		N		64K	64K	N			4		2019	2019		**ten implementations of increasing so**	**missing memory & test bench RTL**
verysimplecpu	https://github.com/MC2S		Abdullah Yıldız	mem	32	32																	verilog			Y	yes	N	N	16K	16K	N	8	2			2014	2019	https://github.con	educational, 2 address, public version is missing processor RTL	
vespa	http://www.arc	untested	David J. Lilja	RISC	32	32																	verilog			Y	asm	N		4G	4G	N	16		32		2005	2005		from book: Designing Digital Computer Systems with Verilog 0-521-82866-X, Un. Minn	
vhdl_cpu	https://github.com/CGras		Charles Grassin	accum	8	16	spartan3	Charles Grass		203	116	4					14.7	0.20	2.0				vhdl	6	computer	Y	asm	N	N	256	256	N	14				2017	2020	http://charleslabs	educational, very simple	case statement program
vhdl_cpu	https://github.com/Harine		Harineko0	RISC	8	16																Y	vhdl	10	cpu	Y		N	N	4K	4K	Y	20					2025		Single-Cycle Processor made with VHI	although cpu-vhdl placed in cpu_vhdl folder
vhdl-cpu	https://github.com/nhewi		Nathan Hewitt	arm9	32	32																	vhdl	11	pipelined	Y	yes	N		4G	4G	Y	14		16			2021		pipelined CPU running a subset of the	both single cycle & pipelined, course project
vhdl-cpu2	https://github.com/lebric		Fabrice Normandin	mips	32	32																					asm	N		4G	4G	Y	29		32	5		2018		McGill Un. Course, MIPS CPU/VHDL	MIPS inst card, pipe hazard notes
vhdl-msp430	https://github.c	mature	Rafael Hideo Toyomot	MSP430	16	16																	vhdl	15	processad	Y	yes	N		64K	64K	N	27		16		2018	2018		course project, inspired by msp430, very little commentary	
vhdl-simple-up	https://github.c	untested	Pietro Lorefice	RISC	16	16	arria-2	James	ran out of memory			A				##	q18.0	0.67	1.0				vhdl	10	processor	Y		N	N	64K	64K	N			16		2014	2014		simple processor using VHDL for logic	based on Gray's xsoc
vhdl-simple-up	https://github.c	untested	Pietro Lorefice	RISC	16	16	kintex-7-3	James	ran out of memory			6				##	14.7	0.67	1.0				vhdl	10	processor	Y		N	N	64K	64K	N			16		2014	2014		simple processor using VHDL for logic	based on Gray's xsoc
vic64-t9k	https://github.com/joachi		joachimdraeger	6502	8	8	tang9k														T	Y	vhdl &	37	top	Y	yes	N		64K	64K	Y						2025	https://hackaday.	Commodore 64 VIC-II Chip, 6502 CPU,	derived from c65_mister & t65
vm80a	https://github.c	untested	1801BM1	8080	8	8	cyclone-3			607		4			104						A		verilog	6	de0_top												2014	2018		Two versions of Soviet i8080a reverse engineered from silicon die, 607 4LUTs, 104MH	
vscale	https://github.c	stable	UC Berkeley	risc-v	32	32	kintex-7-3	James Brakef		3072		6			127	##	14.7	1.00	1.0	41.2	X		verilog	23	vscale_core			N							32		2016	2017		risc-v RV32IM vscale processor, depre	depreciated: not up to date (risc-v)
w11	https://opencor	alpha	Walter Mueller	PDP11	16	16	kintex-7-3	James Brakef		1760		6	1	1	147	##	14.7	0.67	2.0	28.0	X	Y	vhdl	118	pdp11_co	Y	yes	N	N	4M	4M	Y	70	13	8		2010	2023	https://github.con	Boots UNIX, has MMU & cache, retro	PDP-11/70 CPU core and SoC
w450		errors	Ze Long	CISC	8	8	kintex-7-3	James	blocking & non-blockir			6				##	14.7	0.33	3.0				verilog	3	w450					256	256	Y	8		4	3	2012			appears to be class project	3 versions of w450, used latest, patches caused
w65c832	https://github.com/mikea		Michael Kohn	6502	32	8	tang20k														LT	Y	verilog	16	w65c832	Y	yes	N		16M	16M	Y					2024	2025	https://hackaday.	32 bit 6502, 16MB via bank registers	32 bit regs, 24-bit adrs, also see af65k
wb_z80	https://opencor	stable	Brewster Porcella	Z80	8	8	kintex-7-3	James Brakef		2025		6			144	##	14.7	0.33	3.0	7.8	X		verilog	4	z80_core_	Y	yes	N	N	64K	64K	Y					2004	2012		derived from Guy Hutchison TV80	Wishbone High Performance Z80
wb4pb	https://opencor	stable	Stefan Fischer	picoBlaze	13	13	kintex-7-3	James	incomplete port to kcp			6				##	14.7	0.33	3.0			Y	vhdl, v	14	picoblaze_wb_uart				Y								2010	2013	https://en.wikiped	software addon for picoBlazeSoftware	ported to kcpsm6
wb4pb	https://opencor	stable	Stefan Fischer	picoBlaze	13	13	spartan-3	Stefan Fische		309		4		1	102	##	14.7	0.33	3.0	36.2	X	Y	vhdl, v	14	picoblaze_wb_uart				Y								2010	2013	https://en.wikiped	software addon for picoBlazeSoftware	kcpsm3 only works for Spartan 3
whitebeard	https://github.com/Mega		Vuk Đorđević	RISC	8	16	cyclone-3														A		vhdl		cpu			N		64K	64K	Y	20	2	8		2022	2023		simple risc, shift ops, schematic captu	ISA doc only on github web page
whitham_68k	https://www.jw	errors	Jack Whitham	68000	32	16	kintex-7-3	James	no top module							##	14.7	0.67	4.0				vhdl			Y	asm			4G	4G	Y			16		2002	2003		university project, 68020 subset	read thesis, code generator for top modules
x32	http://citeseerx.	stable	Sijmen Woutersen	forth	32	8	kintex-7-3	James	missing defines			6				##	14.7	1.00	1.0				vhdl	32	core	Y	yes	N		4G	4G	Y					2006	2007	https://pdfs.sema	MS thesis, byte code, needs caches	uses preprocessor on VHDL
x9	https://github.com/yehzh		Simon Zhang	RISC	8	9		James Brakef	no IO, will be degenerate														system	24	top_level	Y	asm	N		256	256	Y	13		16		2016	2017		9-bit processor: 4:1:4 op-code, R0, R1 fields	
xgate	https://opencor	alpha	Robert Hayes	RISC	16	16	kintex-7-3	James Brakef		2778		6			159	##	14.7	0.67	1.0	38.3	X		verilog	7	xgate_top	Y		N					42		16		2009	2013		high pin count	Freescale XGATE co-processor compatible
xmega_core	https://opencor	beta	Gheorghiu Iulian	AVR	8	16	kintex-7-3	James Brakef		1116		6			120	##	14.7	0.33	1.0	35.6	X		verilog	34	mega_cor	Y	yes	N		64K	128K	Y	72		32		2017	2018	https://git.morgot	8 AVR cores, 4 sets LUT counts posted	https://git.morgothdisk.com/VERILOG/VERILOG
xproz	http://www.bitl	stable	Herbert Kleebauer	CISC	16	16													1.0				schematic			Y	asm	N		64K	64K						1993	1995		documentation in German	*.1 schematic design
xpu	http://excamera	macros	James Bowman	forth	16	8	kintex-7-3	James	requres preprocessor			6					14.7	0.67	1.0				vhdl	1	c2a												2003	2003		predates J1	uses preprocessor on VHDL
xr16	https://github.c	stable	Jan Gray	RISC	16	16	kintex-7-3	James Brakef		273		6			263	##	14.7	0.67	1.0	644.8	X		verilog	4	xr16	Y		N		64K	64K				16		1999	2001	https://github.con	handcrafted instruction set	tool FPGA P&R, speed mode better
xr16	https://github.c	stable	Jan Gray	RISC	16	16	**zu-2e**	James	needs	346		6			282	##	v20.1	0.67	1.0	547.0	X		verilog	4	xr16	Y		N		64K	64K				16		1999	2001		handcrafted instruction set	tool FPGA P&R, speed mode better
xsoc	http://www.fpg	stable	Jan Gray	RISC	16	16	kintex-7-3	James	very sl	371		6				##	14.7	0.67	1.0		X		verilog	16	xsoc	Y	yes	N	N	64K	64K	Y	16	4	16		2000	2001	https://github.con	very compact, bare core	similar to xr16
xtensa	https://ip.caden	proprietar	tensilica/cadence	RISC	16	16,2	proprietary																proprietary							4G	4G				32	5,7			ch 8, Processor De	upward compatible family, sliding reg	ASIC usage, TIE tool generates RTL & software t
xthundercore	http://forum.ga	alpha	majordomo	RISC	32	16	kintex-7-3	James Brakef		793		6		2	193	##	14.7	1.00	1.0	243.7	X		vhdl	49	xtc	om	yes	N	Y	4G	4G				16	5	2014		http://www.xthur	Gadget Factory Forum thread	in debug, no comments, mostly in simulation
xucpu	https://opencor	alpha	Jurgen Defurne	RISC	16	16	spartan6-	James Brakef		356		6		4	187	##	14.7	1.00	1.0	524.8	X	Y	vhdl	25	system_4k					4K	4K						2015	2017		Experimental Unstable CPU	
xulalx25soc	https://opencor	mature	Dan Gisselquist	RISC	32	32	spartan6-	James	Sparta	7936		6	4	25	87	##	14.7	1.00	1.0	11.0	X	Y	verilog		toplevel			N	N	4G	4G	N	20		16	5	2015				uses ZIP CPU
y80e	https://opencor	stable	Sergey Belyashov	Z80	8	8	cycone-3	Sergey Belyas		2557		4				##	14.7	1.00	3.0				verilog	15	top_level	Y	yes	N	N	64K	64K	Y					2013	2019		Y80e - Z80/Z180 compatible processo	based on Y80 from "Microprocessor Design Usi
yacc	https://opencor	stable	Tak Sugawara	MIPS	32	32	kintex-7-3	James	map e	2220		6	6			##	14.7	1.00	1.0		AX		verilog	10	yacc2	Y	yes	N		4G	4G	Y			32	5	2005	2009		derived from, but independent of plas	YACC Yet Another CPU CPU

_uP_all_soft folder	opencores or prmary link	status	author	style / clone	data	inst sz	FPGA	repor ter	com ents	LUTs ALUT	Dff	LUT?	mults	blk ram	F max	date	tool ver	MIPS /inst	clks/ inst	KIPS /LUT	ven dor	SOC	src code	#src files	top file	doc	tool chai	fltg pt	Hav'd	max dat	max inst	byt ead	# inst	adr mo	# reg	pip e	start year	last revis	secondary web link	note worthy	comments
yafc	https://github.c	alpha	Tim Wawrzynczak	forth	16		kintex-7-3	James Brakef		617		6		4	247	##	14.7	0.67	1.0	268.5	X		vhdl	20	cpu		asm	N	Y	8K	8K		26					2014			influenced by J1, F16 & C18
yard-1	https://github.c	alpha	Brian Davis	RISC	32	16	artix-7	James Brakef		774	271	6		2	89	##	v25.1	1.00	1.0	114.3	LX		vhdl	38	evb	Y		N		4G	4G	Y	60		16		2014	2020		32 bit uP core, intended for embedded	three data sizes, well documented
yari	https://github.c	stable	Tommy Thorn	MIPS	32	32	kintex-7-3	James Brakef		3610		6		15	189	##	14.7	1.00	1.0	52.3	X	Y	verilog	8	top					2M	2M				32		2004	2008		subset of MIPS R3000	
yarvi	https://github.c	beta	Tommy Thorn	risc-v	32	32	cyclone-V					6			100			1.00	1.0		AL		verilog	10	yarvi_soc	Y	yes	N	N	4G	4G				32	8	2016	2022		rewritten for perfomance	
yarvi	https://github.c	beta	Tommy Thorn	risc-v	32	32	kintex-7-3	James Brakef		2152		6		17	122	##	14.7	1.00	2.0	28.3	X		verilog	3	yarvi_soc	Y	yes	N	N	4G	4G				32	3		2016		no multiply or divide	simple implementation of RISC-V
yasep	https://hackada	alpha	Yann Guidon	RISC	16	32	kintex-7-3	James	reduce	632		6			215	##	I14.7	1.00	2.0	170.0	AX		vhdl	3	microYAE	Y	asm	N	N	2G	2G		51		16		2005	2018	www.youtube.com	JavaScript generated VHDL, revisions ongoing	
yfcpu	https://github.c	errors	Cory Walker	RISC	16	16	kintex-7-3	James	degen	18		6				##	14.7	0.67	1.0				verilog	2	yfcpu	Y		N	N	256	256	Y	5	1	16				Colin Mackenzie?	Educational	very simple
ygrec8	https://hackaday.io/proje		Yann Guidon	RISC	8	16																	vhdl					N		256	256	Y	20		8		2017	2023	https://hackaday.i	educational uP with front panel	front panel: one button per op-code
z3	https://opencor	stable	Charles Cole	CISC	8	8	arria-2	James Brakef		3495		A	2		141	##	q18.0	0.33	3.0	4.4	A		verilog	3	boss	Y				128K	128K						2014	2014	https://en.wikipec	Infocom Z-Machine V3, youtube video	http://inform-fiction.org/zmachine/standards/
z8086	https://github.com/nand2		nand2mario	x86	32	8	artix7	James Brakef		1865	573	6		33	75	##	v25.2	0.67	10.0	2.7	AGX		system	6	z8086_top	Y	yes	N	N	1M	1M	Y						2025	https://nand2mar	8086 Core Powered by Original Micro	project files
z386	https://github.com/nand2		nand2mario	x86	16	8	cyclone-V	nand2mario		18000	5000	A		15	85			0.67	10.0	0.3	AG		system	16	z386	Y	yes	N	N	1M	1M	Y						2026	https://hackaday.	Open-Src 80386 using Orig ucode	cache, see z386 and ao486 section
z486	https://github.com/nand2		nand2mario	x86	32	8	de10nanc	nand2mario		24411	#####	A	12	92	58	##	q26.1	0.67	10.0	0.2	AX	Y	system	57	z486	Y	yes	Y		4G	4G	Y			16			2026	https://github.com	core delivers 486DX2-66-class perf	from a0486, both z486 & z486_mister
z486	https://github.com/nand2		nand2mario	x86	32	8	kv260	nand2mario		41290	#####	6	19	26	100	##	v26.1	0.67	10.0	0.2	AX	Y	system	57	z486	Y	yes	Y		4G	4G	Y			16			2026	https://github.com	core delivers 486DX2-66-class perf	see nand2mario.github.io/posts/2026/z486/ fo
z86	https://github.com/nand2		nand2mario	x86	32	8	cyclone-V	nand2mario				A			90			0.67	10.0		X		system	11	z86	Y	yes	N	N	1M	1M	Y			5			2025	https://nand2mar	High-Performance 80286-Class Softco	see z386 and ao486
z80control	https://opencor	alpha	Tyler Pohl	Z80	8	8	kintex-7-3	James Brakef		1483		6			189	##	14.7	0.33	3.0	14.0	X	Y	verilog	55	top_de1	Y	yes	N	N	64K	64K	Y					2010	2012		Microprocessor targeting embedded industrial control systems	
z80-fpga	https://github.com/Obijua		Juan Gonzalez-Gomez	Z80	8	8															L		verilog	5		Y	yes	N	N	64K	64K	Y						2020		Based on iceZ0mb1e by abnoname and TV80, with tinyBasic	
z80soc	https://opencor	stable	Ronivon Costa	Z80	8	8	spartan3e	James Brakef		2474		4	2	19	78	##	14.7	0.33	3.0	3.4	AX	Y	vhdl	19	top_s3e	Y	yes	N	N	64K	64K	Y					2008	2016		based on Daniel Wallner's T80	directory disappeared
z80soc	https://opencor	stable	Ronivon Costa	Z80	8	8	**zu-3e**	James Brakefield				6				##	v21.2	0.33	3.0		AX	Y	vhdl	19	top_s3e	Y	yes	N	N	64K	64K	Y					2008	2016		based on Daniel Wallner's T80	
z80-verilog	https://github.c	ASIC dsgr	Timo Nicolai	z80	8	8												0.33	3.0				verilog	29	z80	Y	yes	N	N	64K	64K	Y						2021		ASIC Thoroughly Tested Verilog Impl c	DRAM interface for both inst & data
zap	https://opencor	alpha	Revanth Kamaraj	ARM7	32	32	arria-2	James	high D	10284		A	2	38	111	##	q18.0	1.00	1.0	10.8	X		verilog	37	zap_top	Y	yes	N	N	4G	4G	Y			16		2017	2022	ddi0100e_armv1-	ARMv4T & Thumbv1	has cache & mmu
zap	https://opencor	alpha	Revanth Kamaraj	ARM7	32	32	kintex-7-3	James Brakef		7558		6	1	9	135	##	14.7	1.00	1.0	17.9	X		verilog	37	zap_top	Y	yes	N	N	4G	4G	Y			16		2017	2022	https://github.com	ARMv4T & Thumbv1	has cache & mmu
zet86	https://opencor	alpha	Zeus Marmolejo	x86	16	8	kintex-7-3	James Brakef		3642		6	1		68	##	14.7	0.67	2.0	6.2	X		verilog	32	fpga_zet_	Y	yes	N	N	1M	1M	Y					2008	2018	https://github.com	equivalent to 80186, boots MS-DOS	Zet The x86 (IA-32) open implementation
zipcpu	https://github.c	stable	Dan Gisselquist	RISC	32	32	kintex-7-3	James Brakef		1687		6		2	218	##	14.7	1.00	1.0	128.9	AX		verilog	7	zipcpu	Y		N	N	4G	4G	Y	35		16	5	2015	2024	http://zipcpu.com	ISA has chnaged, multiple instruction	support for several FPGA boards
zktc	https://github.c	WIP	kkinos	RISC	16	16	spartan-7	James Brakef		1172	690	6		16	112	##	v25.1	0.67	1.0	63.8	X		system	6	zktc	Y	yes	N		64K	64K	N			8			2024		hobby project to design CPU, create compiler, create kernel and run it	
z-machine	https://github.c	stem veril	Robert Baruch	CISC	8	8	arria-2	James Brakefield				A				##	q18.0	0.33	3.0		A		system	15	plugh	Y		N									2016		http://inform-ficti	Z-machine (Zork)	https://www.youtube.com/watch?v=2fNBkUCj
zoom	https://github.com/zenco		Cui Yikai	mips	32	32																	scala	147	cputop	Y	yes	N		4G	4G	Y	90		32			2023		configurable out-of-order MIPS32 uP	thesis, MMU, caches, branch table, AXI
zpu	https://github.c	stable	Oyvind Harboe	forth	32	8	kintex-7-3	James Brakef		1073		6	3		283	##	14.7	1.00	4.0	65.9	X		vhdl	23	zpu_core	Y	yes	N		4G	4G	Y	37				2008	2009	https://grokipedia	zpu4: 16 & 32 bit versions, code size 8	ZPU the worlds smallest 32 bit CPU with GCC to
zpuflex	https://github.c	mature	Alastair M. Robinson	forth	32	8	cyclone-3	Alasta	appro	1000		4											vhdl	4	zpu_core	Y	yes	N		4G	4G	Y	37				2014	2015	https://github.com	addditional instrucitons	
zpuino	http://alvie.com	alpha	Alvaro Lopes	forth	32	8	spartan6-	James Brakef		2547	2200	6	4	11.5	126	##	14.7	1.00	4.0	12.3	X	Y	vhdl		papilio_pr	Y	yes	N		4G	4G	Y	37				2008	2018	https://github.com	SoC version of modified ZPU	pipelined, removed ucf file
zpuino	http://alvie.com	alpha	Alvaro Lopes	forth	32	8	kintex7-3	James	failed in translate			6				##	14.7	1.00	4.0		X	Y	vhdl		papilio_pr	Y	yes	N		4G	4G	Y	37				2008	2018	https://github.com	SoC version of modified ZPU	pipelined, removed ucf file
ztapchip	https://github.c	stable	Vuony Nguyen	MIPS	32	32	cyclone5	James Brakef		31331		A	43	578	100	##	q18.0	1.00	1.0	3.2	AX	Y	vhdl	53	ztachip	Y											2015	2015		multi-core with MIPS master	files no longer available, was under developme
ztapchip	https://github.c	stable	Vuony Nguyen	MIPS	32	32											q18.0	1.00	1.0		AX	Y	vhdl	53	ztachip	Y											2015	2022		vexriscv uP, AXI crossbar	Intel & Xilinx support, runs tensor flow

133 # usable(beta, stable or m 28 143 292 blank 725 # 691 # 63 535 verilog 499 non-blank 864 ## 765 40 30

51 "B" or "X" of limited interest 1203 917 876 vhdl 447 asm 180 Web page DMIPS pe en.wikipedia.org/wiki/Instructions_per community.freesc www.eembc.org/coremark/index.php

80 zu-3e sys verilog 96 forth 14 DMIPS per clock for many microprocessors: http://en.wikipedia.org/wiki/Instructions_per_second

MIPS/MHz Pro-rating for data size:

1-bit	0.04	16-bit	0.67	64-bit	2.00
4-bit	0.17	24-bit	0.80		
8-bit	0.33	32-bit	1.00		
12-bit	0.40	48-bit	1.50		

Under the assumption that the core is capable of one instuction per clock

Silicon Area equivalents 6LUT or ALUT ~= 1.5 4LUT

LUTS/DSP48	16:1
LUTS/Block RAM	32:1

proprietary	38
scala	29
schematic	33
vhdl, verilog	19

773 Unique folders in this sheet

77	_paper_only
60	educational
25	_weak_start
8	_up_cores
27	in limbo
10	planning
81	simulation
573	main+sim
492	net main
639	total

428	VHDL
476	Verilog
82	System Verilog
28	Spinal/Scala
19	VHDL, Verilog
3	MyHDL
38	proprietary
14	other
33	Schematics
####	total

https://github.com/fayalalebrun/awesome-spinalhdl (17) scala/spinal CPUs

418 designs with FOM (KIPs/LUT) results (some duplicates due to multiple FPGA runs)

385 designs with best FOM (likely true measure of # of usable designs)

Column Titles	Details
"A"	A: 1st choice clone, B: 2nd choice clone, W: 1st choice original, X: 2nd choice original
"B"	used to indicate best KIPS/LUT for a given design, usually using fast FPGA family
cat	main, educational, planning, simulation, paper, in limbo or weak
_uP_all_soft folder	if opencores design is their folder name, otherwise my folder name
opencores or primary link	about 200 designs in open cores, about 100 in github
status	ASIC, paper (detailed in), planning (no source), alpha, beta, stable, mature, proprietary, untested; incomplete, educational typically <16 instructions, simulation
author	First Name, Last Name or university or corporation
style / clone	part number or "forth", RISC, accumulator, etc. "asic" indicates: avail as asic & fpga, an asic netlist source or a hard core within fpga chip
data size	data register size in bits
inst size	shortest instruction size in bits
FPGA	FPGA family for compile, place, route & timing, usually using fastest part grade
reporter	First Name, Last Name
comments	compile, place, route & timing problems
LUTs ALUT	total number of LUTs, ALUTs or tiles used including route-thrus & otherwise unavailable
Dff	total number of DFFs
LUT?	4-LUT, 6-LUT, Altera ALUT, Actel Tile
mults	total number of multipliers/DSPs used; 9x9 multiplier counts divided by two and rounded up
blk RAM	total # of block RAMs used, Xilinx half block RAM counts divided by two and rounded up
Fmax	maximum primary clock speed from compile, place & route run with best clock constraint, fastest part, best die temp
date	date of compile, place & route; serves to identify source version
tool ver	Altera (**Q**uartus), Xilinx (ISE, **V**ivado), Lattice Semiconductor(**D**iamond) or MicroSemi(**L**ibero) tool version number
MIPS /inst	prorated DMIPS per instruction, reduced for data word sizes under 32-bits, greater than one for multiple issue processors
clks/ inst	number of clocks per instruction, typically 1.0 for modern pipelined processors, subjective for older uP
KIPS /LUT	figure of merit, does not include effects of memory capacity, floating point or instruction set quality
Vendor	Vendors for which design builds: Actel: Libero, Intel(Altera): Quartus; Latticesemi: Diamond & iCEcube, Xilinx: ISE & Vivado
SOC	B: bare core (no RAM connections or memory access delay), Y: System on a Chip (has peripherals)
src code	VHDL or Verilog or System Verilog or schematic or gates or Proprietary or Scala etc
# src files	number of source files for compile, place, route & timing; includes test benches
top file	top file for compile, place, route & timing run, multiple versions of same design distinguished here
doc	is documentation provided?
tool chain	is there a compiler or assembler provided or available
fltg pt	does the compile, place, route & timing run include floating point?
Hav'd	H: separate instruction and data memory(s), 2C: # caches, M: MMU, N: von Neuman (single memory bus)
max data	maximum data address
max inst	maximum instruction address
byte adrs	is byte addressing provided
# inst	number of unique instructions, conditionals count as one instruction, somewhat subjective
# adr modes	abs, imm, PC rel, indexed, reg-reg indexed; stack, indir, indir++, --indir; (indir), (indir++), (--indir), (indexed), abs-short/direct page, scaled
# reg	number of registers in register file
pipe len	number of pipeline stages
start year	year of first design activity
last revis	last year for revisions or web page updates
secondary web link	secondary web address
note worthy	anything special about the design